GEK-36431

INSTRUCTIONS

3S7513TC300A11

ELECTROHYDRAULIC CONTROL

FOR

BOILER FEEDPUMP TURBINE

GENERAL ELECTRIC

AUGUST, 1973

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

3S7513TC300A11
ELECTROHYDRAULIC CONTROL
FOR
BOILER FEEDPUMP TURBINE

INTRODUCTION

The electronic portion of an Electrohydraulic Control system for a boiler feedpump turbine drive is described in these instructions.

Any adjustment or maintenance of this equipment should be performed either by qualified turbine personnel or supervised by such personnel.

This equipment is an electronic speed governor whose speed set point can be varied by an automatic boiler controller or by an operator from a remote control unit. Electronic redundancy and mechanical overspeed trip are used to provide a high degree of reliability and safety.

The elementary diagram and alignment diagram for this unit are included in these instructions. Since both the number and arrangement of auxiliary components vary with particular application, it is necessary to refer to the drawings furnished with each equipment to determine the correct connections and auxiliary components furnished in each instance.

RECEIVING, HANDLING AND STORAGE

Receiving and Handling

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

Storage

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be placed in long term storage, the interior cabinet temperature should be 5°C above ambient.

DESCRIPTION

Speed Control

The speed control loop will be described first. Starting at zone B6 and B7 of the Alignment Drawing, the coil of the electrohydraulic servo valve receives a low power signal from the electronic unit and controls the flow of oil to a double acting hydraulic cylinder which positions the turbine steam control valve. Turbine speed is sensed by three magnetic sensors mounted near the periphery of a shaft driven gear. Pulses whose frequency is directly proportional to turbine speed are generated as each gear tooth passes a sensor. Primary and secondary speed channels in the electronic control produce a DC voltage proportional to the frequency of the sensor outputs. Outputs of two speed channels are fed into high value gate number one, which rejects the lower of the two speed signals and passes

on the higher valued speed signal. This redundancy permits one channel to fail without significantly affecting turbine speed. The secondary channel is adjusted for an output, 1% less than the primary channel. It is therefore uncontrolling, except in case of failure of the primary channel or if a "Backup Amplifier Test" switch in the cabinet is actuated which increases the output of the secondary channel to 1% greater than the primary channel. A test meter in the cabinet can be used to determine the controlling channel.

The speed signal from high value gate one is summed with the speed set point to obtain a speed error. See zone B3 of the alignment diagram. The speed error is passed through a stability compensation and error amplifier, which includes a regulation adjustment to obtain a valve lift demand signal. This is summed with a valve position signal to obtain a valve position error which is passed through the servo amplifier to the servo valve. Increasing the speed set point increases valve lift demand, valve position error, and valve opening. Speed will increase until the rise matches the set point increase.

The valve position signal is obtained from a Linear Variable Differential Transformer (LVDT) whose core is positioned by the control valve. Three kilohertz excitation for the LVDT is generated in the electronic cabinet. A demodulator in the cabinet produces a DC signal proportional to the AC output of the LVDT which varies with core position.

A control valve closing bias is applied directly to the servo coil of the control valve in the tripped condition. Before the EHC can reset the feedpump turbine, a valve closing signal from the servo amplifier (2SA) must be present to operate the Close Signal Check (CSC) relay and remove the bias signal that had been previously applied directly to the servo valve coil. Relay circuitry also requires that the electrical connectors must be plugged into the servo valve, the LVDT, and the speed sensors before starting. The LVDT signal is biased to close the control valve if the LVDT connector is removed with the turbine running. The servo valve contains a 5% mechanical bias which will close the control valve in the absence of an electrical signal.

Speed Indication

At zone B9 of the alignment diagram, a third magnetic sensor and circuitry idential to the previously described speed signal is used for speed indication. An analog signal is obtained for speed indication on the EHC Remote Control Panel, local panel, and customer use.

A clamp signal from the 10% speed voltage comparator holds the turbine to less than 10% speed if both the primary and secondary speed channels show less than 10% speed.

At 105% speed a voltage comparator operates a relay that unlocks the lock out valve in the emergency trip exerciser scheme. At 110% speed another voltage comparator operates the 110SX relay to indicate that 110% speed has been reached. A contact of this relay is provided for the customer.

Speed Reference or Set Point

Identical motor driven LVDTs are used to generate the manual and automatic speed references. The core position is varied by a

two-phase stepping motor which moves in discrete steps and locks in position when not running. Raise and Lower pushbuttons on the operator's manual control panel cause the manual reference to run from the LSS stop to the HSS stop in 80 to 90 seconds. Raise Fast and Lower Fast pushbuttons will run the manual reference from stop to stop in approximately 10 seconds. The automatic speed reference circuits contain interface circuitry for compatibility with a specific boiler controller. The auto speed reference is adjusted to run from the LSS to the HSS in 10 seconds. The automatic and manual speed references are fed into a low value gate which rejects the higher reference and passes on the lower reference to control turbine speed. A relay operated from this gate indicates which of the two references is controlling.

An overspeed test can be performed after first running both manual and automatic references to the high speed stop. The overspeed test pushbutton is then actuated causing an additional speed reference to be generated which increases linearly with time at approximately 12% speed per minute. This is added to both manual and automatic references. If turbine speed rises to 120% of rated, a speed signal will override the ramp generator and prevent further speed increase. Upon release of the Test pushbutton, the reference ramps back to normal at the same rate as it increased.

High Pressure/Low Pressure Stop Valve Indication

The EHC cabinet contains circuitry for driving HPSV and LPSV position indicators located on the control panel supplied with the EHC cabinet. The signals for these position indicators are obtained from LVDTs mounted on their respective stop valves. This function is for position indication only and is not part of the control circuitry for the feedpump. Refer to zone C7, C8 of the alignment drawing.

Low Speed Switch

An electronic Low Speed Switch (LSS) is provided in the EHC cabinet to automatically, if selected, engage the turning gear on decreasing turbine speed less than 2 rpm.

If at turbine speeds greater than 200 rpm, pulses are lost from the LSPU shown at zone A9 of the alignment drawing, the LSS will prevent the LSSX relay from energizing and prevent automatic engagement of the turning gear by the EHC.

The LSS provides an alarm under the following conditions:

1. If the turbine is at standstill, 0 rpm, and then is increased in speed and pulses are not detected from the LSPU, the LSS 2K contacts will energize the LSS Alarm X relay when 200 rpm is reached.

2. If the pick-up pulses are detected from the LSPU but lost before 200 rpm is reached, the LSS 2K contacts will energize the LSS Alarm X relay when 200 rpm is reached.

3. If pick-up pulses are lost from the LSPU at turbine speeds greater than 200 rpm, the LSS 2K contacts immediately energize the LSS Alarm X relay.

4. Once the LSS Alarm X relay is energized, the automatic

engagement of the turning gear is prevented until pulses from the LSPU are obtained.

Power Supplies

Two redundant power supplies, referred to as Line A and Line B, are provided. They are the full wave rectifier, capacitor filtered variety. Each receives its 120 VAC from the customer's power.

The 120 VAC is fed to a transformer that has three secondary windings. The output of each of the rectified, filtered supplies is connected to three identical series regulators that are adjusted for 30 VDC, 24 VDC and -22 VDC.

The outputs of the series regulators on the Line A supply are adjusted slightly higher so they are carrying load and the Line B supply is the backup power supply.

Series Regulator

Operation of the series regulators is described as follows. Output of the voltage sensing bridge consisting of 15R, 16R, 17R, 1P and 2ZD is fed to 7Q and 8Q. An increase in output voltage raises the voltage on 8Q, increasing its conduction and lowering conduction of 7Q. Decreased current in 7Q decreases current in 4Q, 5Q, and 6Q dropping the output voltage until balance is restored. Excessive current output through 10R turns 3Q on, 2Q off, and 1Q on, grounding the base of 7Q. With 7Q off, series regulators 4Q, 5Q, and 6Q are off and the output goes to zero. When 1C discharges, 3Q is turned off and output is restored. The circuit will oscillate between off and on until the cause of the high current is removed.

MONITORING

Located in the front of the cabinet are two meter panels that contain several alarms, meters, indicating lights and test meter that will allow checking or monitoring of the power supplies and most amplifiers in the cabinet.

Power Supply Alarm

For example, an alarm will occur if the 30 volt Line A power supply should exceed the 95% or 105% set point of the meter relay in the cabinet. There will be three lights that will be illuminated. First, the Power lamp on the control panel supplied with the cabinet will change from a steady glow to a flashing indication, the Alarm lamp in the cabinet will be illuminated and the illuminated pushbutton above the Line A 30 volt meter will indicate either a high or low voltage.

The three Alarm lamps can be cleared after the malfunction is corrected by depressing the lighted pushbutton above the Line A 30 volt meter. Identical functions are provided for both the Line A and Line B 30 V, 24 V, and -22 V power supplies and 3 KHz 6 V RMS power supply in the EHC cabinet. If any of these power supplies are not at their proper voltage level when the 120 V 60 Hz is applied, startup of the feedpump will be prevented.

Ground Alarm (24 VDC)

If a ground were to occur on the positive or negative side of either the Line A or Line B 24 volt power supply, the Power lamp on the control panel would flash, the +24 V GND/-24 V GND lamp in

the cabinet would be illuminated, and the Alarm lamp in the cabinet would be illuminated. These lamps will be cleared when the ground is removed. Depressing the +24, -24 V lamp in the cabinet is not required to clear the lamp.

HSS/LSS Alarm

HSS and LSS indication is provided both in the cabinet and on the control panel for both the Manual and Auto Speed reference travel limits. These lamps will be cleared when the respective speed reference is driven off the limit.

Cable Check Alarm

The Cable Check will prevent startup of the feedpump if the cables are not properly connected. The Power lamp on the control panel will flash, the Cable Check lamp in the cabinet will be illuminated, and the Alarm lamp in the cabinet will be illuminated until the cable is properly reconnected.

Lamp Test

A lamp test can be performed on the indicating lamps in the cabinet by depressing the Alarm pushbutton. There aren't any lamps in the Backup Speed Amplifier Test pushbutton.

Depressing the Power pushbutton on the control panel will test the lamps on the control panel. There aren't any lamps in the Lower Fast, Lower, Raise, Raise Fast pushbuttons.

Test Switch

The amount of current being supplied by each series regulator can be observed by the Test Switch and meter in the cabinet. When measuring series regulator current the meter is calibrated to read approximately 5 amps full scale.

The voltage output of amplifiers 1 thru 12 can be observed by using the Test Switch and meter. When measuring amplifier outputs the meter is calibrated to read 10 volts full scale. The amplifier numbers and function are shown on the alignment drawing and elementary diagram.

INSTALLATION AND ADJUSTMENT

The electronic control cabinet should preferably be mounted in a clean dry air-conditioned environment. Ambient temperature should not exceed 104°F unless otherwise noted on the outline drawing for a specific unit.

Equipment adjustment and the required test equipment are covered in detail in an alignment diagram and alignment instructions furnished with each unit.

The location of sub assemblies within the cabinet is shown on the elementary diagram furnished with each control. Printed circuit board locations are shown on sheets 1 and 2 of the elementary diagram.

OPERATION

Refer to the instruction book for the turbine.

MAINTENANCE

No periodic maintenance is required. Outage time due to component failure

is best minimized by maintaining a complete set of spare parts. Elementary diagrams of the printed circuit boards are included in this instruction book. The assembly drawings showing component location on the printed circuit boards are furnished with each order except where component identifications are marked on the boards, in which case assembly drawings are not furnished.

TROUBLESHOOTING

Most troubleshooting problems can be quickly isolated to a particular section of the control by use of the built-in test meter, the meter relays, and alarm lights. These used in conjunction with the alignment diagram indicate the operating condition of each section. An oscilloscope is recommended for pinpointing a defective component or components. Each printed circuit board can be removed from the rack for servicing without disconnection due to the use of terminal boards.

RENEWAL PARTS

When ordering renewal parts, address the nearest Apparatus Sales Office of General Electric Company and supply the following information.

a. Catalog number stamped on the part and a complete description of the part, including its use and location.

b. Complete nameplate data appearing on the assembly of which the part is a component.

c. If possible, data on original order on which equipment was first supplied, including all numerical references.

Since operation of a steam turbine-generator is dependent upon operation of the feedpump control, it is suggested that an adequate stock of spare parts be carried on hand to minimize outage time due to component failure.

METER RELAY LAMP REPLACEMENT

Type 195 Meter Relays have been made with two versions of lamps; one without leads attached to the lamp (GE Pt. No. 4868K32P001) and the other with leads attached to the lamp and lamp holder assembly (GE Pt. No. 4872K12G700). These replacement parts can be ordered directly from: the nearest General Electric Sales Office or Utility & Process Automation Products Department, General Electric Company, 40 Federal Street, West Lynn, Mass. 01910.

In the event of lamp failure, the applicable lamp or lamp assembly can be replaced when the control unit is detached from the rear of the indicator set-point unit. Then, the lamp-access plate (see Figure) must be removed (screws are beneath adhesive labels on some models) to expose the lamp holder which can be pulled out.

Pt. No. 4868K32P001 Lamp

When installing the bayonet-type Pt. No. 4868K32P001 (without leads), twist the lamp in the holder assembly until the lamp filament will be in the horizontal position when the lamp holder is replaced in the indicator set-point unit.

Pt. No. 4872K12G700 Lamp & Holder

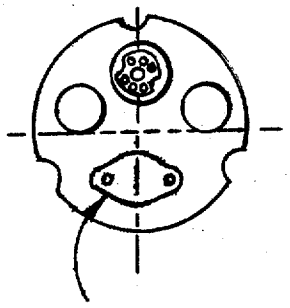
The Pt. No. 4872K12G700 lamp has been fixed in the lamp holder at the factory such that the filament position will be in the correct position when the keyed assembly

is inserted in the indicator set-point unit. Make sure the lamp-lead clips are engaged by the screws when the lamp access cover is replaced.

Note: If relay operation is not within ±2% of set-point index, the lamp holder position must be rotated 180°.

Reassembly

After the new lamp or lamp assembly has been installed, make sure the lamp access plate is replaced (and screws are tight) before control unit is reassembled on rear of indicator set-point unit.



LAMP ACCESS PLATE

DRAWING LIST

Description	Drawing Number
Diode Board, 1DB	44A300243
Reference Low Value Gate, 2LVG	44A300281
Compensation and Regulation, 1CR	44A301400
Overspeed Integrator, 1OSI	44A302692
Signal Low Value Gate, 1LVG	44A304763
Alignment Instructions	44A306302
Series Regulator Board #1, 1REC	44A306392
Renewal Parts	44A335415
Cable Wiring Diagram, Signal & Low Voltage, 1CO.	44B300423
Control Panel Outline	44B306123
Low Speed Switch Power Supply, LPS	44B306144
Cable Wiring Diagram Power, 2CO & 3CO	44B306148
Series Regulator Board #2, 1SR	44B306197
Detector Indicator, 1DI	44C300609
Operational Amplifier, 10A	44C300610
Analog Output, 1A02	44C301216
Analog Output, 3A02	44C301622
Frequency to Voltage Converter, 2FVC	44C301682
Clock and Stepping Motor Drive, CL&SMD	44C303567
Servo Amplifier, 2SA	44C304444
3 kHz Oscillator, 6 OCS	44C304462
Signal Converter, 3SC	44C304898
Voltage Comparator, 1VC	44C305259
Elementary Drawing, Master	44C306284
Connection Drawing	44C306285
Low Speed Switch, LSS	44C306444
Boiler Control Interface, 7BCI	44C306492
Line A & Line B Power Supply	44C306526
Cabinet Outline	44D300043
Alignment Drawing	44D305954
Control Panel Connection Diagram	44D305988
Relay Panel Connection Diagram	44D305989
Summary	44X303673

GENERAL ELECTRIC

44A300243
CONT ON SHEET SH NO.

REV NO. 0

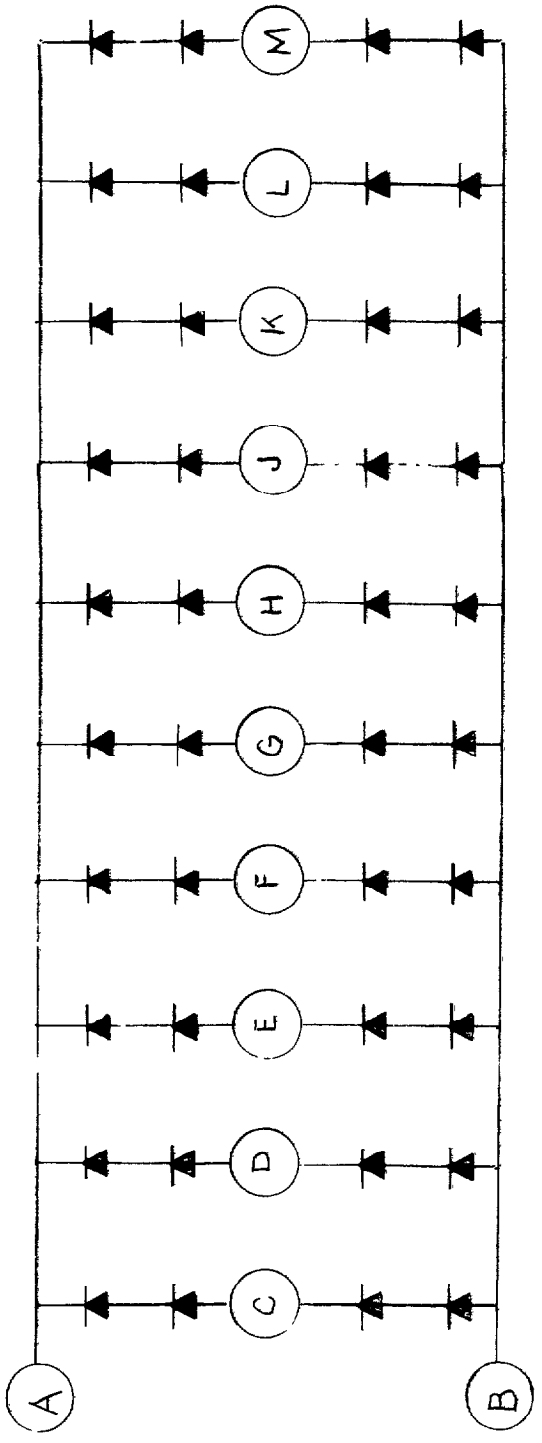
CONT ON SHEET SH NO.

TITLE

DIODE BOARD 1DB

FIRST MADE FOR 44D236370-G01

REVISIONS



ALL DIODES 200 MA; 200 V.

11

DFB 12P

PRINTS TO

MADE BY R. Chambers 1/15/68

ISSUED MAR 12 1968

APPROVALS MBC

SPECIALTY CONTROL DIV OR DEPT.

WAYNESBORO, VA LOCATION

44A300243
CONT ON SHEET SH NO.

FF-803-WF (2-67)
PRINTED IN U.S.A.

CODE IDENT NO.

2

GENERAL ELECTRIC

44A300281

CONT ON SHEET SH NO

REV NO. (1) C KENNEDY

CONT ON SHEET SH NO.

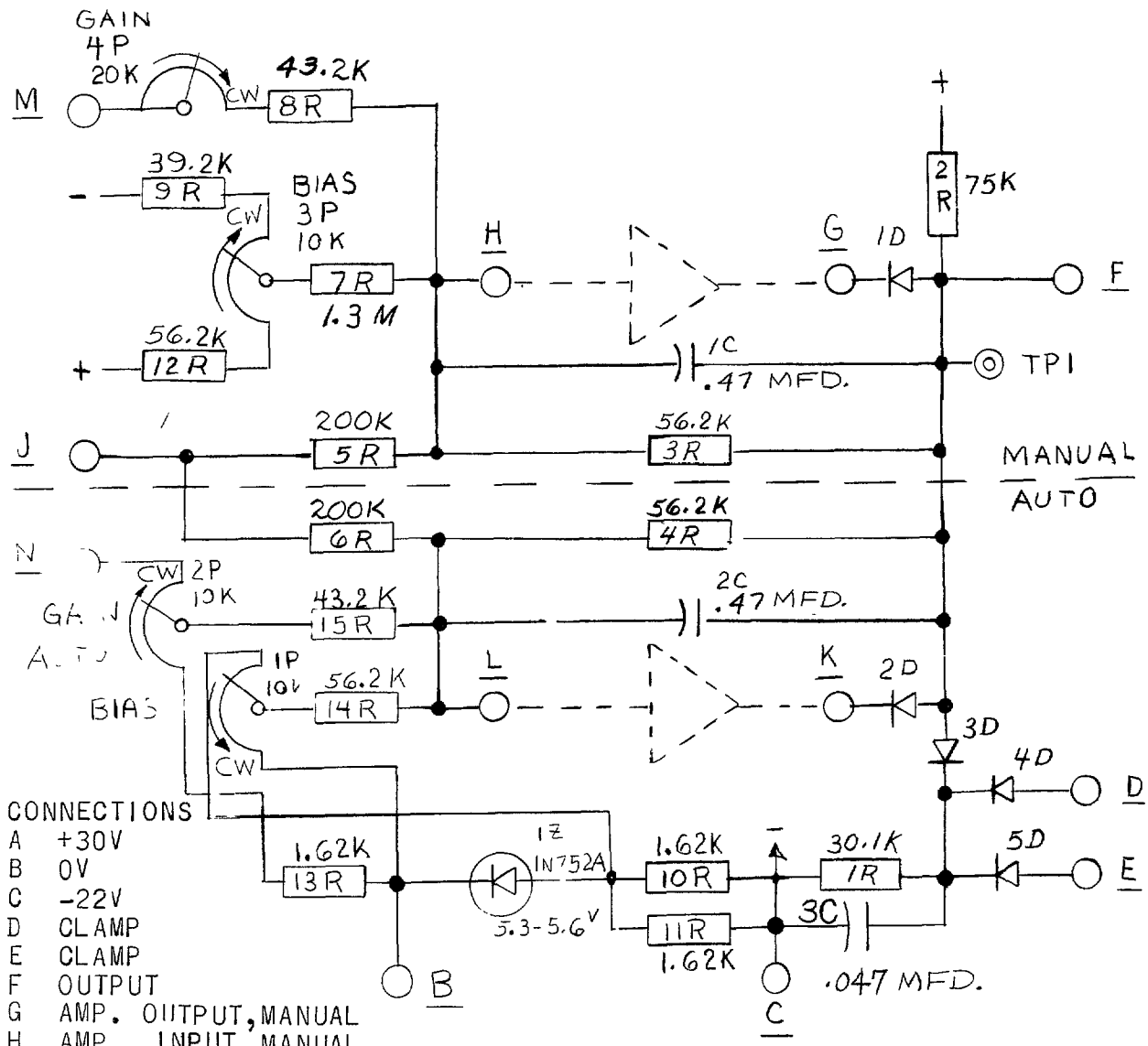
TITLE

REFERENCE LOW VALUE GATE (2LVG)

FIRST MADE FOR 44D236378-G01

REVISIONS

1	CHANGED 5R, 6R, 8R VALUES BUKOSKEY	MAY 5 1969
2	C.A. Lewis	OCT 19 1970
3	Wadsworth	Nov 17, 1971



CLAMP NOTE: ZERO VOLTS ON D OR E WILL HOLD OUTPUT (F) AT ZERO.

ADJUSTMENT:

1: WITH AUTO INPUT AT HSS AND MANUAL AT LSS, ADJUST BIAS FOR ZERO OUTPUT AT F.

2: WITH BOTH GAINS MAX. (CW) AND INPUTS AT HSS, ADJUST MANUAL GAIN FOR 5 VOLTS OUTPUT.

3: DECREASE AUTO GAIN UNTIL OUTPUT OF THE ASSOCIATED AMPLIFIER SWITCHES NEGATIVE TO POSITIVE.

4: AUTO BIAS IS SET CCW INITIALLY. FINAL SETTING PER APPLICATION REQUIREMENT.

11

12P DFB

FG-W

PRINTS

MADE BY B Chambers March 11 1968

ISSUED APR 17 1968

APPROVALS R W Simmons

SPECIALTY CONTROL DIV OR DEPT

WAYNESBORO, VA LOCATION

44A300281

CONT ON SHEET SH NO.

CODE IDENT NO

FF-803-WF (9-66)
PRINTED IN U S A.

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GENERAL ELECTRIC

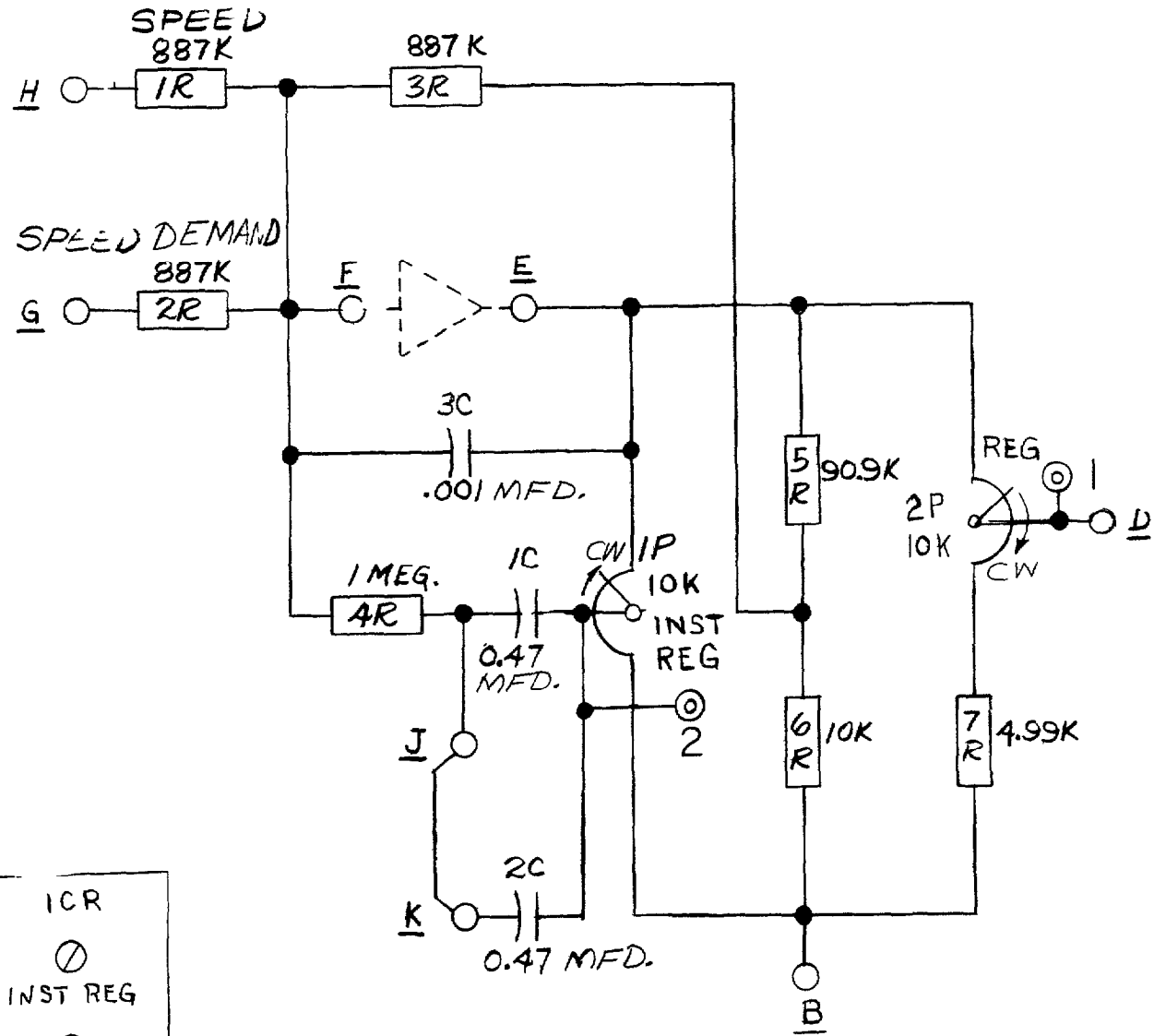
44A301400
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REV NO.

44A301400
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TITLE
COMPENSATION & REGULATION (ICR)

FIRST MADE FOR 44D236388-G01



ICR
INST REG
REG

REVISIONS

1	R M. BUKOSKEY APR 1 1969	2	E.A. Ellington Sept. 2, 1971
3	J. Ardizzone Nov 17, 1971	4	J. L. Brooks Aug. 28, 1972

11

PRINTS TO

MADE BY R.M. Bukoskey
ISSUED FEB 18 1969

APPROVALS R W

SPECIALTY CONTROL DIV OR DEPT
WAYNESBORO, VA. LOCATION

44A301400
CONT ON SHEET SH NO.

CODE IDENT NO.

FF-803-WF (9-67)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44A302692
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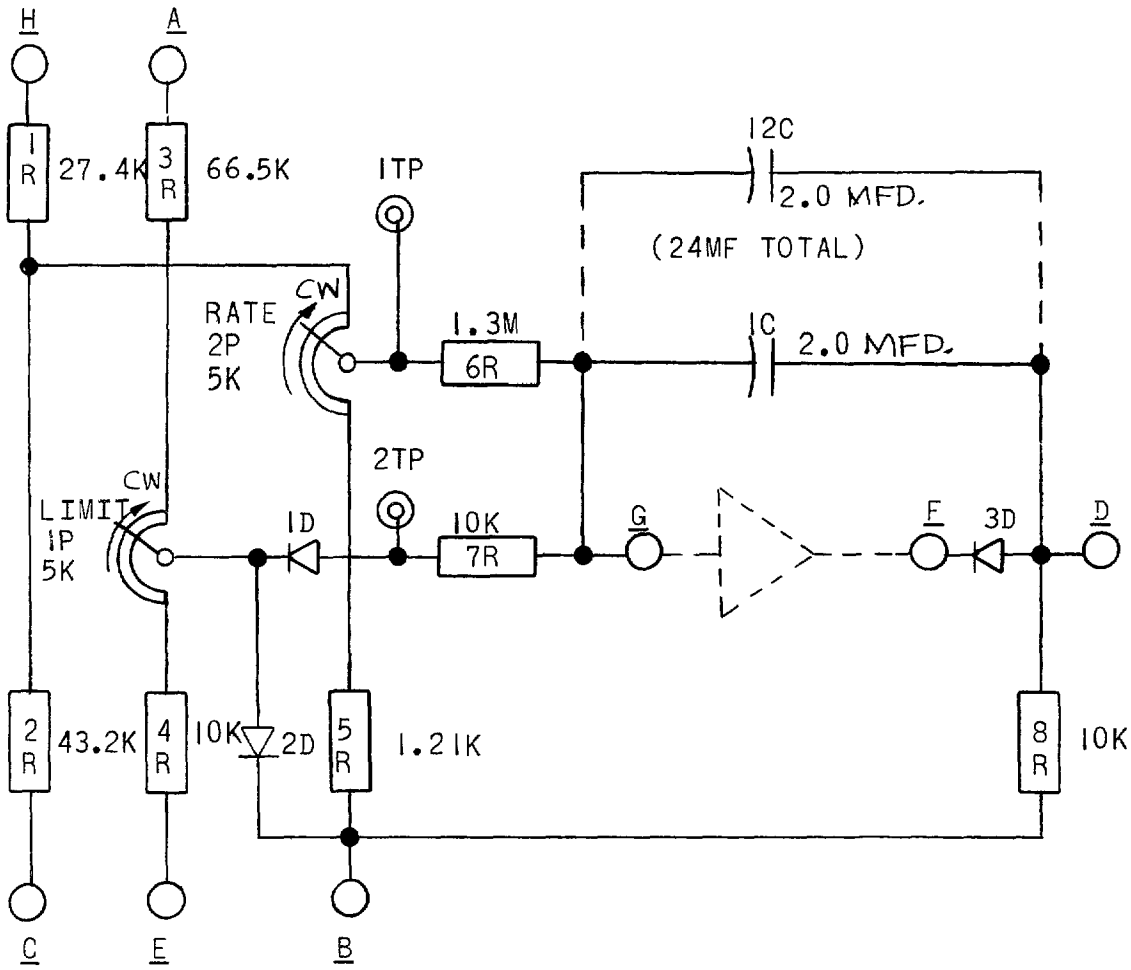
REV NO. ① C. KENNEDY

44A302692
CONT ON SHEET SH NO.

TITLE
OVERSPEED INTEGRATOR IØSI

FIRST MADE FOR 44D236373-G02

REVISIONS



CONNECTIONS

A + 30V
B 0V
C -22V
D OUTPUT
E SPEED SIGNAL
F AMPLIFIER OUTPUT
G AMPLIFIER INPUT
H POS. SUPPLY THROUGH RELAY
J N.C.

PRINTS

MADE BY A. Templeton 3-16-70
ISSUED APR 8 1970

APPROVALS R. Culver

Comm & Gen. Devices DIV OR DEPT
Waynesboro, Va. LOCATION

44A302692
CONT ON SHEET SH NO

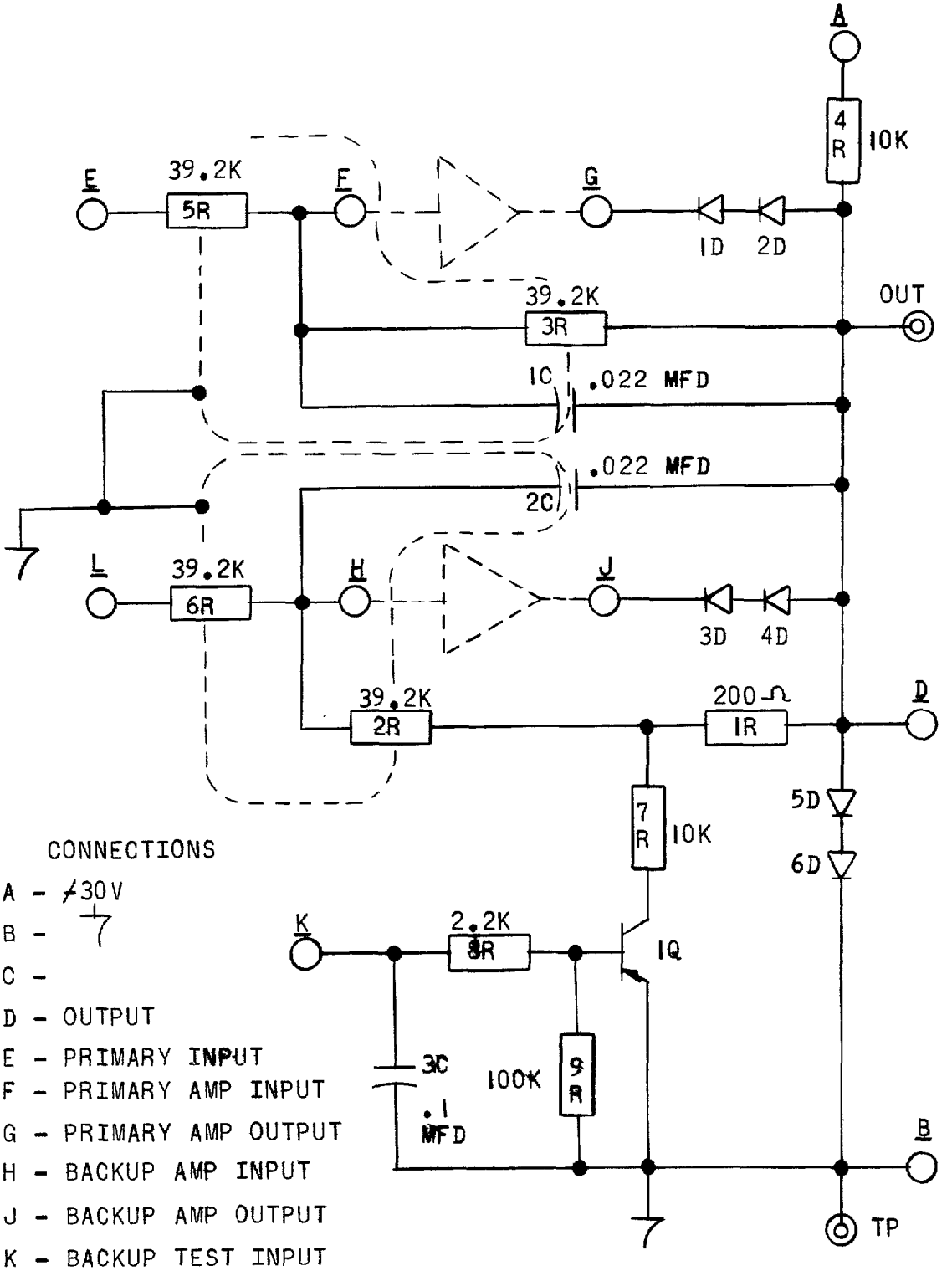
FF-803-WF (10-67)
PRINTED IN U S A.

CODE IDENT NO
2+1T

GENERAL ELECTRIC

REV NO.	TITLE	44A304763
44A304763	SIGNAL LOW VALUE GATE	CONT ON SHEET SH NO.
CONT ON SHEET SH NO.	FIRST MADE FOR 44D236372-G02	ILVG

REVISIONS



CONNECTIONS

A - +30V

B - ⏚

C -

D - OUTPUT

E - PRIMARY INPUT

F - PRIMARY AMP INPUT

G - PRIMARY AMP OUTPUT

H - BACKUP AMP INPUT

J - BACKUP AMP OUTPUT

K - BACKUP TEST INPUT

L - BACKUP INPUT

Nov. 13, 1971

11

12P DFB

FG-W

PRINTS TO

MADE BY	APPROVED	COMM. & CONT. DEVICES DIV OR DEPT	44A304763
H. Fogelson 8-13-71	R. Culver	WAYNESBORO, VA. LOCATION	CONT ON SHEET SH NO.
ISSUED Aug. 16, 1971	D.L. McKnight		

FF-803-WF (2-62)
PRINTED IN U.S.A.

TITLE TAMPA ELECTRIC
ALIGNMENT INSTRUCTIONS
ELECTRONIC FEED PUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A11 TURBINE 122881

REVISIONS

REFERENCE ALIGNMENT DIAGRAM 44D305954

TEST EQUIPMENT

1. AUDIO OSCILLATOR: HEWLETT PACKARD 201C, 200CD; GENERAL RADIO 1210C, OR EQUIVALENT. OUTPUT MUST BE 10 VOLTS RMS MINIMUM INTO 600 OHMS.
2. AC/DC DIGITAL VOLTMETER: 0.1 MV TO 100 V MAX.
3. DIGITAL FREQUENCY COUNTER: 6 VOLT RMS SENSITIVITY. 30 TO 5000 HZ.
4. TEN TURN POTENTIOMETER; NOT MORE THAN 5000 OHMS RESISTANCE.
5. TWO INPUT OSCILLOSCOPE, CALIBRATED SWEEP: X1 AND X10 PROBES.
6. OPTIONAL: BRUSH RECORDER SPECIAL TESTS ARE DESIRED.
7. VOLT - OHM - MILLIAMMETER
8. TWO 100K RESISTORS, TWO 1 MFD CAPACITORS (TO ADJUST OSCILLATOR OFFSET)

I. PRELIMINARY

1. CHECK INTERCONNECTION OF CONTROL CABINET, OPERATOR'S CONTROL DEVICES, TURBINE DEVICES, ETC.
2. MECHANICALLY ZERO TEST METER IN CABINET AND OPERATOR'S TACHOMETER AND OTHER INSTRUMENTS.

II. LINE "A" AND LINE "B" POWER SUPPLIES LOC. 12 AND 13

1. SET UP CONNECT 120V, 60 HZ TO LINE "A" POWER SUPPLY TERMINALS T24 AND T25. REMOVE POWER FROM LINE "B" TERMINAL T22 & 23.
2. DC SERIES REGULATORS, ADJUST POWER SUPPLIES AND METER CALIBRATIONS PER CHART. IF THE VOLTAGES LISTED CANNOT BE OBTAINED IT MAY BE NECESSARY TO ADJUST THE TAPS ON TRANSFORMER 1T, LOC. 12 & 13.

LINE "A" SUPPLY	TEST POINTS	ADJUST
+30V	BUS BARS	0308
-22V	BUS BARS	0316
24V	+F LOC. 0322; -G, LOC. 0324	0324

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12P-DFB 2
FG-W

PRINTS TO

REVISIONS

3. METER RELAY ALARMS VARY ALARM SET POINT OF LINE "A" POWER SUPPLY METER RELAYS TO CHECK ALARM OPERATION. SET ALARMS FOR 95% AND 105%.

4. CONNECT POWER TO LINE "B" POWER SUPPLY BY JUMPERING TERMINAL T24 TO T22 AND T25 TO T23, SO THAT POWER IS APPLIED TO BOTH SUPPLIES.

5. ADJUST THE INDIVIDUAL LINE "B" SERIES REGULATOR BOARDS SO THAT THEY ARE CARRYING ONLY 0.1 AMP AND THE LINE "A" SUPPLIES THE REMAINING LOAD. WHILE NOTING THIS CONDITION IS MAINTAINED MAKE FINAL ADJUSTMENTS IF NECESSARY PER CHART.

SUPPLY	LINE B	LINE A	TESTPOINTS
+30	0310	0308	BUS BAR
-22	0318	0316	BUS BAR
+24	0326	0324	LINE A, + F, 0322; -G, 0324 LINE B, + F, 0328; -G, 0326

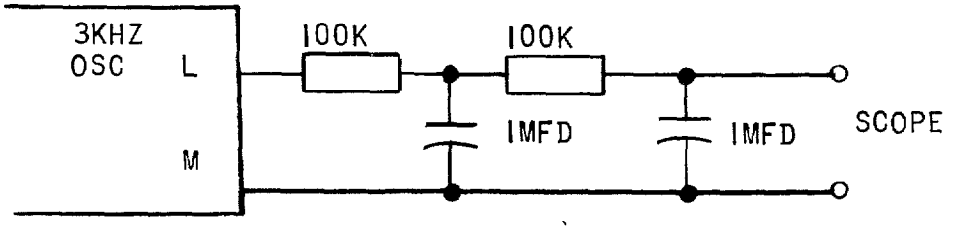
6. TEST AND SET LINE "B" METER RELAY ALARMS AT 95% AND 105%

III. 3KHZ OSCILLATOR, -0217

1. VERIFY OUTPUT FREQUENCY AT 0217 IS 3000HZ. ± 300 HZ. USING COUNTER AT TERMINALS L&M.

2. ADJUST VOLTAGE LEVEL FOR 6.000 ± .006V. SET METER FOR 100%. CHECK METER RELAY ALARMS AND SET AT 95 AND 105%.

3. OFFSET ADJUSTMENT, CONNECT A FILTER CIRCUIT AS SHOWN OR DIGITAL VOLTMETER TO L & M. VOLTMETER ON MILLIVOLT RANGE.



3.1 REMOVE 3KHZ OSC BD FROM RACK AND ADJUST OFFSET FOR ZERO D.C. VOLTS ± 1MV.

IV. SSPU SPEED INDICATION CIRCUITS

1. SET UP, REMOVE LEAD FROM T28 AND T29 AND CONNECT AUDIO OSCILLATOR. SET OSCILLATOR TO SSPU RATED HZ.

//
12P-DFB 2
FG-W

PRINTS TO

REVISIONS

2. FREQUENCY TO VOLTAGE CONVERTER

2FVC LOC. 0224. ADJUST COARSE AND FINE POTS TO OBTAIN ⌿5.00V AT TERMINAL J.

3. VOLTAGE COMPARATORS IVC LOC., 0218, 0219, 0220, 0221. SET EACH VOLTAGE COMPARATOR (IVC) PER CHART ON ALIGNMENT DRAWING. RED LIGHT ON IVC SHOULD BE ON WHEN RELAY ENERGIZED. CHECK EACH IVC SETTING BY SLOWLY VARYING FREQUENCY ABOUT THE OPERATING POINT.

4. TACHOMETER 3AØ2 LOC. 0107 WITH OSCILLATOR AT ZERO FREQUENCY ADJUST 2A BIAS FOR ZERO INDICATION ON REMOTE CONTROL PANEL. SET OSCILLATOR TO RATED SSPU FREQUENCY AND ADJUST GAIN AND I OUT AS REQUIRED FOR CORRECT INDICATION OF TURBINE RPM.

4.1 REPEAT ZERO FREQUENCY AND RATED SPEED SETTINGS AND MAKE FINAL ADJUSTMENTS IF REQUIRED.

5. SECOND TACHOMETER 3AØ2 LOC. 0109 WHERE REQUIRED ADJUST 1A & 2A USING PROCEDURE IN STEPS 4 AND 4.1 ABOVE.

6. REMOVE OSCILLATOR FROM T28 AND T29 AND RECONNECT INCOMING LEADS.

V. SSPU SPEED MEASURING CIRCUITS

1. SET-UP JUMPER T4 TO T7 AND T5 TO T8. CONNECT OSCILLATOR AND COUNTER TO T4 AND T5.

2. FREQUENCY TO VOLTAGE CONVERTERS 2FVC, LOC. 0203 AND 0206 HIGH SPEED GATE ILVG, LOC. 0209.

2.1 PRIMARY AT RATED SHAFT SPEED PICKUP FREQUENCY, ADJUST 2FVC LOC. 0203 FOR -5.000 VOLTS AT OUTPUT TESTPOINT OF ILVG, 0209. (OBSERVE WITH TEST METER IN POSITIONS 6 AND 7 THAT AMPLIFIER 7 IS MORE POSITIVE THAN 6; IF NECESSARY ADJUST 2FVC, 0206, TO OBTAIN THIS CONDITION).

2.2 BACKUP DEPRESS BACKUP SPEED AMPLIFIER TEST SWITCH AND ADJUST 2FVC, 0206, FOR -5.05 VOLTS AT OUTPUT TP, 0209.

3. REMOVE TERMINAL POINT JUMPERS AND OSCILLATOR AND COUNTER LEADS.

VI. CONTROL VALVE RAM SERVO LOOP

1. CHECK LVDT CORE PER LEVER DIAGRAM.

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PRINTS TO

2. DETECTOR INDICATOR 1DI, LOC. 0216. WITH CONTROL VALVE FULL OPEN ADJUST NULL OF 1DI FOR 0.00 ± 0.01 VOLTS AT OUT TP. NOTE VOLTAGE READING ON ALIGNMENT DIAGRAM. WITH VALVE IN CLOSED OVERTRAVEL POSITION RECORD OUT TP VOLTAGE ON ALIGN. DIAG. ADJUST ZERO FOR 0 INDICATION OF CONTROL VALVE POSITION ON OPERATOR'S CONTROL CONSOLE. RETURN VALVE TO FULL OPEN POSITION AND ADJUST CALIBRATE FOR 100% POSITION INDICATION.

3. DETERMINE GAIN OF DETECTOR CIRCUIT IN VOLTS PER INCH BY DIVIDING THE DIFFERENCE IN OUT TP VOLTS AT EXTREME VALVE POSITIONS BY THE TOTAL VALVE TRAVEL IN INCHES, AND RECORD THIS FIGURE ON THE ALIGNMENT DIAGRAM.

3.1 ANALOG CONTROL VALVE POSITION SIGNAL

1AØ2 LOC. 0105. WITH VALVE CLOSED ADJUST 1A BIAS FOR ZERO POSITION ON OPERATORS METER.

3.2 WITH VALVE OPEN ADJUST 1A GAIN AND I OUT FOR FULL OPEN INDICATION.

3.3 REPEAT 3.1 AND 3.3 AS NECESSARY.

4. MAKE ELECTRICAL CONNECTIONS TO SERVO VALVE.

5. SERVO AMPLIFIER, 2SA, LOC. 0214.

5.1 PS10 TRIP CHECK FOR APPROX. -0.7 VOLTS AT "I" TP UNDER TRIP CONDITION.

5.2 CLOSE SIGNAL CHECK WITH TURBINE TRIPPED, DISCONNECT DEMAND TERMINAL E AND LIFT TERMINAL D OF 2SA. CONNECT TEST POT WITH NEGATIVE OUTPUT TO TERMINAL D AND SET POT FOR ABOUT -3 VOLTS AT LIFT TP. THE OUTPUT OF AMPLIFIER 9 ON TEST METER POSITION 9 SHOULD BE ABOUT -10 VOLTS. TURN BIAS ADJUSTMENT CCW UNTIL AMP. 9 GOES TO ≠10 VOLTS. THERE SHOULD BE LITTLE OR NO CONTROL OF 9 AT INTERMEDIATE VOLTAGES. LEAVE 9 AT ≠10 VOLTS AND RESET TURBINE. SLOWLY REDUCE TOWARDS 0 THE TEST POT SETTING AND NOTE THAT 9 SWINGS FROM ≠10 TO A MINIMUM VALUE OF APPROX. -6.2 VOLTS. NOW APPLY MORE NEGATIVE VOLTAGE FROM THE TEST POT AND NOTE THAT 9 HAS A MAXIMUM VALUE OF APPROX. ≠6.2 VOLTS.

5.3 GAIN REFER TO ALIGNMENT DRAWING TO OBTAIN LIFT TEST POINT VOLTAGE REQUIRED TO PRODUCE 1 VOLT AT I TEST POINT, WHEN GAIN IS PROPERLY ADJUSTED.

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PRINTS TO

5.31 VARY THE TEST POT SETTING UNTIL I TEST POINT IS 0.000 VOLTS AND NOTE LIFT TEST POINT VOLTAGE. CHANGE THIS VOLTAGE WITH THE TEST POT BY PLUS LIFT TEST POINT VOLTAGE FROM STEP 5.3, ADJUST GAIN UNTIL I TEST POINT READS MINUS 1.000 VOLTS.

5.32 RECHECK ZERO POINT BY RETURNING TEST POT SO THAT ZERO VOLTS IS ONCE AGAIN AT I TEST POINT AND TEST POT VOLTAGE IS THE SAME AS NOTED IN 5.31.

5.33 ADJUST TEST POT FOR MINUS LIFT TEST POINT VOLTAGE AND NOTE THAT I TEST POINT IS PLUS 1.000 VOLT.

5.4 BIAS DISCONNECT TEST POT AND RECONNECT THE LEAD REMOVED FROM D. WITH THE VALVE RAM AT ITS INTERNAL STOP AND ZERO VOLTS A DEMAND TEST POINT SET BIAS FOR -0.35 ± .01V AT I TEST POINT. RECONNECT TERMINAL E.

6. AFTER REMOVING THE ELECTRICAL CONNECTOR, MOUNT THE SERVO VALVE AND VERIFY THAT THE MECHANICAL BIAS IS IN THE VALVE CLOSING DIRECTION.

VII. REGULATION AND COMPENSATION ICR, LOC. 0212

1. REGULATION CONNECT THE OUTPUT OF THE TEST POT TO SUPPLY A NEGATIVE VOLTAGE TO TERMINAL E. ADJUST REGULATION TO OBTAIN VALVE TRAVEL FROM NO LOAD LIFT TO FULL LOAD LIFT FOR 3.00 VOLTS CHANGE IN THE OUTPUT OF THE TEST POT.

2. COMPENSATION VERIFY THAT JUMPER J-K IS NOT CONNECTED.

3. INSTANTANEOUS REGULATION SET TEST POT FOR -1 VOLT AT E AND ADJUST INST. REG. FOR VOLTAGE EQUAL TO 0.00 VOLTS AT INST REG. T.P.

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VIII. MANUAL SPEED REFERENCE (STAND-STILL CHECKS)

1. THIS REFERENCE IS OBTAINED FROM THE UPPER OF THE TWO MOTOR DRIVEN LVDT'S. WHEN POWER IS APPLIED IT SHOULD RUN TO LSS AND LIGHT THE LSS INDICATING LIGHT.

2. SLOW REFERENCE SPEED, CL & SMD, LOC. 0117. SHUT OFF HIGH PRESSURE OIL SUPPLY OR DISCONNECT SERVO VALVE BEFORE RAISING MANUAL REFERENCE. ACTUATE RAISE PUSHBUTTON AND TIME TRAVEL TO HSS LIGHT INDICATION. TIME SHOULD BE 80-90 SECONDS.

3. FAST REFERENCE SPEED, CL & SMD, LOC. 0117. ACTUATE LOWER FAST PUSHBUTTON AND TIME MOTOR TRAVEL FROM HSS LIGHT TO LSS LIGHT. IT SHOULD BE 8 TO 10 SECONDS.

4. BIAS AND GAIN, MANUAL REFERENCE, 2LVG, LOC. 0115. DISCONNECT K OF 2LVG. RAISE MANUAL REFERENCE OFF LSS AND BACK INTO IT AT SLOW RATE WITH LOWER PUSHBUTTON. ADJUST MANUAL BIAS FOR 0.000 VOLTS AT OUTPUT TP, 0113. RAISE TO HSS AT SLOW RATE AND ADJUST MANUAL GAIN FOR OUTPUT VOLTAGE AT TP 0113 AS SPECIFIED BY ALIGNMENT DIAGRAM.

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IX. AUTO SPEED REFERENCE, (STAND-STILL CHECK)

1. BOILER CONTROLLER INTERFACE, 7BCI, LOC. 0121.
CONFIRM THAT THE INTERCONNECTING CABLE BETWEEN THE FEEDWATER CONTROLLER AND EHC CABINET IS A SHIELDED CABLE, AND THE SHIELD OF THE CABLE IS CONNECTED TO COMMON ON THE FEEDWATER CONTROLLER AND TO T34 ON THE EHC CABINET.

   THE ADJUSTMENT OF IP (CMR) IS TO ELIMINATE ANY NOISE THAT MAY BE PICKED UP BETWEEN THE FEEDWATER CONTROLLER AND THE EHC CABINET.

2. USING A SCOPE, DETERMINE IF NOISE IS PRESENT BETWEEN TERMINALS B AND D OF 7BCI, LOC. 0121. IF A NOISE SIGNAL (PROBABLY 60 HZ OR 120 HZ) EXISTS, SET FEEDWATER CONTROLLER SIGNAL FOR LSS AND ADJUST IP FOR MINIMUM NOISE AT 1A OUT.

3. IF THERE ISN'T ANY NOISE SIGNAL BETWEEN TERMINALS D AND B OF 7BCI LOC. 0121, THEN DISCONNECT CUSTOMER'S INCOMING SHIELDED CABLE AT EHC TERMINALS T32, T33, AND T34 AND TEST EHC AS FOLLOWS. PLACE A JUMPER BETWEEN T32 AND T33 AND CONNECT OSCILLATOR BETWEEN T32 AND T34. SET OSCILLATOR AT 6 V RMS, 60 HZ AND ADJUST IP FOR MINIMUM AT 1A OUT. REMOVE OSCILLATOR AND JUMPER AND RECONNECT CUSTOMER LEADS TO T32, T33 AND T34.

4. WITH FEEDWATER CONTROLLER AT LSS, ADJUST 2P BIAS FOR 0VDC AT 1A OUT.

5. SIGNAL CONVERTER 3SC, LOC. 0125

   WITH FEEDWATER CONTROLLER AT LSS. CHECK THAT 2LS LIMIT SWITCH CONTACTS BETWEEN 1TB(D)& 1TB(E) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, REDUCE INPUT FROM FEEDWATER CONTROLLER UNTIL 2LS LIMIT SWITCH CLOSES. JUMPER SIGNAL FAILURE HOLD CONTACTS T77 AND T78 WITH 2LS LIMIT SWITCH CLOSED AND FEEDWATER CONTROLLER SIGNAL AT LSS. CONNECT VOLTMETER TO 1TP OF 3SC LOC. 0125 AND CHECK FOR ZERO VOLTS. IF NECESSARY, OBTAIN ZERO VOLTS BY WITHDRAWING 3SC FROM RACK AND ADJUSTING 5P.

6. REMOVE T77-T78 JUMPER AND SET FEEDWATER CONTROLLER SIGNAL TO HSS. CHECK THAT 1LS LIMIT SWITCH CONTACTS BETWEEN 1TB(D) AND 1TB(C) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, ADJUST IP CW ON 3SC LOC. 0125 AND/OR 2P GAIN UNTIL 1LS LIMIT SWITCH CLOSES. WITH FEEDWATER CONTROLLER AT HSS AND 1LS LIMIT SWITCH CLOSED, JUMPER SIGNAL FAILURE HOLD CONTACTS T77-T78 AND ADJUST 2P GAIN FOR MINUS 5.00 V ±.01 VDC AT 1TP.

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7. WITH T77-T78 STILL JUMPERED, ADJUST IP B.C. REF. FOR ZERO VOLTS AT 2TP. REMOVE JUMPER.

8. WITH FEEDWATER CONTROLLER AT LSS CHECK FOR ZERO VOLTS AT ITP AND WITH FEEDWATER CONTROLLER AT HSS CHECK FOR ZERO VOLTS AT 2TP.

9. ADJUST 3P DEADBAND FOR MINUS 2VDC (0.1% DEADBAND) AT 3TP. DISCONNECT WIRE AT OUTPUT TERMINAL F OF 3SC LOC. 0125.

10. ADJUST "BOILER CONTROLLER" INPUT TO LSS (-9±1VDC AT 2TP). CONNECT AN OSCILLOSCOPE TO TERMINAL F AND CHECK FOR A PULSE WIDTH OF 10 MSEC. IF NECESSARY, WITHDRAW BOARD AND ADJUST FREQUENCY CONTROL POTENTIOMETER 4P. RECONNECT WIRE TO TERMINAL F. AUTO SPEED REFERENCE SHOULD DRIVE FROM LSS TO HSS IN 10 SECONDS.

11. AUTO SPEED REFERENCE STOPS, LSS, 2LVG, LOC. 0115 DISCONNECT G, 0115 AND WITH AUTO REFERENCE AT LSS ADJUST SC BIAS, 0115 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM.

12. HSS, 2LVG, LOC. 0115. WITH AUTO REFERENCE AT HSS ADJUST SC GAIN, 0115 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM. RECONNECT G.

13. LOCAL/REMOTE CONTROL LIGHT, IVC, LOC. 0116. ADJUST REFERENCE, 0116 FOR 6.5 VOLTS AT L TP. CHECK THAT LOCAL CONTROL INDICATION IS OBTAINED WHEN AUTO REFERENCE IS AT LSS AND MANUAL REFERENCE IS LOWERED SO THAT 2LVG OUTPUT IS LESS THAN APPROXIMATELY 1 VOLT.

14. AUTO SPEED REFERENCE INDICATION, IDI LOC. 0128. ADJUST ZERO AT LSS AND CAL AT HSS TO OBTAIN 0 AND 100 INDICATIONS ON OPERATORS POSITION INDICATOR.

X. OVERSPEED TEST INTEGRATOR, 10SI, LOC. 0111

1. RATE CONNECT VOLTMETER TO OUTPUT TEST POINT 2LVG, 0115. WITH MANUAL AND AUTO REFERENCE AT HSS JUMPER T54 TO T58. VOLTMETER READING SHOULD INCREASE 0.6 ± 0.1 VOLTS PER MINUTE FOR A (12% ± 2%) SPEED INCREASE PER MINUTE. ADJUST RATE, 0111 AS REQUIRED.

2. LIMIT CONNECT OSCILLATOR AND COUNTER TO T4 AND T5. SET FREQUENCY FOR 120% SSPU FREQUENCY. OBSERVE OUTPUT OF AMPLIFIER 5 WITH TEST METER. JUMPER T54 TO T58 AND ADJUST LIMIT, 0111 FOR POINT AT WHICH AMPLIFIER SWITCHES FROM POSITIVE SATURATION TO ZERO VOLTS. CHECK BY REDUCING OSCILLATOR FREQUENCY AND SLOWLY INCREASING IT TO THE SWITCHING POINT FREQUENCY.

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XI. LP STOP VALVE POSITION INDICATOR 1DI, LOC. 0102. WITH STOP VALVE FULLY CLOSED ADJUST DIFF. TRANS. NULL FOR -3 VOLTS AT OUT TP AND METER ZERO FOR 0 POSITION. ADJUST CAL. FOR 100% WITH VALVE FULL OPEN.

XII. HP STOP VALVE POSITION INDICATOR 1DI, LOC. 0104. SAME AS XI.

XIII. LOW SPEED SWITCH POWER SUPPLY, LPS, LOC 0226

1. CHECK FOR 5.6 VDC AT TERMINAL D.

2. CHECK FOR 12VDC AT TERMINAL E.

XIV. LOW SPEED SWITCH LSS LOC. 0225

1. STANDSTILL CHECKS - CHECK THAT TERMINALS F&G ARE +5.6VDC WITH RESPECT TO B. THIS WILL CONFIRM 1TDX & 200RPM RELAYS ARE ENERGIZED.
2. RELAY CONTACTS AT TERMINAL D AND E SHOULD BE CLOSED.
3. 1TP SHOULD BE 5.6 VDC.
4. REMOVE WIRE FROM F AND JUMPER F TO B. RELAY CONTACTS AT H & K CLOSE. REMOVE JUMPER AND CONNECT WIRE TO F.
5. OPERATE TURNING GEAR TO OPEN CONTACTS AT H & K.

OPERATIONAL CHECKS

1. CHECK FOR 0-5 VOLT PULSE AT TERMINAL N WHEN TURNING GEAR OPERATED.
2. POSITIVE PULSES SHOULD BE PRESENT AT 2TP WHEN TURNING GEAR OPERATED.
3. 1TP SHOULD BE ZERO VOLT WHEN TURNING GEAR OPERATED.
4. RELAY CONTACTS AT D & E SHOULD BE OPEN WHEN TURNING GEAR OPERATED.
5. WITH TURNING GEAR OFF AND TURBINE AT STANDSTILL RELAY CONTACTS AT D & E SHOULD BE CLOSED.

XV. SEE LEVER DIAGRAM FOR RUNNING CHECKS OF SPEED REGULATION.

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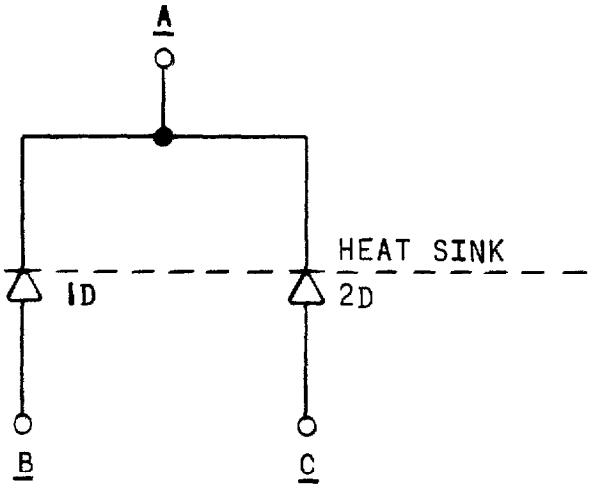
PRINTS TO

GENERAL ELECTRIC

44A306392
CONT ON SHEET SH NO.

REV NO.	TITLE
44A306392	RECTIFIER 1 REC
CONT ON SHEET SH NO.	FIRST MADE FOR 44B317827-G01

REVISIONS



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FG-W
12F-DFJ

PRINTS TO

MADE BY	APPROVALS		
Eleanor Ellison 4/24/73	R.G. Sd Hoff	DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.	44A306392
ISSUED D. Moyer April 26, 1973	R.J. Culver	LOCATION	CONT ON SHEET SH NO.

FF-803-WF (8-71)
PRINTED IN U.S.A.

CODE IDENT NO.
3

TITLE	GENERAL ELECTRIC	44A335415
PRINCIPAL RENEWAL PARTS LIST TAMPA ELECTRIC BIG BEND 3 EHC RFTB 122881 091-40778		SH. NO 1

3S7513TC300A11 CONTROL PANEL

QTY	PART NUMBER	DESCRIPTION
2	3S7513KF300A5	LINE POWER PANEL
1	3S7513KF306A9	METER PANEL
1	3S7513KF306A11	METER PANEL ASM
1	3S7513KF307A14	RELAY PANEL
5	44B331761-G01	PWB VOLT COMPARATR
1	44D236389-G02	PWB SERVO AMPLI
5	44D381585-G01	PWB DETECTOR IND
9	44D381589-G01	PWB OPER AMPLIFIER
1	44B331717-G01	PWB 3KHZ OSC
1	44D236379-G01	PWB ANALOG OUTPUT
2	44D236379-G03	PWB ANALOG OUTPUT
2	44D331918-G01	PWB CL MOTOR DRIVE
3	44D236365-G02	PWB 2FREQ VOLT CNV
1	44B331743-G01	PWB SIG CONV 3SC
1	44B331746-G03	PWB BOILR CON INTF
3	44D236370-G01	PWB DIODE BOARD
6	44B317827-G01	PWB SERIES REG 1
6	44D236371-G01	PWB SERIES REG 2
1	44D236373-G02	PWB OVERSPD INTEG
1	44D236372-G02	PWB SIG LOW VALUE
1	44D236388-G01	PWB COMP+REG ICR
2	44D315178-G01	DIFF TRANS DRV STD
1	44D236378-G01	PWB REF LOW VALUE
1	44B337328-G01	PWB LPF PWR SUPPLY
1	44B337327-G01	PWB LOW SPD SW LSS
3	44C319112-G01	RACK ASM PWB
2	K9774719L50	CAP 500UF 50VDC

3S7513KF300A5 LINE POWER PANEL

QTY	PART NUMBER	DESCRIPTION
2	44B334726-G01	SUPPLY TRANSF
18	44A316724-M19	CAP 1900UF 75VDC
6	44A333453-D36	RES 360HM 100W
6	44B214518-G01	REACTOR

REVISIONS	FINISH
	11
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3S7513KF306A9 METER PANEL

QTY	PART NUMBER	DESCRIPTION
5	44A317759-018	SWITCH
20	GE387	LAMP
4	44A316746-001	METER RELAY, DC
1	44A316747-001	METER RELAY, AC

3S7513KF306A11 METER PANEL ASM

QTY	PART NUMBER	DESCRIPTION
1	128A4057H20	RES 200K OHMS 5 W
2	44A317759-018	SWITCH
2	44A317759-019	SWITCH
1	44A317759-012	SWITCH
3	44A317759-020	SWITCH
20	GE387	LAMP
2	44A316746-001	METER RELAY, DC

LAMPS & HOLDER REPLACEMENT PART NO. 4872K12G700 FOR METER RELAYS MAY BE ORDERED THRU NEAREST G.E. SALES OFFICE FROM INSTRUMENT DEPT. G.E. CO. LYNN, MASS.

ISSUED D. Mercer March 1973	CCD 549	DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.	SH NO 3

3S7513KF307A14 RELAY PANEL

QTY	PART NUMBER	DESCRIPTION
22	3S2791G138D4	MINIATURE RELAY
2	3SAA1343A2	RELAY
1	12HMA11B3	RELAY
1	44B312431-003	TIME DELAY RELAY
1	44A230536-001	SIGNAL FLASHER
36	44B232019-004	RECT 1A 400V
2	44A317994-G01	COMP BD ASM
4	3S2791G138D3	MINIATURE RELAY
2	K9774723K40	CAP 40UF 450VDC
2	44B314000-G01	D C REACTOR
2	K590830G1	T B ASM
28	9338545P2	RELAY SOCKET
10	44B232011-001	RECT 200MA 200VDC

44D315178-G01 DIFF TRANS DRV STD

QTY	PART NUMBER	DESCRIPTION
4	44A316424-001	SPRING
2	44A316456-G01	TRANSFORMER ASM
2	K2412593P1	SPRING
2	44A316738-001	STEPPING MOTOR DC
4	44A218404-001	MICRO SW. ASSEMBLY
2	44A317738-001	SPUR GEAR
2	44A315989-003	GEAR
2	44A315989-001	SPUR GEAR
2	44A315989-004	SPUR GEAR
2	44A315988-001	SPUR GEAR
2	44A315988-002	SPUR GEAR
12	44B232214-007	BEARING
2	K9774719L50	CAP 500UF 50VDC

REVISIONS

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44A316456-G01 TRANSFORMER ASM

QTY	PART NUMBER	DESCRIPTION
2	44A317777-001	TRANSF LVDT

3S7513KL300A9 CONTROL PANEL

QTY	PART NUMBER	DESCRIPTION
3	CR2940U207	CONTACT BLOCK
2	CR2940UA202C	PUSH BUTTON
1	44A317759-007	PUSHBUTTON SWITCH
4	44A317759-001	PUSHBUTTON SWITCH
3	44A317759-008	PUSHBUTTON SWITCH
1	44C392721-G01	DIODE BOARD
10	GE387	LAMP
3	CR2940UA200C	OPERATOR
1	6105700G43G	IND,LIGHT

GENERAL ELECTRIC

44B300423 CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO 44B300423 CONT ON SHEET SH NO

TITLE CABLE WIRING

FIRST MADE FOR 3S7513TC300AR2

INITIAL ELOW VOLTAGE

PLUG TERMINAL	WIRE	SOCKET TERMINAL
V	SHIELD	V
M	WHITE/BLACK/VIOLET	M
N	BLACK	N
W	BROWN	W
d	WHITE/BLACK/GRAY	d
c	SHIELD	c
U	RED	U
T	WHITE/BROWN/RED	T
L	SHIELD	L
F	ORANGE	F
G	WHITE/BROWN/ORANGE	G
H	SHIELD	H
P	WHITE	P
X	WHITE/BLACK	X
e	SHIELD	e
k	WHITE/BROWN/YELLOW	k
J	WHITE/BROWN/GREEN	J
h	SHIELD	h
b	WHITE/BROWN	b
g	WHITE/RED	g
a	SHIELD	a
S	WHITE/ORANGE	S
K	WHITE/YELLOW	K
E	WHITE/GREEN	E
A	WHITE/BLUE	A
B	WHITE/VIOLET	B
C	WHITE/GRAY	C
D	WHITE/BLACK/BROWN	D
J	WHITE/BLACK/RED	J
R	WHITE/BLACK/ORANGE	R
Z	WHITE/BLACK/YELLOW	Z
f	WHITE/BLACK/GREEN	f
m	WHITE/BLACK/BLUE	m
s	WHITE/BROWN/BLUE	s
r	WHITE/BROWN/VIOLET	r

SHIELDED CABLE

SHIELDED CABLE

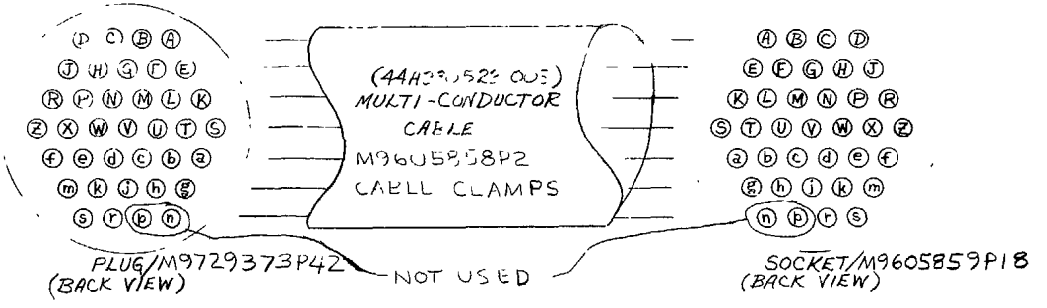
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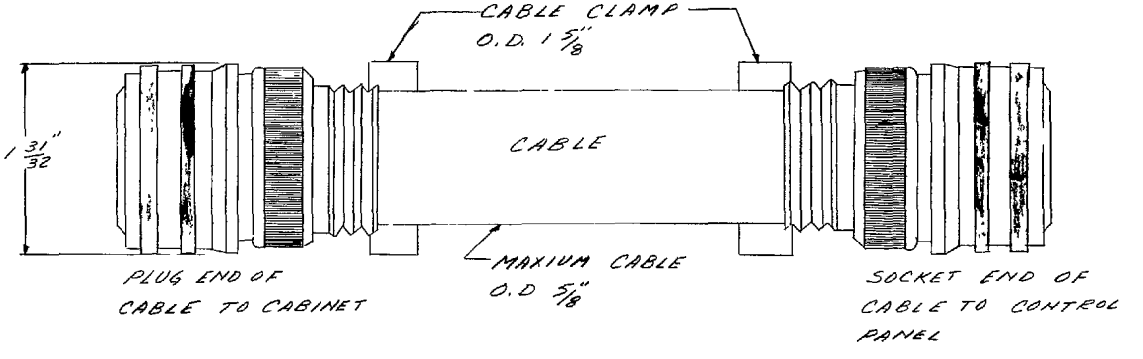
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SHIELDED CABLE





NOTES:

1. USE PLASTIC SLEEVING OVER WIRE AND TERMINALS TO INSULATE CONNECTIONS FROM EACH OTHER.

REVISIONS			PRINTS TO	
1	[illegible]	AUG 19 1969		11
2	H. Templeton		12P-0	EB
3	ADDED NOTE: D.K MOYER	Feb. 13 1973	EG-W	

MADE BY R M Sukoshey ISSUED MAY 22 1968

APPROVALS R M S

SPECIALTY CONTROL DIV OR DEPT WAYNESBORO, VA. LOCATION

44B300423 CONT ON SHEET SH NO

FF-806-F (12-66) PRINTED IN U S A

GENERAL ELECTRIC

44B306123
CONT ON SHEET FNL. SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

REV NO

CONT ON SHEET FL. SH NO 1

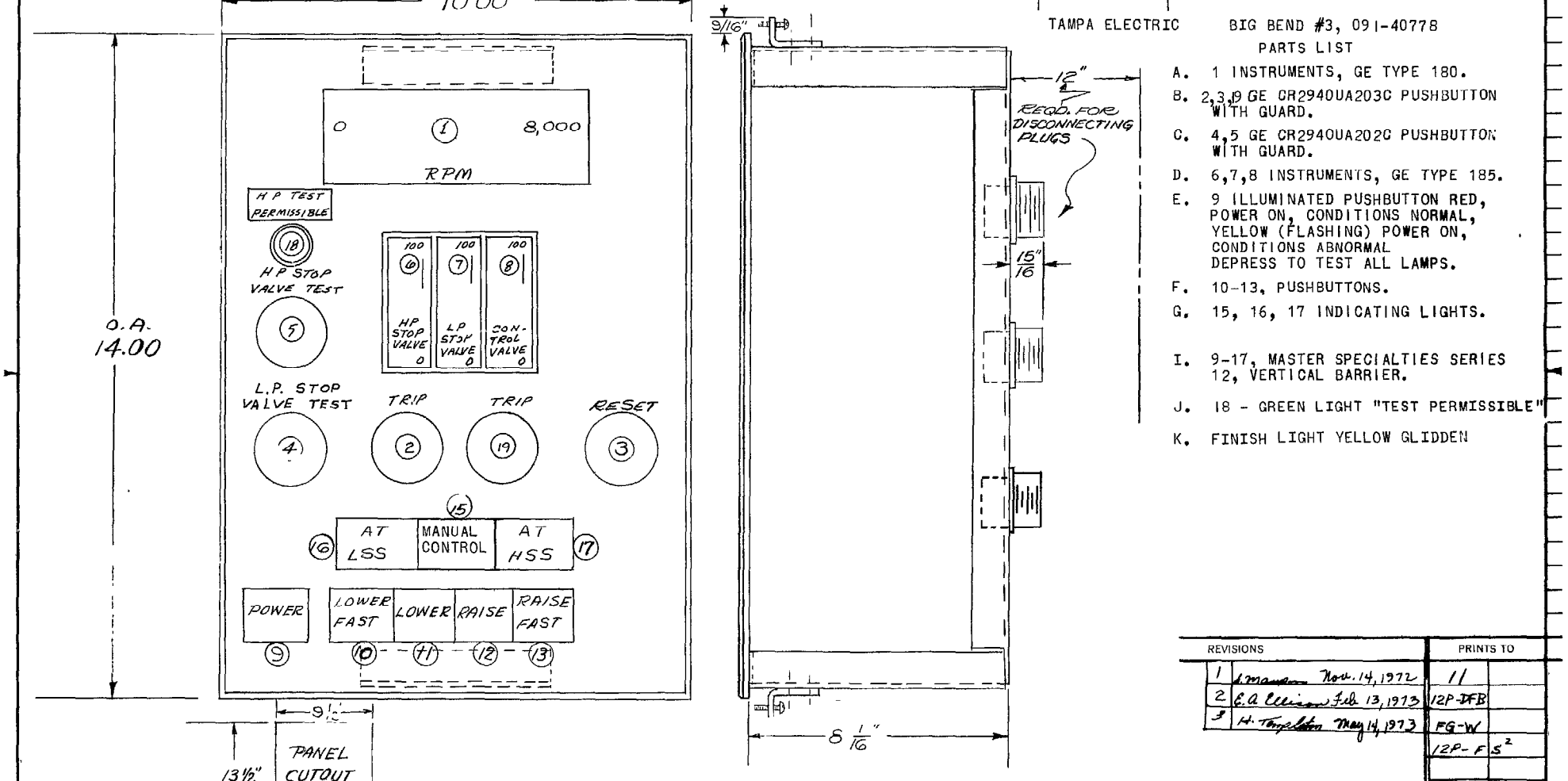
TITLE (OUTLINE)
FEEDPUMP TURBINE CONTROL PANEL

FIRST MADE FOR 3S7513KL300A9 TURBINE 122881

TAMPA ELECTRIC BIG BEND #3, 091-40778

PARTS LIST

A. 1 INSTRUMENTS, GE TYPE 180.
B. 2,3,9 GE CR2940UA203C PUSHBUTTON WITH GUARD.
C. 4,5 GE CR2940UA202C PUSHBUTTON WITH GUARD.
D. 6,7,8 INSTRUMENTS, GE TYPE 185.
E. 9 ILLUMINATED PUSHBUTTON RED, POWER ON, CONDITIONS NORMAL, YELLOW (FLASHING) POWER ON, CONDITIONS ABNORMAL DEPRESS TO TEST ALL LAMPS.
F. 10-13, PUSHBUTTONS.
G. 15, 16, 17 INDICATING LIGHTS.
I. 9-17, MASTER SPECIALTIES SERIES 12, VERTICAL BARRIER.
J. 18 - GREEN LIGHT "TEST PERMISSIBLE"
K. FINISH LIGHT YELLOW GLIDDEN



REVISIONS			PRINTS TO	
1	L. Mauer	Nov. 14, 1972	11	
2	E.A. Ellison	Feb 13, 1973	12P-DFB	
3	H. Templeton	May 14, 1973	FG-W	
			12P-F	S²

MADE BY H. Templeton 6-6-72
ISSUED Dot Moyer June 12, 1972
APPROVALS
DRIVE SYSTEMS PROD. DEPT.
DIV. OR DEPT
WAYNESBORO, VA.
LOCATION
44B306123
CONT ON SHEET FNL. SH NO 1

FF 806 F (12 68)
PRINTED IN U S A.

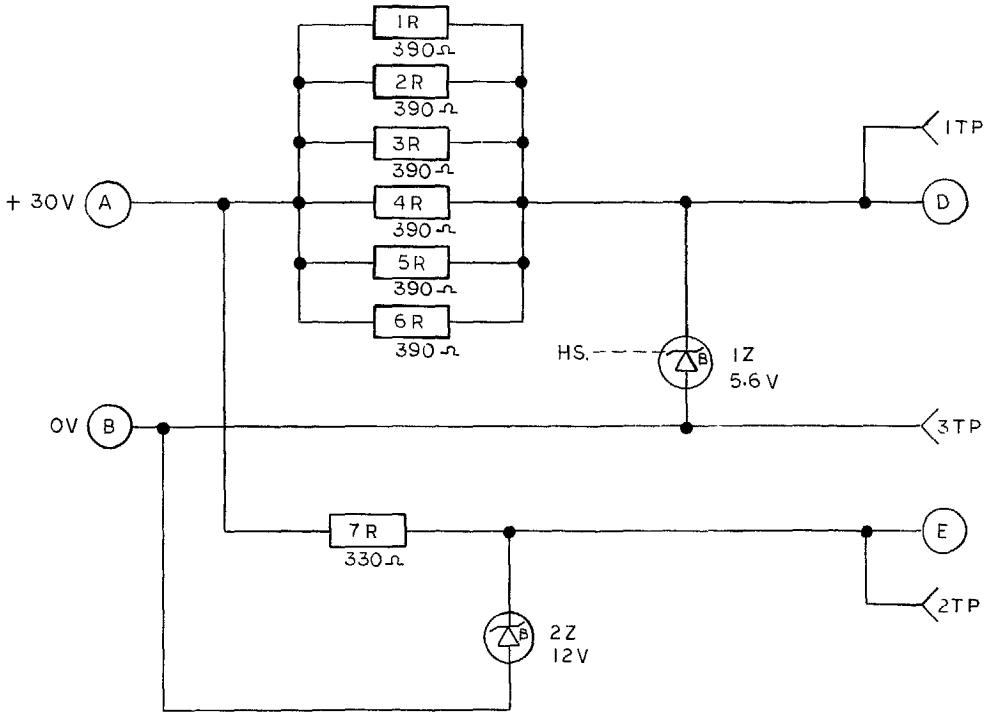
5

GENERAL ELECTRIC

TITLE

FIRST MADE FOR

SCHEM DIAGRAM



REVISIONS	PRINTS TO
	//
	IEP-DTB
	LCW

DRIVE SYSTEMS PROD DEPT.
WAYNESBORO, VA.

44B3[illegible]144

D.K. Moyer Sept. 13, 1972

FF 806 F (10 71)
PRINTED IN U.S.A.

3

GENERAL ELECTRIC

44B306148 CONT ON SHEET SH NO

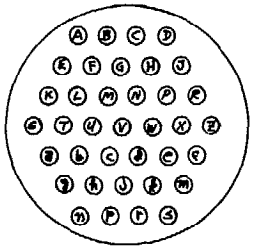
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±
REV NO
CONT ON SHEET SH NO

TITLE
POWER CABLE WIRING, ELECTRONIC BOILER FEED PUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A10
2C0,3C0 POWER CABLE

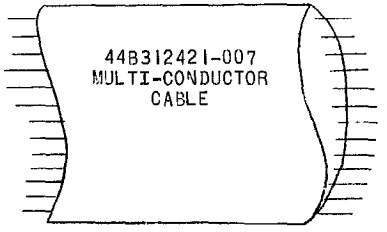
ASSEMBLY NOTES

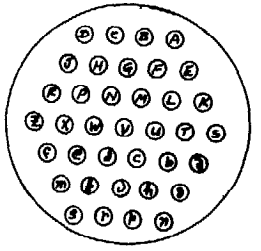
THE SPECIFIC COLOR OF WIRE CONNECTED TO A PIN IS NOT IMPORTANT, BUT THE SAME WIRE SHOULD BE CONNECTED TO THE SAME PIN AT THE OTHER CABLE END AND CONNECTOR.
EXAMPLE: IF A RED WIRE IS CONNECTED TO PIN "A" ON THE CONNECTOR PLUG, THE OTHER END OF THAT RED WIRE MUST BE CONNECTED TO PIN "A" OF THE SOCKET CONNECTOR AT THE OTHER END.

CONNECT SAME COLOR WIRE TO THESE PINS AT BOTH CABLE ENDS		DO NOT CONNECT ANY WIRES TO THESE PINS	
A		B	b
	h	D	d
C		G	g
T		L	l
E	t	M	m
F	m	N	n
	n	P	p
H	X	R	r
J	Z	S	s
K	a	U	u
		V	v
		W	w

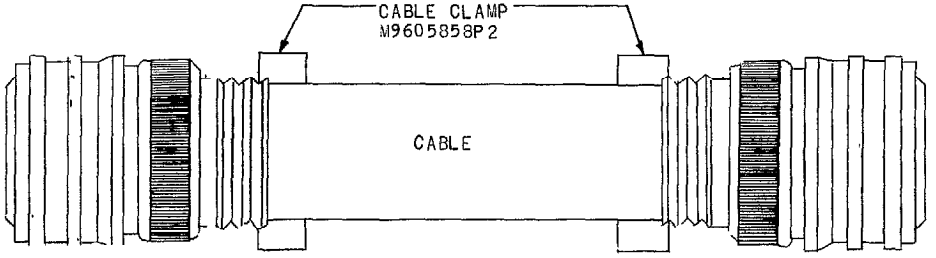


SOCKET
M9605859P18





PLUG
M9605859P42



SOCKET END OF
CABLE TO EHC REMOTE CONTROL PANEL.

PLUG END OF
CABLE TO EHC CABINET

REVISIONS

PRINTS TO
11
12PDF2
FG-W

MADE BY A. Tompkins 10-18-72
APPROVALS D. Moyer Oct. 20, 1972
COMM & CONT. DEVICES DIV OR DEPT
WAYNESBORO, VA. LOCATION
44B306148 CONT ON SHEET SH NO

FF 806 F (12 69)
PRINTED IN U S A

DRAWING NO 44B306197 SH NO
CONT ON SHEET

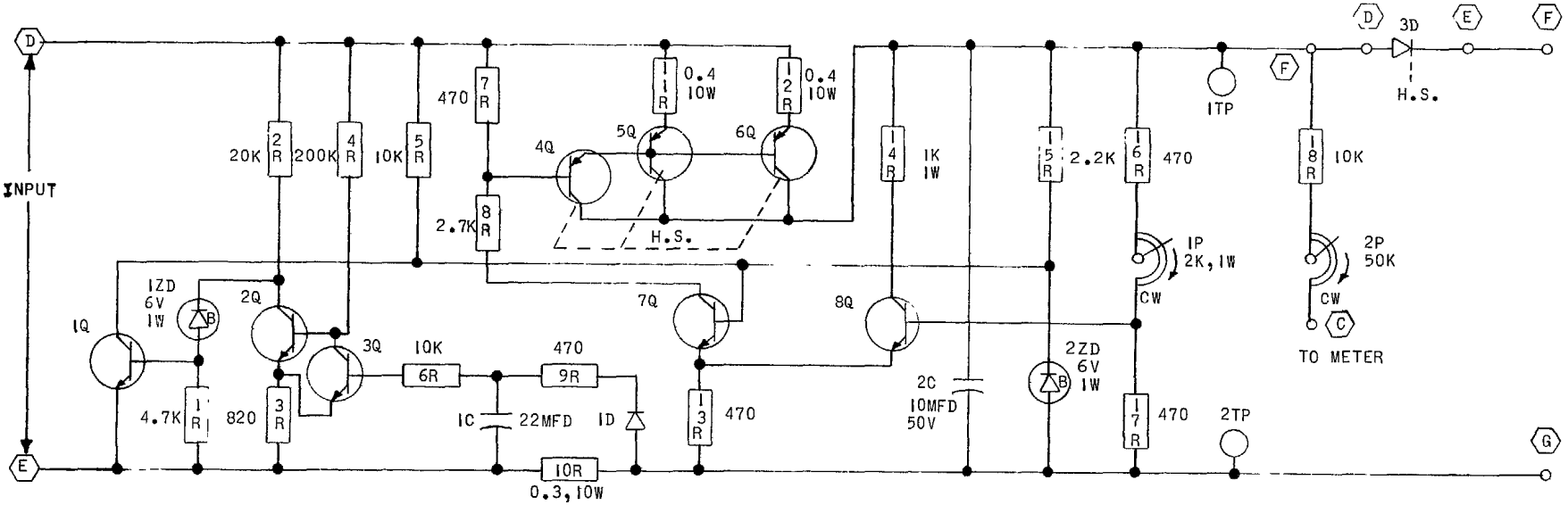
GENERAL ELECTRIC

44B306197
CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —					REV NO
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS			44B306197
		FRACTIONS	DECIMALS	ANGLES	
	✓	±	±	±	CONT ON SHEET SH NO

TITLE
SERIES REGULATOR ISR

FIRST MADE FOR 44B236371-G01



REVISIONS

PRINTS TO
11
Ty W
12C-DFC2

MADE BY
ISSUED D. Moyer April 26, 1973

APPROVALS

DRIVE SYSTEMS PROD. DEPT. DIV. OR DEPT
WAYNESBORO, VA. LOCATION

44B306197
CONT ON SHEET SH NO

FF 806 F (5-72)
PRINTED IN U S A

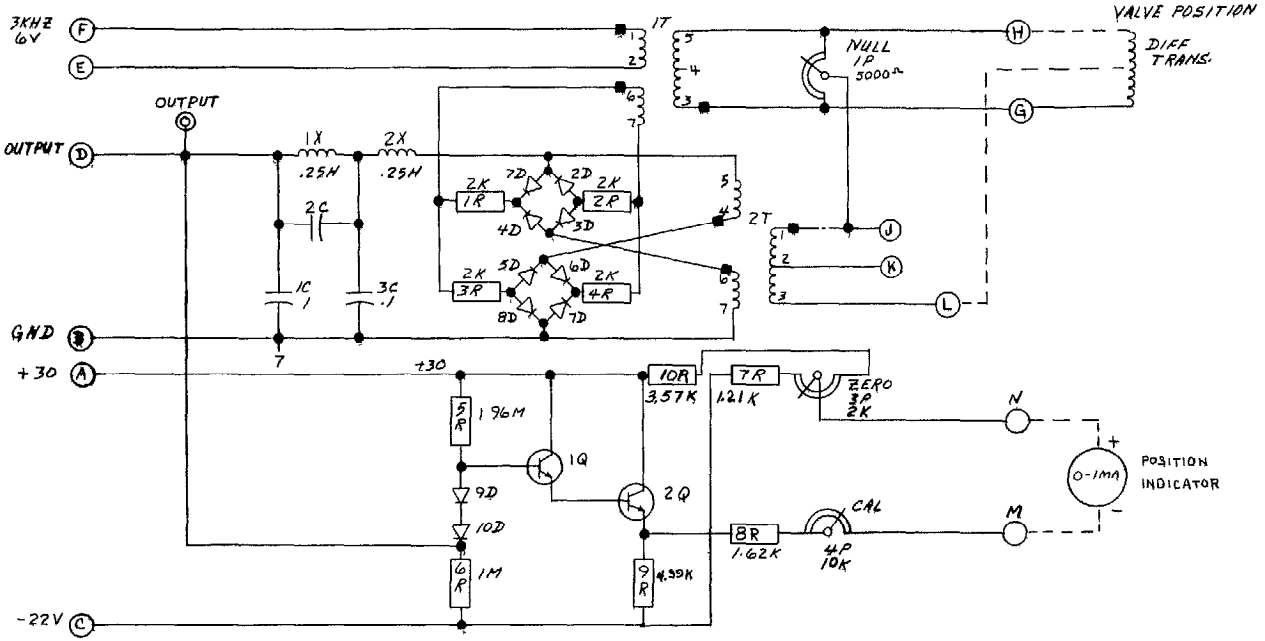
5

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS, DECIMALS, ANGLES

TITLE
DETECTOR INDICATOR
1DI

FIRST MADE FOR 44D381585-G01



NOTES:

1. CONNECTIONS

A	+30V	G, H, J, K, L, M	DIFFERENTIAL TRANSFORMER CONNECTIONS
B	GROUND		
C	-22V		
D	OUTPUT		
E, F	6V, 3 KHZ	N	OUTPUT METER 0-1MA

44C300609 REV NO 0

REVISIONS		PRINTS TO
1		K
2		EG-M
3	DCT 15 1968	12P-D&B
4		29VN
5	MAY 18 1970	
6	OCT 19 1970	

R. W. Templeton JAN 26 1968 ISSUED
APR 17 1968
APPROVALS
SPECIALTY CONTROL DIV OR DEPT
WAYNESBORO, VA LOCATION
44C300609 CONT ON SHEET

1T + 1D

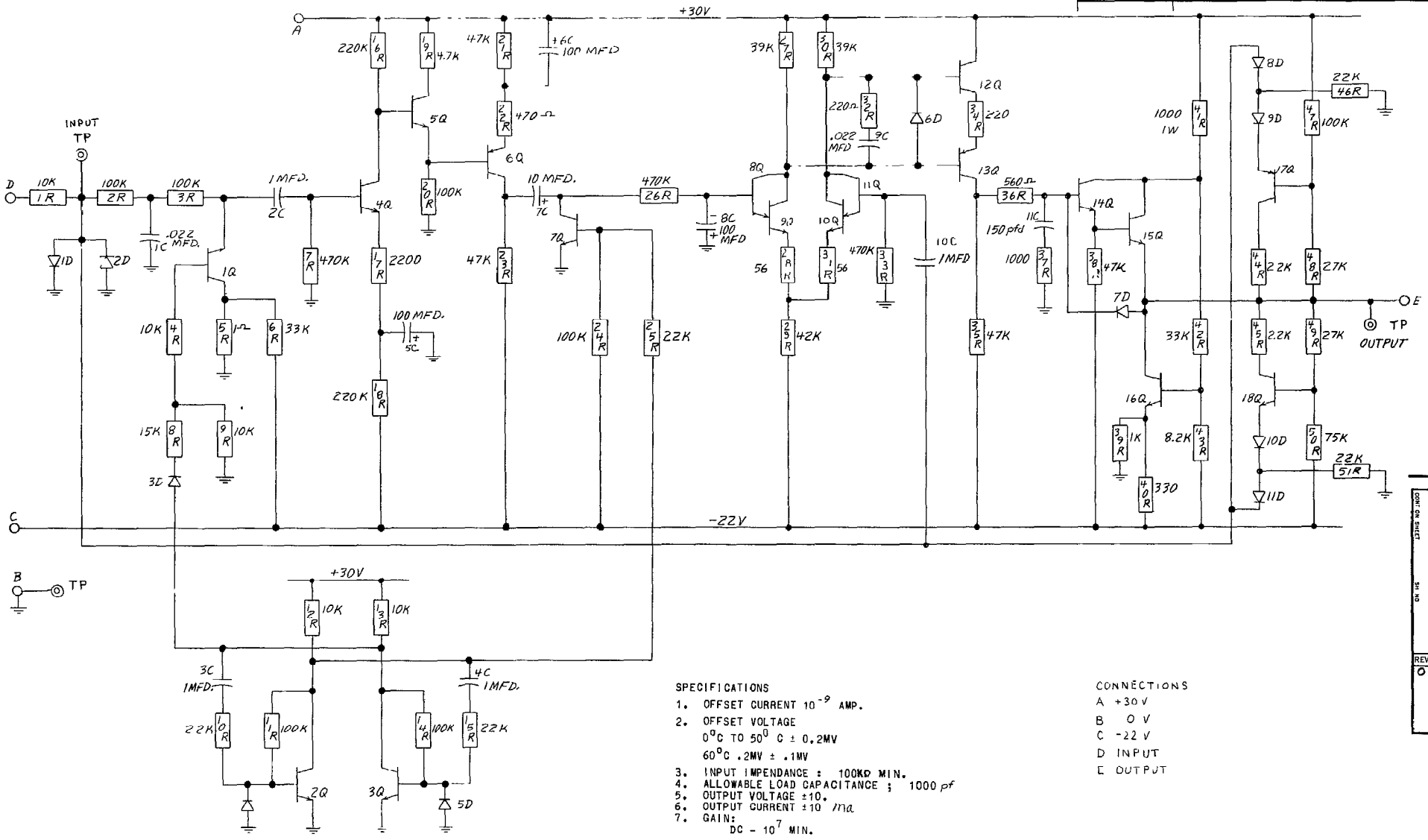
GENERAL ELECTRIC

44C300610 CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
CHOPPER STABILIZED
OPERATIONAL AMPLIFIER
FIRST MADE FOR 44D381589-G01

10A



SPECIFICATIONS

1. OFFSET CURRENT 10^{-9} AMP.
2. OFFSET VOLTAGE
   0^0C TO 50^0 C ± 0.2MV
   60^0C .2MV ± .1MV
3. INPUT IMPENDANCE : 100KΩ MIN.
4. ALLOWABLE LOAD CAPACITANCE ; 1000 pf
5. OUTPUT VOLTAGE ±10.
6. OUTPUT CURRENT ±10 ma
7. GAIN:
   DC - 10^7 MIN.
   AC - 5000 MIN. AT 100 CPS
   CHARACTERISTIC: BREAK FREQUENCY - 250 CPS
   20 db PER DECADE TO
   CROSSOVER.
8. CHOPPER STABILIZER RECOVERY TIME: 1 MINUTE
9. POWER REQUIREMENTS: 30 VOLTS - 25 ma
   -22 VOLTS - 20 ma
10. AMPLIFIER PROTECTION:
    A. OUTPUT CURRENT LIMITED TO APPROX 10 ma
    B. OUTPUT LIMITED TO 10 VOLTS
    C. ALLOWABLE INPUT VOLTAGE BEFORE DAMAGE - 70 VOLTS.

CONNECTIONS
A +30 V
B 0 V
C -22 V
D INPUT
E OUTPUT

REVISIONS	PRINTS TO
1. [illegible] NOV 17, 1967	11
2. [illegible] AUG 7, 1968	12P-DFB
	26 VN

MADE BY J C Glenn JAN 22 1968
ISSUED MAR 12 1968
APPROVALS
SPECIALTY CONTROL DEPT WAYNESBORO, VA. LOCATION
44C300610 CONT ON SHEET SH NO

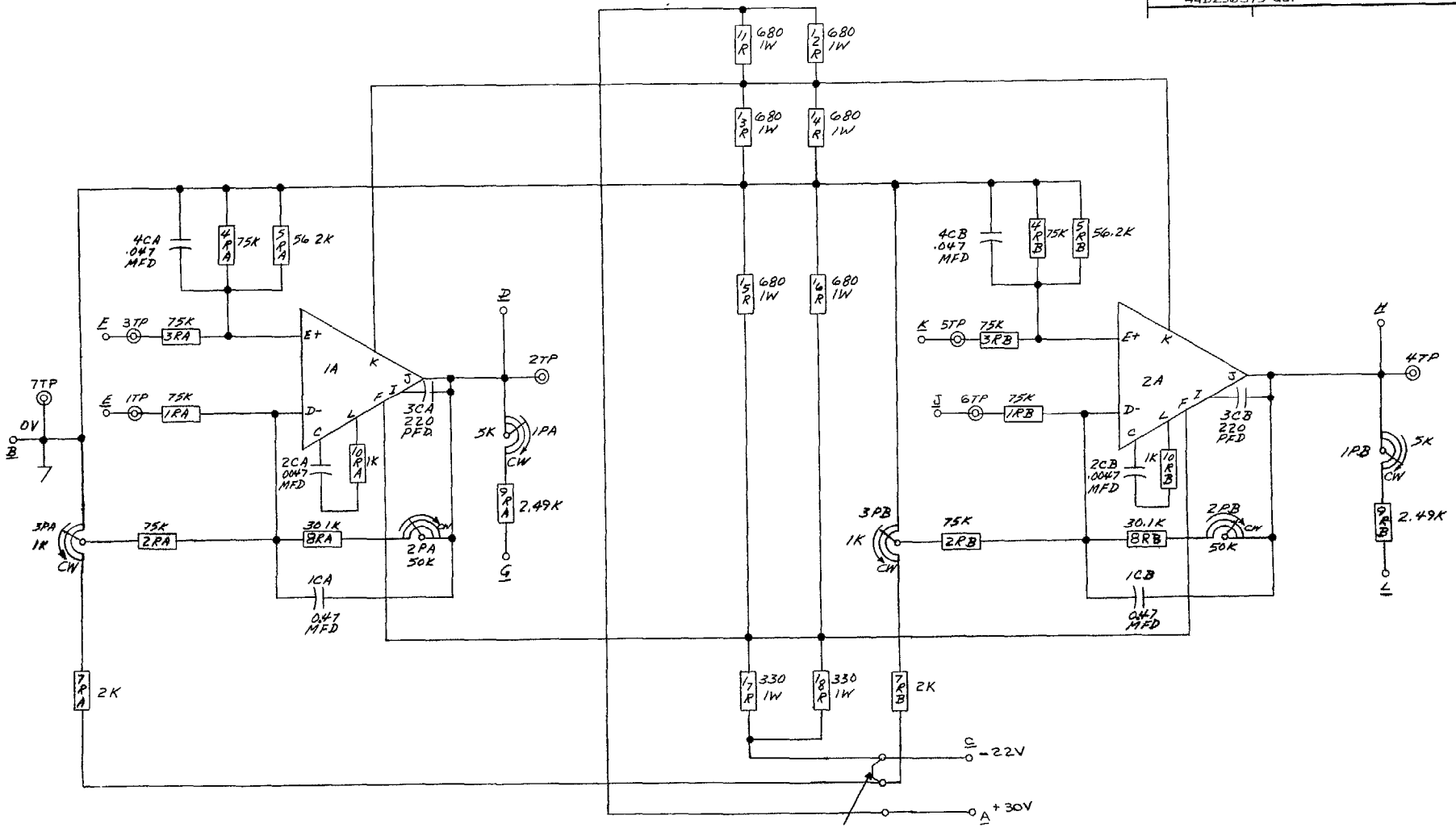
17 + 2

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ANALOG OUTPUT 1Ø2

FIRST MADE FOR
44D236379-G01



JUMPER POINTS PROVIDED ON THE PRINTED CIRCUIT BOARD SO RESISTORS 7RA & 7RB MAY BE CONNECTED TO EITHER +30V OR -22V.

CONNECTIONS

A = +30V
B = 0V
C = -22V
D = 1A VOLTAGE OUTPUT
E = 1A INPUT
G = 1A CURRENT OUTPUT
H = 2A VOLTAGE OUTPUT
J = 2A INPUT
L = 2A CURRENT OUTPUT

44C301216 REV NO.

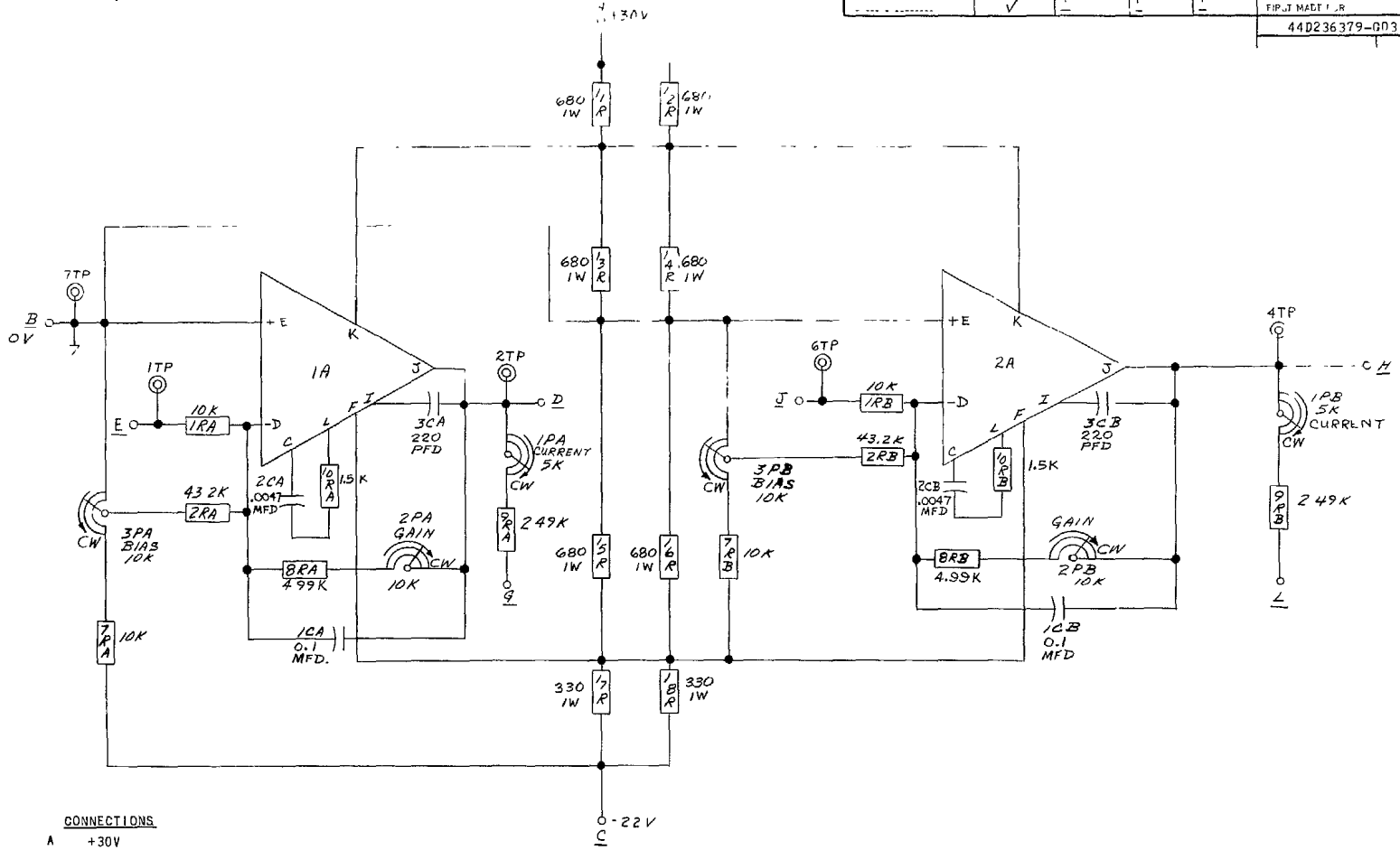
REVISIONS		PRINTS TO
1	R.M. SEP 30 1968	11
2	R.M. OCT 9 1969	12P-DF
3	[illegible] NOV 17, 1971	FG-W
4	[illegible]	29VN

DRAWN R.W. Templeton SEP 16 1968 | APPROVALS | SPECIALTY CONTROL | DIV OR DEPT | 44C301216
ISSUED SEP 24 1968 | WAYNESBORO, VA | LOCATION | CONT ON SHEET

GENERAL ELECTRIC 44C301622

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES FRACTIONS | DECIMALS | ANGLES

TITLE
ANALOG OUTPUT 3A0∠
FIRST MASTER
44D236379-003



CONNECTIONS

A	+30V
B	0V
C	-22V
D	1A VOLTAGE OUTPUT
E	1A INPUT
G	1A CURRENT OUTPUT
H	2A VOLTAGE OUTPUT
J	2A INPUT
L	2A CURRENT OUTPUT

DRAWING NO 44C301622 CONT ON SHEET SH NO REV NO

REVISIONS
1 am SEP 30 1968 Nov 17, 1971
Aug 7, 1972

PRINTS TO

R. W. Templeton SEP 17 1968
SEP 19 1968
APPROVALS

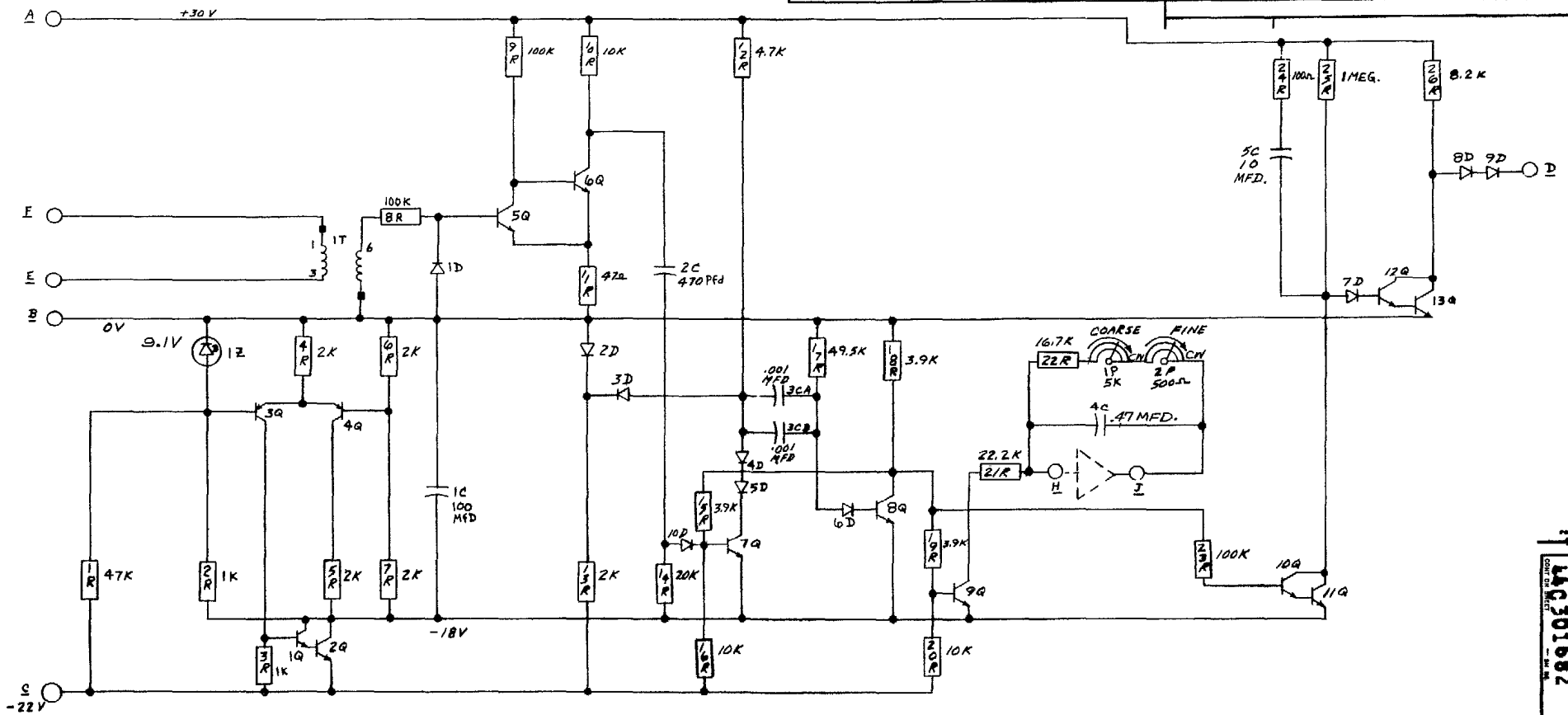
44C301622

GENERAL ELECTRIC

44C301682

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
FREQUENCY-VOLTAGE CONVERTER (2FVC)
FIRST MADE FOR 44D236365-G02



REVISIONS

1	[illegible] MAY 17 1971	
2	C.A. Card Dec. 5 1972	12P - DFB
3	E.A. Ellerman Aug. 7, 1974	FG-W

PRINTS TO
11.

DRAWN BY R.N. Templeton NOV 15 1968
ISSUED OCT 3 1969
APPROVALS R.L. Lubey
SPECIALTY CONTROL DIV. OR DEPT. WAYNESBORO, VA LOCATION
44C301682 CONT ON SHEET — SH NO

REV NO.

FF 809 F 69 671 PRINTED IN U.S.A.

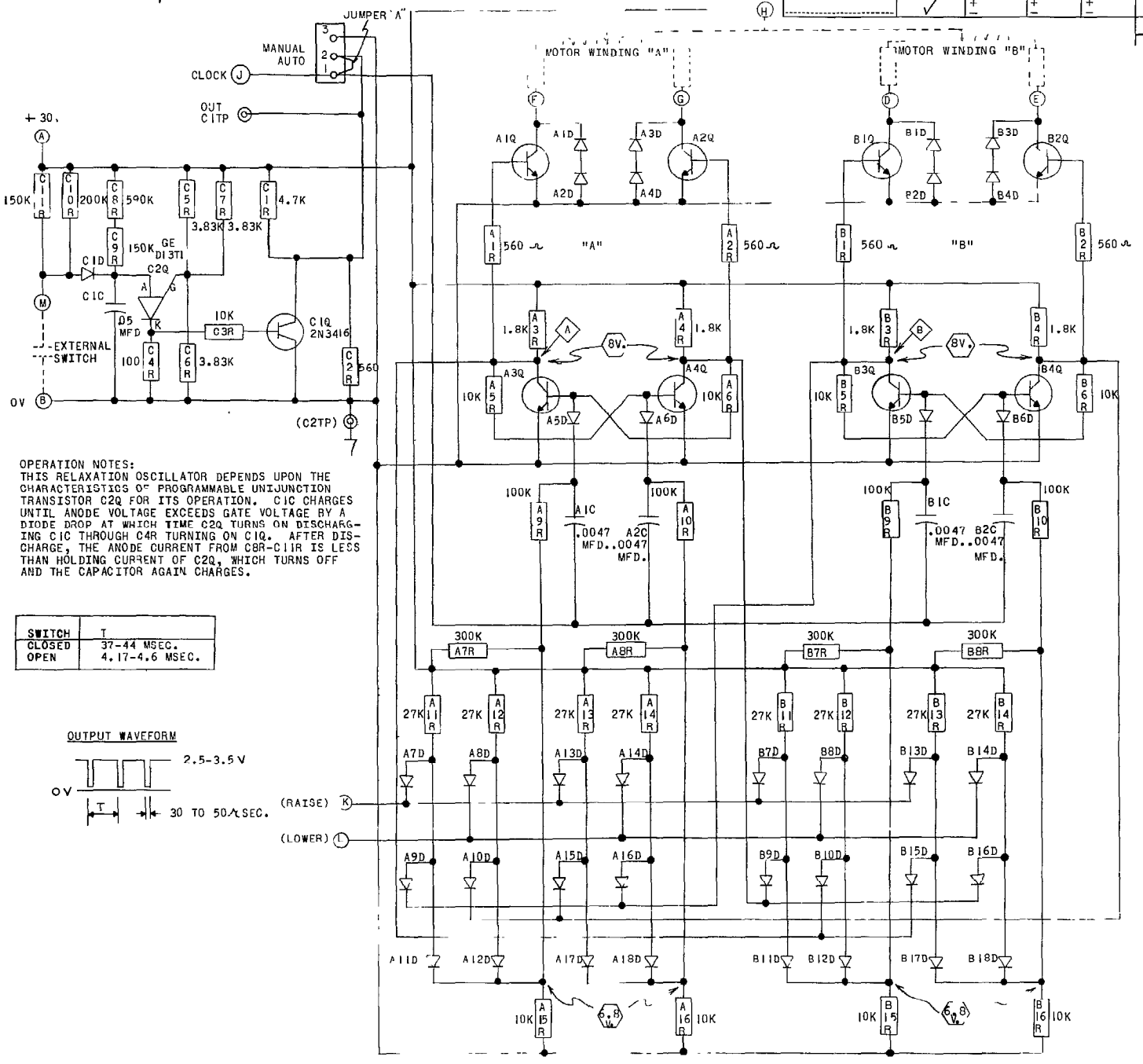
GENERAL ELECTRIC

44C303567

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSION: FRACTIONS, DECIMALS, ANGLES

TITLE: CLOCK & STEPPING MOTOR DRIVE

FIRST MADE FOR

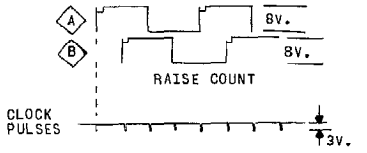


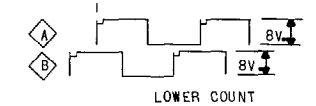
OPERATION NOTES:
THIS RELAXATION OSCILLATOR DEPENDS UPON THE CHARACTERISTICS OF PROGRAMMABLE UNIJUNCTION TRANSISTOR C2Q FOR ITS OPERATION. C1C CHARGES UNTIL ANODE VOLTAGE EXCEEDS GATE VOLTAGE BY A DIODE DROP AT WHICH TIME C2Q TURNS ON DISCHARGING C1C THROUGH C4R TURNING ON C1Q. AFTER DISCHARGE, THE ANODE CURRENT FROM C8R-C1IR IS LESS THAN HOLDING CURRENT OF C2Q, WHICH TURNS OFF AND THE CAPACITOR AGAIN CHARGES.

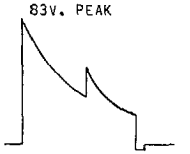
SWITCH	T
CLOSED	37-44 MSEC.
OPEN	4.17-4.6 MSEC.

OUTPUT WAVEFORM

1. THIS BOARD CONSISTS OF A RELAXATION OSCILLATOR CLOCK IDENTIFIED BY "C" COMPONENTS AND ALSO TWO FLIP-FLOPS "A" AND "B" WITH DIODE LOGIC TO FORM A REVERSIBLE COUNTER. OUTPUT OF THE FLIP-FLOPS FEEDS POWER AMPLIFIERS A1Q, A2Q, B1Q & B2Q. THE STEPPING MOTOR WINDINGS ARE CONNECTED DIRECTLY TO THE COLLECTORS OF THESE FOUR TRANSISTORS.
2. THE CLOCK PULSES ARE THREE VOLTS AMPLITUDE WITH FAST FALL TIME. NORMAL CLOCK FREQUENCY IS 240 PULSES PER SECOND.
3. RAISE COUNTING RESULTS FROM 0 TO 1 VOLT ON TERMINAL (K) & +6 VOLTS OR MORE ON TERMINAL (L). REVERSING THE VOLTAGE ON TERMINALS (K) & (L) RESULTS IN LOWER COUNTING. A POSITIVE VOLTAGE ON TERMINAL (K) & (L) WILL INHIBIT COUNTING. 0 TO 1 VOLT ON BOTH (K) & (L) IS NOT PERMISSIBLE.
4. CONNECT JUMPER "A" TO 1 & 2 FOR MANUAL SPEED REFERENCE; CONNECT JUMPER "A" TO 2 & 3 FOR AUTO SPEED REFERENCE.







REVISIONS		PRINTS TO
1	MAR 9 1970	11
2	July 26, 1971	12P-DFB²
3	Nov 17, 1971	FS W

DRAWN BY: 2-26-70 — ISSUED: MAR 3 1970 — APPROVALS: MC

COMM & CONT DEVICES — WAYNESBORO, VA — DIV OR DEPT — LOCATION

44C303567

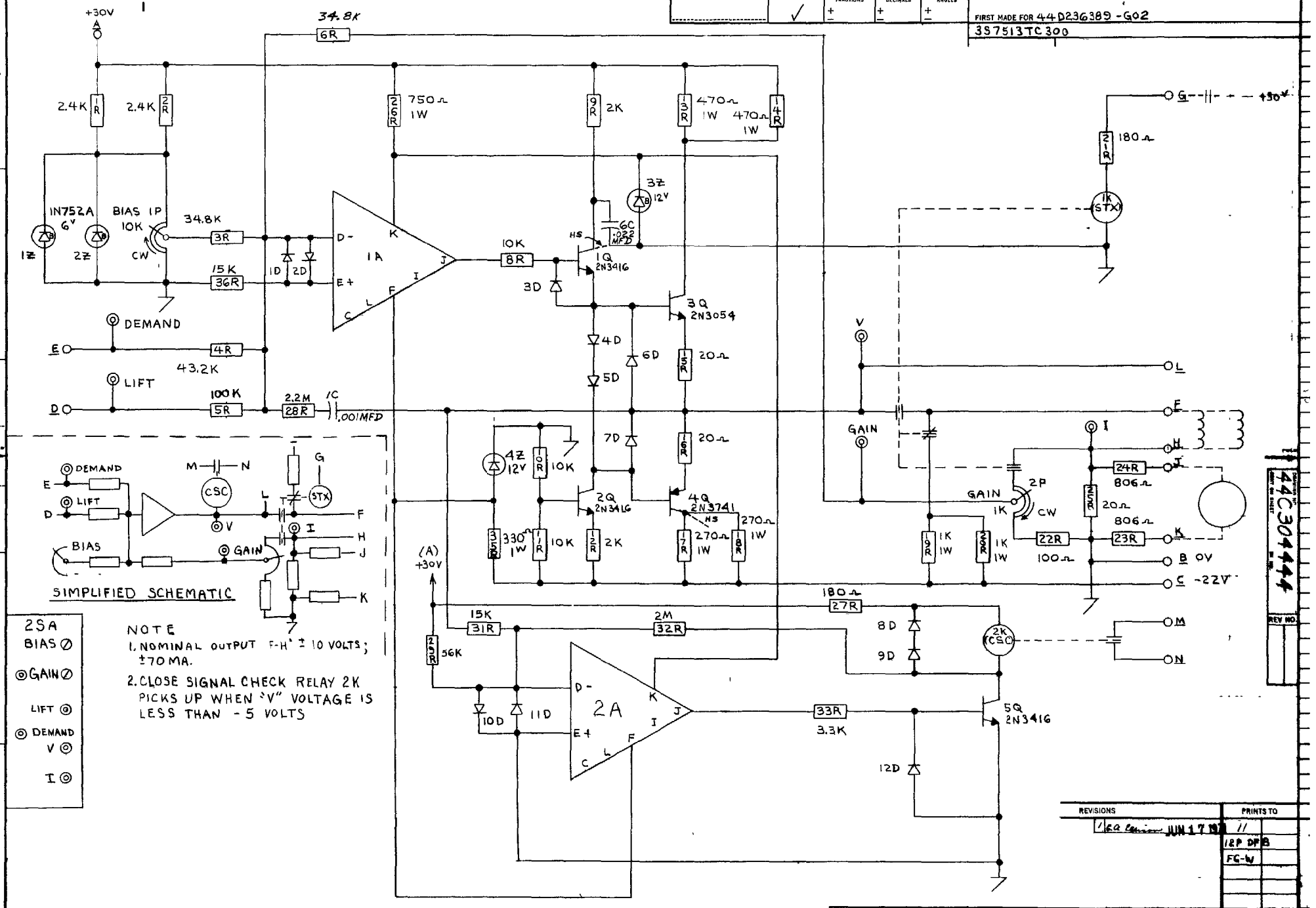
DRAWING NO 44C303567 — CONT ON SHEET — REV NO

FORM 8081 P 10 68 PRINTED IN U.S.A.

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
SERVO AMPLIFIER (2SA)
FIRST MADE FOR 44D236389-G02
3S7513TC300



SIMPLIFIED SCHEMATIC

2SA
BIAS ⊘
GAIN ⊘
LIFT ◎
DEMAND ◎
V ◎
I ◎

NOTE
1. NOMINAL OUTPUT F-H ± 10 VOLTS; ±70 MA.
2. CLOSE SIGNAL CHECK RELAY 2K PICKS UP WHEN "V" VOLTAGE IS LESS THAN -5 VOLTS

REVISIONS
1 EO [illegible] JUN 17 19[illegible]

PRINTS TO
12P DF B
FC-W

ISSUED MAR 26 1971
COMM. & CONT. DEVICES WAYNESBORO, VA
44C304444

GENERAL ELECTRIC

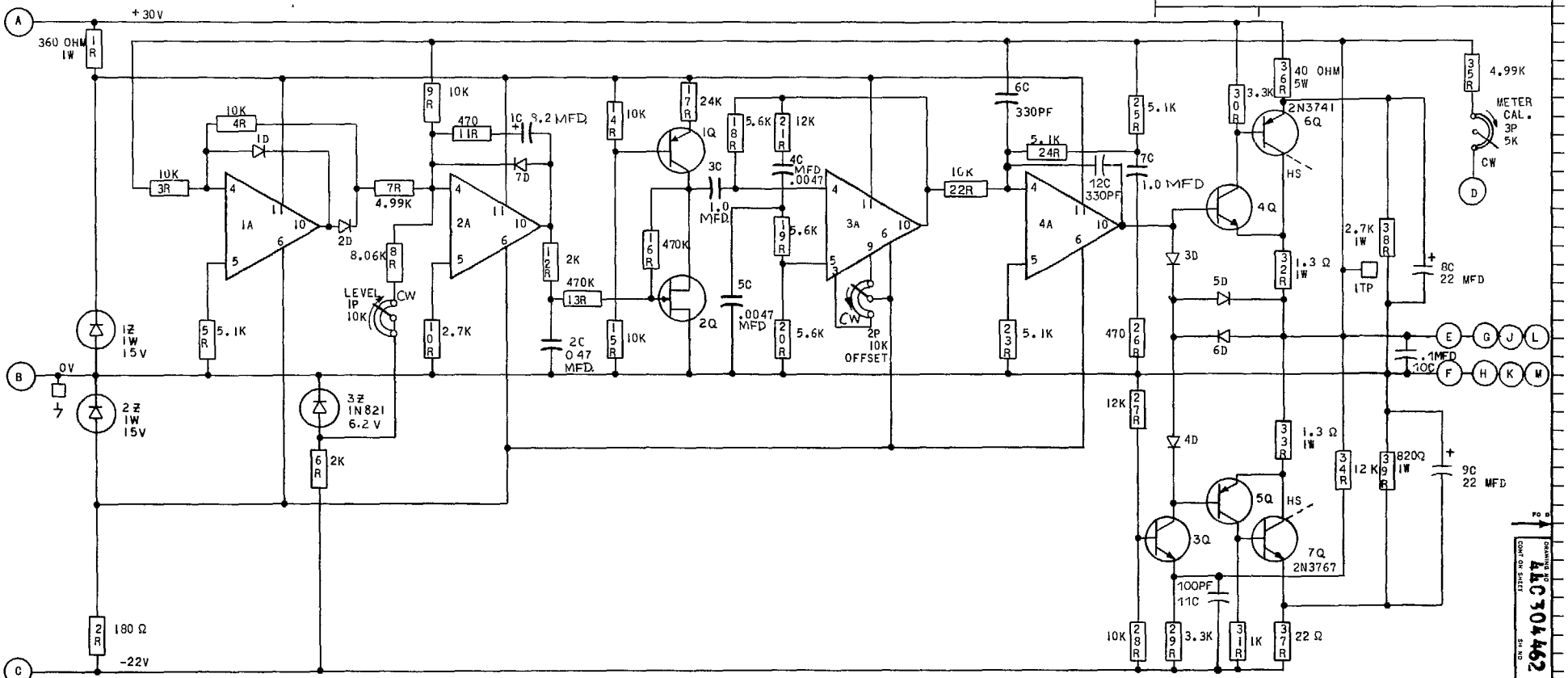
44C304462

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMAL	ANGLES
	✓	±	±	±

TITLE: 3KHZ OSCILLATOR

FIRST MADE FOR: EHC 6-OSC 44B331717-G01



FREQUENCY: 3000HZ± 300HZ
AMPLITUDE: 6.00 VRMS (17V PEAK TO PEAK)
DC OFFSET: ADJUSTABLE TO ZERO
MINIMUM LOAD IMPEDANCE: 25Ω
DRIFT: 0.1% FIFTEEN HOURS
AMBIENT TEMPERATURE

	V	%
DISTORTION 2ND HARMONIC	.056	0.93
3RD HARMONIC	.0083	0.14
4TH HARMONIC	.0006	0.01
5TH HARMONIC	.0022	0.04

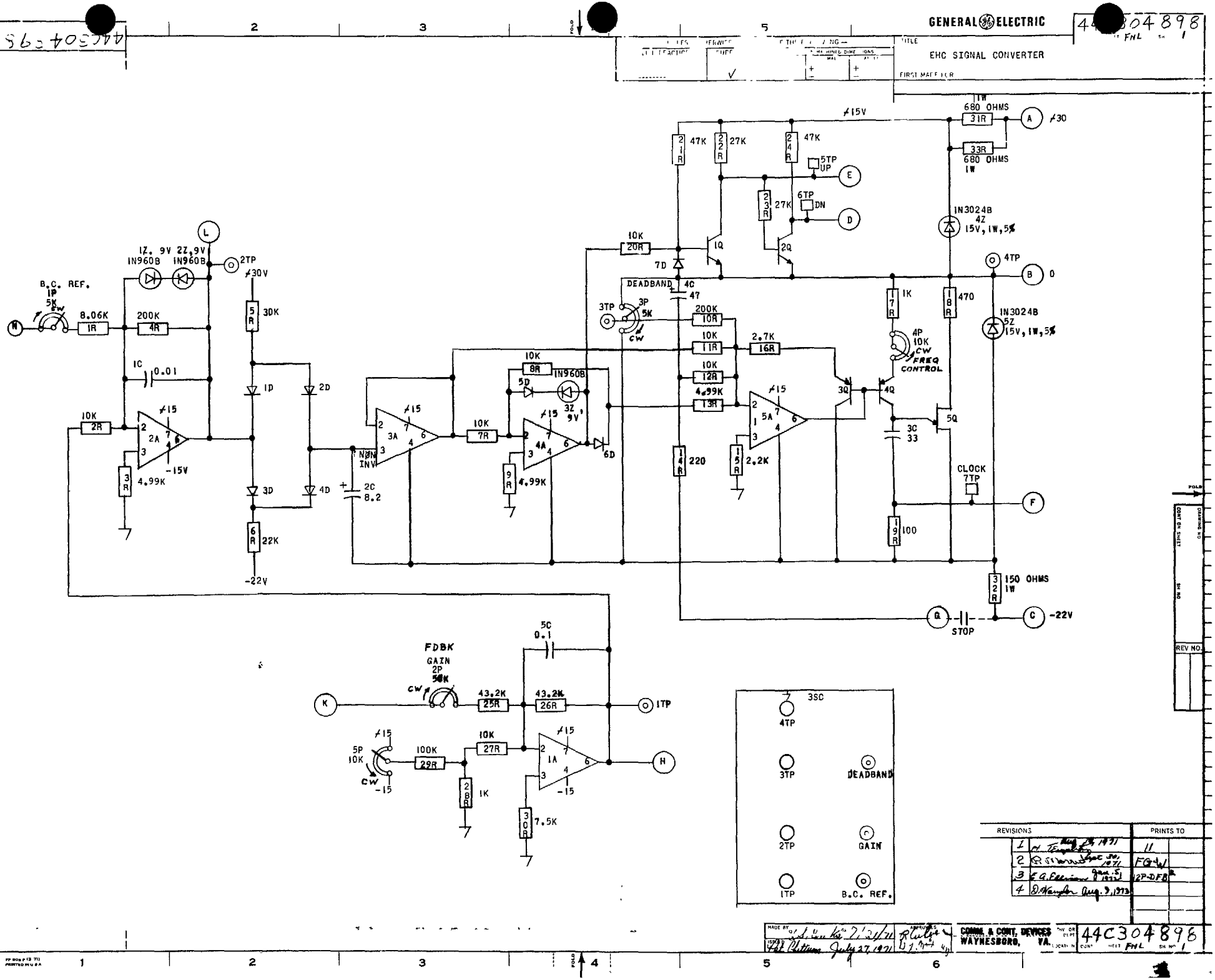
DRAWING NO. 44C304462 REV NO

REVISIONS			PRINTS TO
1	Nov 17, 1971		11
2	Oct 2, 1972	12-DFR	
3	Nov 9, 1972	FG-W	
4	May 24, 1973		

APPROVED: MAY 20 1974

COMM. & CONT. DEVICES, WAYNESBORO, VA

44C304462


GENERAL ELECTRIC
44C304898
EHC SIGNAL CONVERTER
+15V
-15V
-22V
1W 680 OHMS
33R 680 OHMS 1W
IN3024B 4Z 15V, 1W, 5%
IN3024B 5Z 15V, 1W, 5%
150 OHMS 1W
B.C. REF.
8.06K
200K
0.01
30K
22K
2.7K
4.99K
2.2K
220
100
470
1K
CLOCK
STOP
DEADBAND
FDBK GAIN
43.2K
7.5K
NON INV
4P 10K CW FREQ CONTROL
5P 10K CW
REVISIONS
PRINTS TO
FGW
2P-DFB
44C304898

DRAWING NO 44C305259 SH NO
CONT ON SHEET

GENERAL ELECTRIC

44C305259
CONT ON SHEET | SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

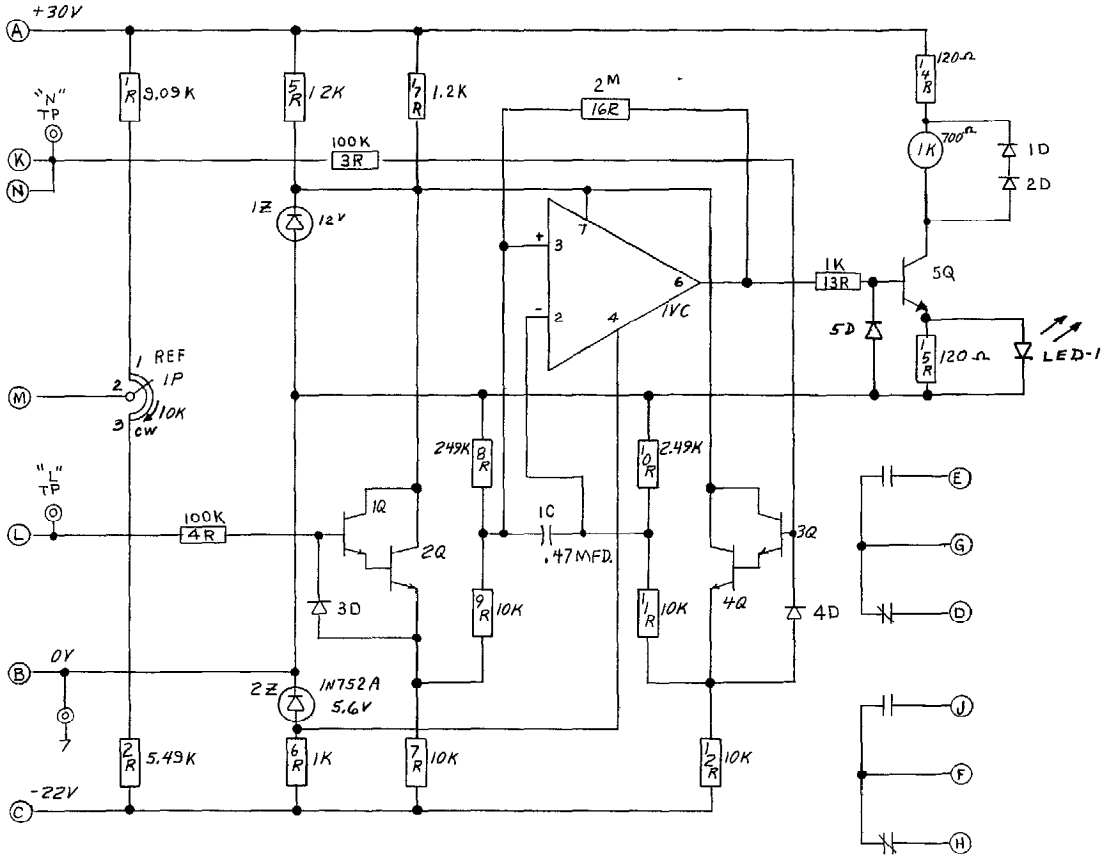
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
VOLTAGE COMPARATOR
1VC

FIRST MADE FOR 44B331761-G01

CONNECTIONS

A	+30V	
B	0 V	
C	-22V	
D		
E		
F		
G		
H		
J		
N, K		INPUT
L		SEE NOTE 2
M		

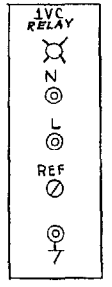


NOTES:

1. IN NORMAL OPERATION THE RELAY IS ENERGIZED. IT CAN BE ADJUSTED TO DROPOUT ON EITHER INCREASING OR DECREASING SIGNALS BETWEEN PLUS AND MINUS 10 VOLTS.
2. DROPOUT ON — INPUT — JUMPER — REF. TP — DROPOUT

INCREASING SIGNAL	N	L-M	L	N>L
DECREASING SIGNAL	L	N-M	N	

3. RELAY CONTACTS RATED 2A, 28VDC.
4. RELAY ENERGIZED WHEN LIGHT IS ON.



FACEPLATE

DRAWING NO 44C305259 SH NO
REV NO

REVISIONS	PRINTS TO
V 60 Clemson Aug 7, 1972	11
	12P DFB
	FG-W

MADE BY	APPROVALS	SPECIALTY CONTROL ... DIV OR DEPT	44C305259
Clemson 10/21/71	R. Culver	WAYNESBORO, VA. LOCATION	CONT ON SHEET / SH NO
ISSUED C. Lotts Dec 8, 1971			

FF 608 F 19 673
PRINTED IN U.S.A.

2

GENERAL ELECTRIC

44C306284 CONT ON SHEET 1A SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 357513TC300A11

ELEMENTARY DIAGRAM

091-40778, TB122881

TAMPA ELECTRIC
BIG BEND #3

PRINTED CIRCUIT BOARD LOCATIONS

RACK 1

LP STOP VALVE		HP STOP VALVE		CONTROL VALVE		SPEED SIGNAL		SPEED SIGNAL		OVERSPEED TEST		REF. LVG.				LOCAL	MANUAL REFERENCE		AUTO REFERENCE							
1DI		1DI		1AØ2		3AØ2		3AØ2		1ØSI		1ØA	1ØA	1ØA	2LVG	1VC	CL & SMD	1DI	7BCI		3SC		CL & SMD	1DI	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

RACK 2

PRI SPEED			SEC. SPEED			SIGNAL LVG				REGULATION & COMPENSATION		SERVO VALVE LOOP				3KHZ OSC	SPEED SENSING 10% 105% 110%			200 RPM	SPEED INDICATION AND SWITCHING			LOW SPEED SWITCH		LSS PWR SUP	
1ØA	2FVC		1ØA	2FVC		1ØA	1ØA	1LVG		1ØA	1CR	2SA		1DI		60SC	1VC	1VC	1VC	1VC	1ØA	2FVC		LSS		LPS	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

RACK 3

			30 VOLT SUPPLY LINE A				LINE B				22 VOLT SUPPLY LINE A				LINE B				24 VOLT SUPPLY LINE A				LINE B			
1DB	1DB	1DB		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

FRONT VIEW

REVISIONS			PRINTS TO
1	G.R. Bard Nov. 27, 1972		11
2	E.A. Cannon May 14, 1973		12P-DFB2
3	D. Naughton Aug 23, 1973		FG-W
			12P-FS2

MADE BY R. Templeton 6-6-72
ISSUED Bob Mayer June 12, 1972
APPROVED R. Culliver 6-6-72

COMM. & CONT. DEVICES DIV. OR DEPT. WAYNESBORO, VA LOCATION

44C306284 CONT ON SHEET 1A SH NO 1

FORM 828 P 12-68 PRINTED IN U.S.A.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11

ELEMENTARY DIAGRAM

091-40778, TB122881 TAMPA ELECTRIC BIG BEND #3

PRINTED CIRCUIT BOARD AND SUB. ASM. LIST

PRINTED CIRCUIT BOARD

SYMBOL	TITLE	ELEM. DIAG.	ORDER NO.	QTY.
1AØ2	ANALOG OUTPUT BOARD	44C301216	44D236379-G01	1
CL& SMD	CLOCK & STEPPING-MOTOR DRIVE	44C303567	44D331918-G01	2
1CR	COMPENSATION & REGULATION	44A301400	44D236388-G01	1
1DB	DIODE BOARD	44A300243	44D236370-G01	3
1DI	DETECTOR INDICATOR	44C300609	44D381585-G01	5
1LVG	SIGNAL LOW VALUE GATE	44A304763	44D236372-G02	1
2FVC	FREQUENCY TO VOLTAGE CONVERTER	44C301682	44D236365-G02	3
1REC	SERIES REG. BD. #1	44A306392	44B317827-G01	6
1SR	SERIES REG. BD. #2	44B306197	44D236371-G01	6
1ØA	OPERATIONAL AMPLIFIER	44C300610	44D381589-G01	9
1ØSI	OVERSPEED INTEGRATOR	44A302692	44D236373-G02	1
2SA	SERVO AMPLIFIER	44C304445	44D236389-G02	1
3SC	SIGNAL CONVERTER	44C304898	44B331743-G01	1
3AØ2	ANALOG OUTPUT BOARD	44C301622	44D236379-G03	2
LPS	LOW SPEED SWITCH PWR. SUPPLY	44B306144	44B337328-G01	1
LSS	LOW SPEED SWITCH	44C306144	44B337327-G01	1
1VC	VOLTAGE COMPARATOR	44C305259	44B331761-G01	5
7BCI	BOILER CONTROL INTERFACE	44C306492	44B331746-G02	1
2LVG	REF. LOW VALUE GATE	44A300281	44D236378-G01	1
60SC	3KHZ. OSC. BOARD	44C304462	44B331717-G01	1

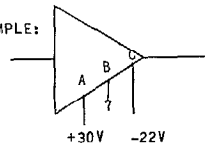
SUB. ASSEMBLIES - PANELS -

TITLE	ORDER NO.	CONNECTION DIAGRAM
POWER PANEL	3S7513KF300A5	44C306526
METER PANEL	3S7513KF306A9	44C305834
METER PANEL	3S7513KF306A11	44C306262
RELAY PANEL	3S7513K307A14	44D305989
DIFF. XFMR. DRIVE	44D315178-G01	44C300882
CONTROL PANEL	3S7513KL300A9	44D305988

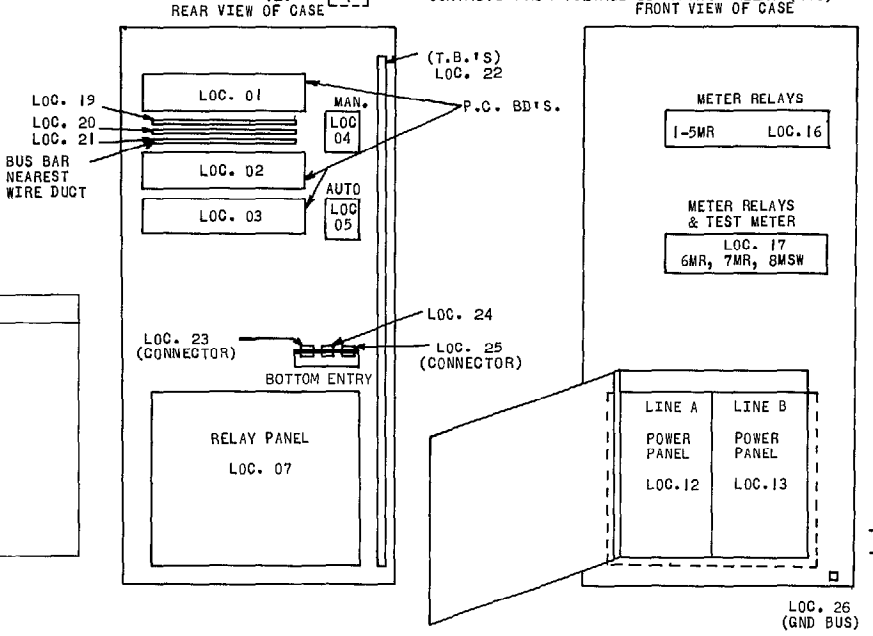
CUSTOMER NOTES:

1. ⏚ INDICATES ZERO REFERENCE POTENTIAL, NOT CASE GROUND.
2. ⏚ INDICATES STATION GROUND.
3. ALL OPERATIONAL AMPLIFIERS HAVE ADDITIONAL CONNECTIONS NOT SHOWN ELSEWHERE

EXAMPLE:



4. THE LOCATION OF THE PRINTED CIRCUIT BOARDS IS SHOWN BY THE NUMBERS APPEARING IN OR NEAR THE SQUARE WHICH REPRESENTS THE P.C. BD. EXAMPLE 0215 WOULD BE RACK #2 AND SLOT #15.
5. 1MR TO 7MR (SHEET 10) ARE METER RELAYS.
6. * INDICATES ITEMS NOT FURNISHED BY WAYNESBORO.
7. INDICATES SHIELDED CABLE.
8. INDICATES TWISTED WIRES.
9. INDICATES TERMINALS AT EHC CASE FOR CUSTOMER CONNECTION
10. INDICATES TERMINALS AT TURBINE
11. INDICATES SUB-PANEL & PC BOARD TERMINALS IN EHC CASE
12. CONTACTS FROM VOLTAGE COMPARATOR RELAY (1VC)



REVISIONS			PRINTS TO
1	C.A. Carol Nov 22, 1972		11
2	E.A. Leeson May 14, 1973		12P-DRB2
3	Dale Wangler Aug 23, 1973		FG-W
			12P-FS2

MADE BY L. Templeton 6-6-72 APPROVED W. Gilley ISSUED Bob Mayes June 12, 1972

COMM. & CONT. DEVICES DIV OR DEPT WAYNESBORO, VA LOCATION

FV 809 P 12-68 PRINTED IN U S A

GENERAL ELECTRIC

44C306284 CONT ON SHEET 3 SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

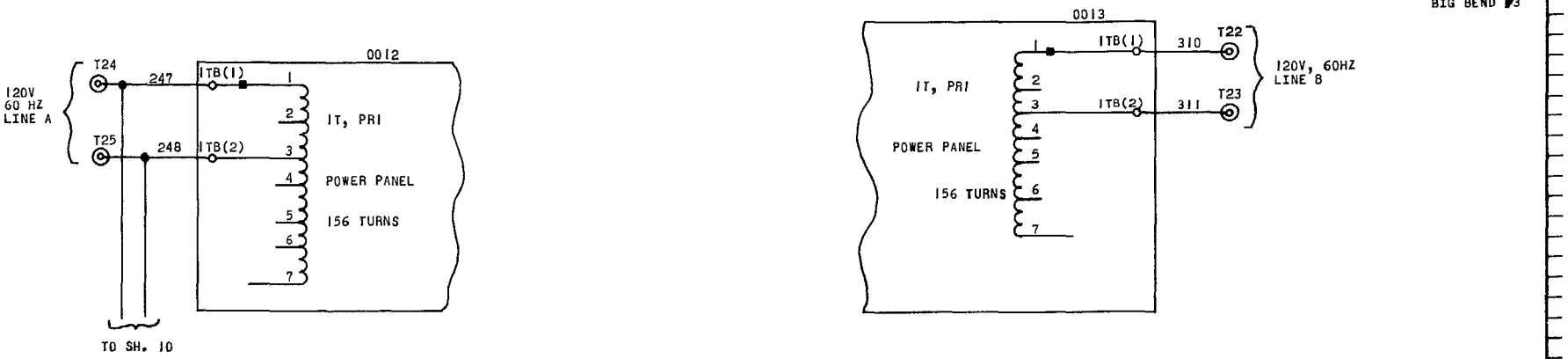
TITLE

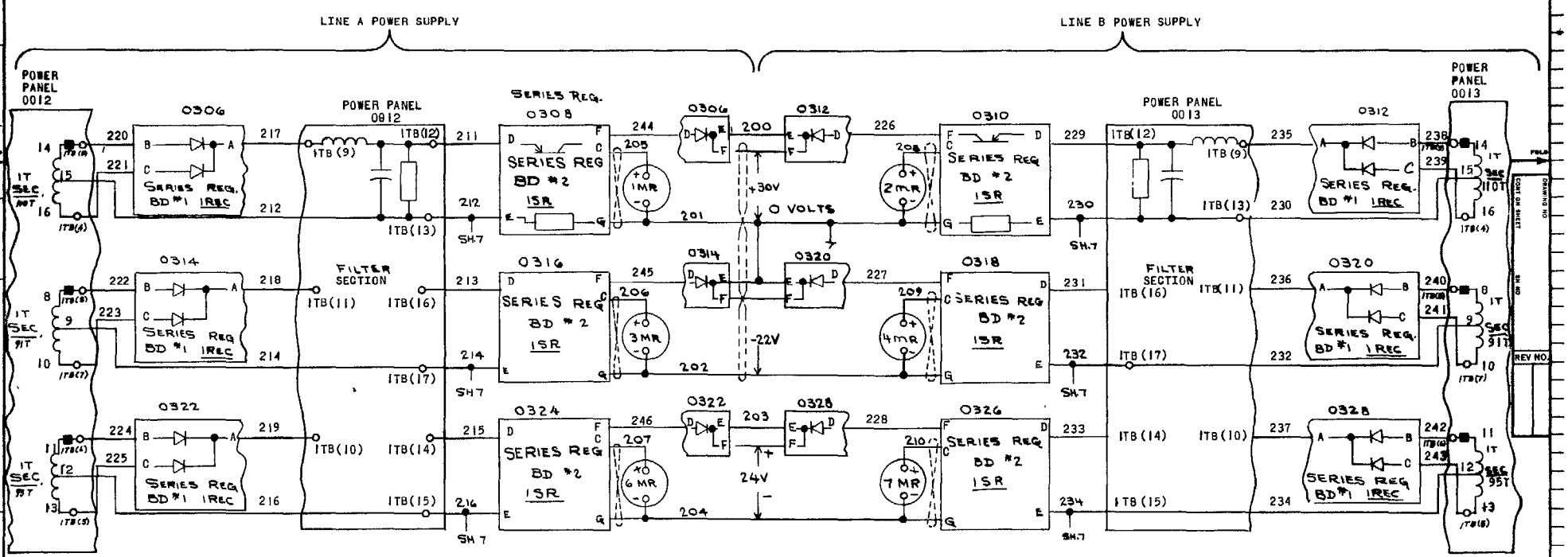
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881

ELEMENTARY DIAGRAM

TAMPA ELECTRIC
BIG BEND #3





REVISIONS		PRINTS TO
1	B.D. Camp Nov. 22, 1976	11
2	G.A. Clem May 14, 1976	12P-DFB 2
		FG-W
		12P-F 5 2

MADE BY ISSUED: Tampa Elec 6-6-72; Not Major June 12, 1972

APPROVALS: R.D. Culver 12-5-72

COMM. & CONT. DEVICES DIV. OR DEPT, WAYNESBORO, VA LOCATION

44C306284 CONT ON SHEET 3 SH NO. 2

FY 809 P 12-691 PRINTED IN U.S.A.

GENERAL ELECTRIC

44C306 284
CONT ON SHEET 4 SH NO 3

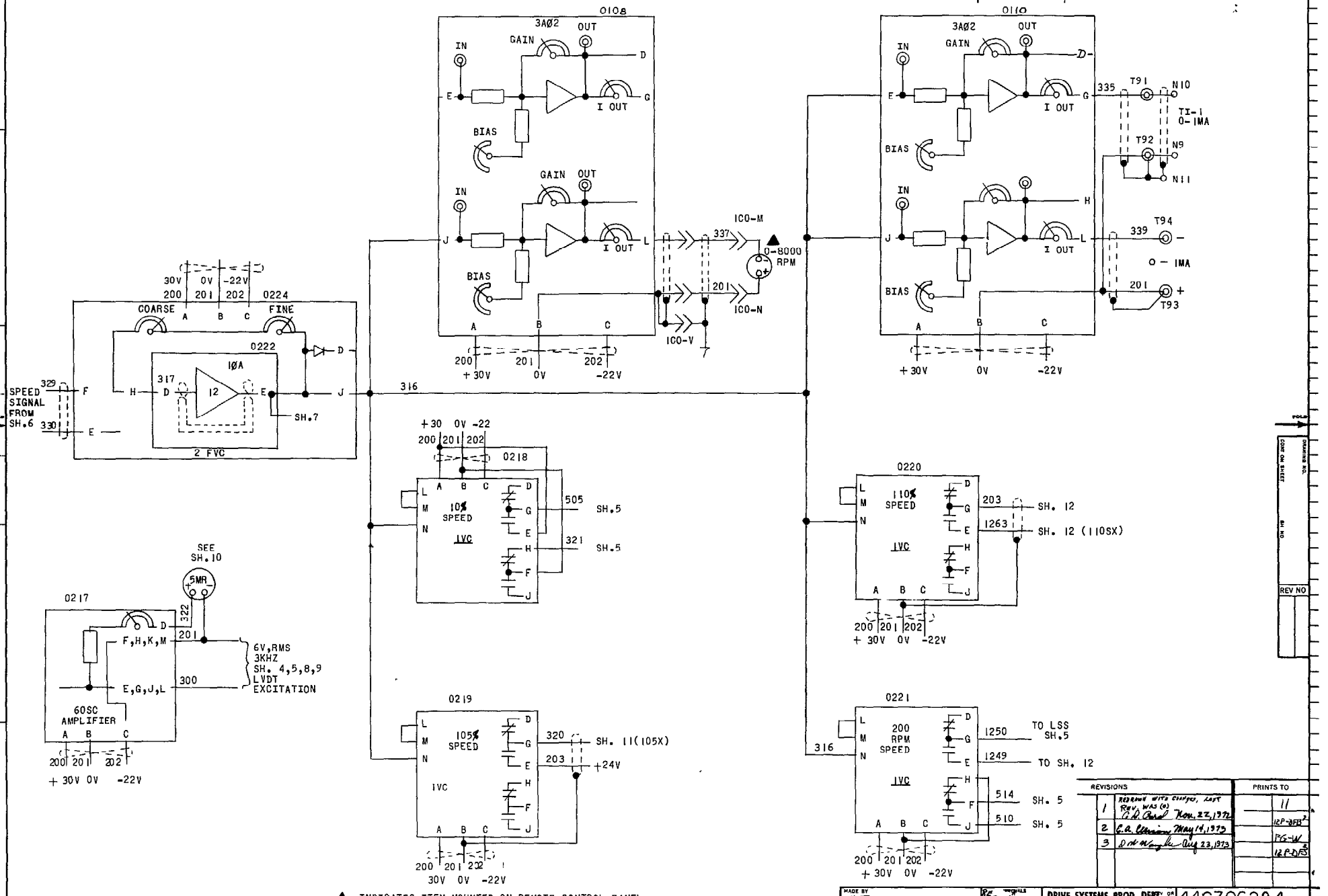
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 091-40778 TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3



▲ INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL

REVISIONS		
1	REDRAWN WITH CHANGES, LAST REV. WAS (0) C.A. Carol Nov. 22, 1972	
2	C.A. Carol May 14, 1973	
3	D.W. Wright Aug 23, 1973	

PRINTS TO
11
12P-BFB
PG-W
12P-D/B

MADE BY A. Trogdon 6-6-72
CHECKED D. Moyer Nov 22, 1972
RELEASED R.G. Pulson R.G. Pulson

DRIVE SYSTEMS PROD DEPT. OR DEPT
WAYNESBORO, VA. LOCATION

44C306284
CONT ON SHEET 4 SH NO 3

FP-809-P (4 72)
PRINTED IN U.S.A.

5

GENERAL ELECTRIC

44C306284 CONT ON SHEET 5 SH NO 4

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S751350300A11 091-40778, T2122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3

MANUAL REFERENCE

SPEED REFERENCE

SH.11 416 417 203

0118
CLOCK PERIOD
FAST 4.17 M
SLOW 40.0 msec
RAISE & LOWER FAST
M
L — DN
K — UP
RELAY PNL. (ITB-1)
CL & ISMD
C B A
202 201 200
-22 +30

0120
1DI
3KHz
300 201
D
415 SH.5 MAN. REF.
LSS 0V
HSS -7V
F
E
M
K
J
N+
A B C
200 201 202
+30V -22V

1A 1B 1C 1D 24V 1E 1F 1G
ILS-HSS
TO MHSS & MLSS RELAYS
2LS-LSS

407 404 406 403 405 402
LF RF L R
12 13
408 200 410 411 412
2H 2K 2J 2L 2M
R W W/R GRN W/G
ISM
BLK 2A
BRN 2B
1DT HSS
GRN 2C
RED 2D
YEL 2E
2F
413 414 201

ILS AT TERMINAL BOARD END
MANUAL 0004

ITB-1 RELAY PANEL

FEEDWATER CONTROLLER
HSS= 20MA
LSS= 40MA
CONTROL SIGNAL
POSITION 0-1MA + −

(T32) (T33) (T34) (T35) (T36)
201 418 201
420 419

0122
4P CURRENT
3A IN
3P BIAS
1A OUT
1P CMR
2P BIAS 78CI
NON INV
LSS 0V
HSS +5V
423

421
AUTO REF.
LSS 0V HSS -5V. TO SH. 7
TO SH. 5
SH.11 439 438

422
+ = DN 0V@ D
− = UP 0V@ E
SH. 7
427
429 430 431 432

0125
K H ITP L 2ITP
GAIN 2P
1A 2A
5P
1P
B.C. REF.
3SC
3P DEAD BAND
4P FREQ. CONTROL
DN D
UP E
CL F
STOP G
425

0126
MOTOR DRIVE
L
K DN
J UP
30 20 10
CL & ISMD
C B A

433 200 435 436 437
2H 2K 2J 2L 2M
ILS AT TERMINAL BOARD END
AUTO 0005

3KHz
300 201
0128
DETECTOR
1DI
440 441 201
444 T76 0-1MA
445 T75
T74
200 201 202
+30 -22

-22V 202
0V 201
+30V 200
428

AUTO REFERENCE
HOLD
RELAY PANEL
(ITB-2)

REVISIONS

1	C.J. Cook Nov. 22, 1972	
2	L.A. Clemens May 14, 1973	
3	D.W. Hampton Aug. 23, 1973	

PRINTS TO

11	
12P-DFB2	
FG-W	
12P-F53	

MADE BY L.L. Templeton 6-7-72
ISSUED Bob Moyer June 12, 1972
APPROVALS R.J. Culver G.S.L. LM
COMM. & CONT. DEVICES DIV OR DEPT WAYNESBORO, VA. LOCATION
44C306284 CONT ON SHEET 5 SH NO. 4

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GENERAL ELECTRIC

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TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881
ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3

REFERENCE LOW VALUE GATE 0116

2 LVG

REF. LOW VALUE GATE AND OVERSPEED TEST INT.

RELAY PANEL (ITB-2)

200 — 9 — ØTT — 10 — 201 — 512

OVERSPEED TEST INTEGRATOR 0112 SH.4

1ØSI

2P RATE

509

1P LIMIT

0113 1ØA

MANUAL REF 415
LSS 0V
HSS-7V

0114 1ØA

BIAS — GAIN

502 — 3 — 501 — SH.7

500 — 501 — SH.7

AMPLIFIER NOT CONTROLLING AT +10V

507 OVERSPEED REF.
-5V≈20% SPEED

SH.4 AUTO REF 421
LSS 0V
HSS-5V

0115 1ØA

504 — 4 — 503

503 — SH.7

505 — SH.6
321 — SH.3

0227 LSS PWR SUP. LPS

-22V — 202 — 201 — 0V, 200 — +30V

SPEED SIGNAL SH. 6 — 610

12 VDC — 511 — 5.6VDC

RELAY PANEL ITB-7 — 201 — 510

200 RPM — 514 — 0225

1TDX

LSPUI

M20 T106 — A
M21 T107 — B — 513
M22 T108 — C — 201
M19 T109

LOW SPEED SWITCH LSS

ALARM 2K

LSS 1K

1250 — 200 RPM, SH.3
203 SH. 12
SH. 2

1259 SH. 12

3K HZ SH. 3 — 30V — 300

0V, +30V-22V

0117

N B A C
M L
REM
1VC LEVEL +6↑
D G E H F J

RELAY PANEL (ITB-4) — 201

203 — SH.12 (AX RELAY)
506

DROPOUT=LOCAL, MANUAL CONTROL
PICKUP=REMOTE, AUTO CONTROL (FEEDWATER)

REVISIONS			PRINTS TO	
1	C.A. Card	Nov. 22, 1972		11
2	G.G. Allen	May 14, 1973	12P-DFB²	
3	D.A. Hayden	Aug. 27, 1973	FG-W	
			12PFS²	

MADE BY C.A. Allen 6/7/72
ISSUED Dot Morgan June 12, 1972
APPROVING R.J. Culler

COMM. & CONT. DEVICES DIV OR DEPT — WAYNESBORO, VA LOCATION

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GENERAL ELECTRIC

44C306284 CONT ON SHEET 7 SH NO 6

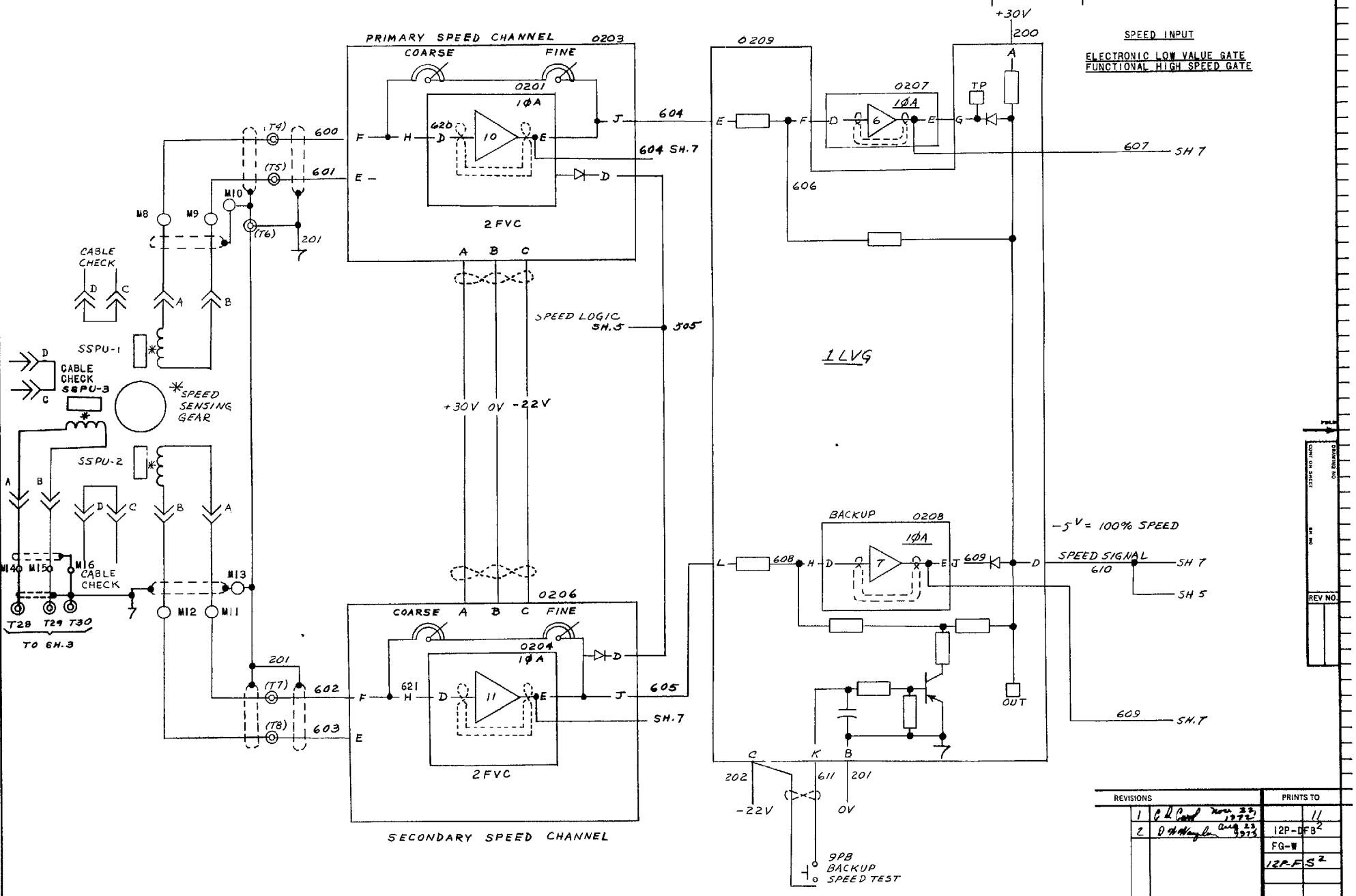
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APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEN #3



REVISIONS			PRINTS TO	
1	C.L. Cook Nov 22, 1972			11
2	D.H. Mayle Aug 23, 1973		12P-DFB	2
			FG-W	
			12P-F	S 2

MADE BY R. Templeton 6-7-72 ISSUED Dist Mgr June 12, 1972 APPROVAL R.J. Culver

COMM. & CONT. DEVICES DIV. OR DEPT. WAYNESBORO, VA. LOCATION

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GENERAL ELECTRIC

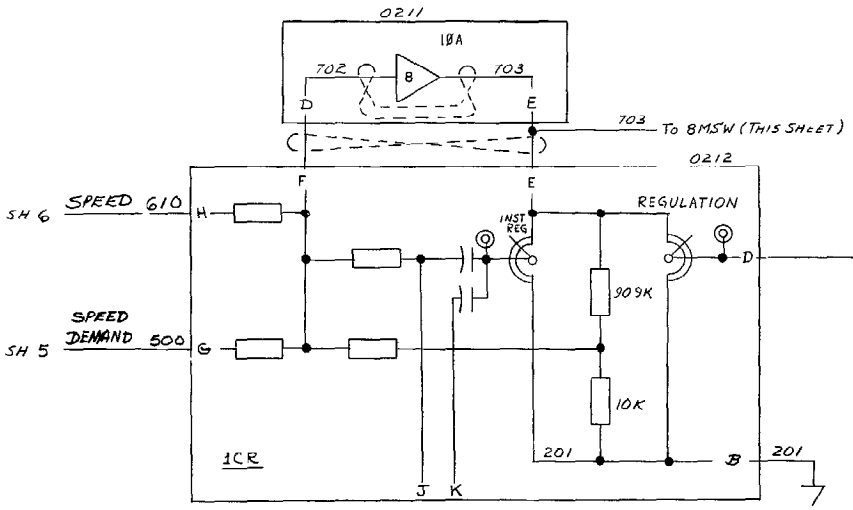
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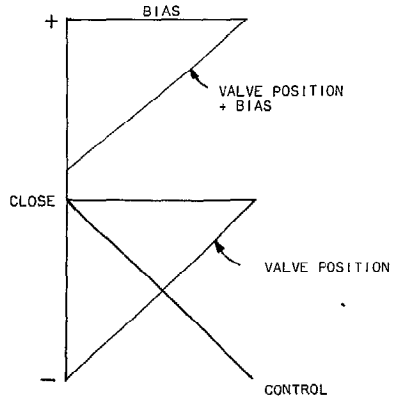
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TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

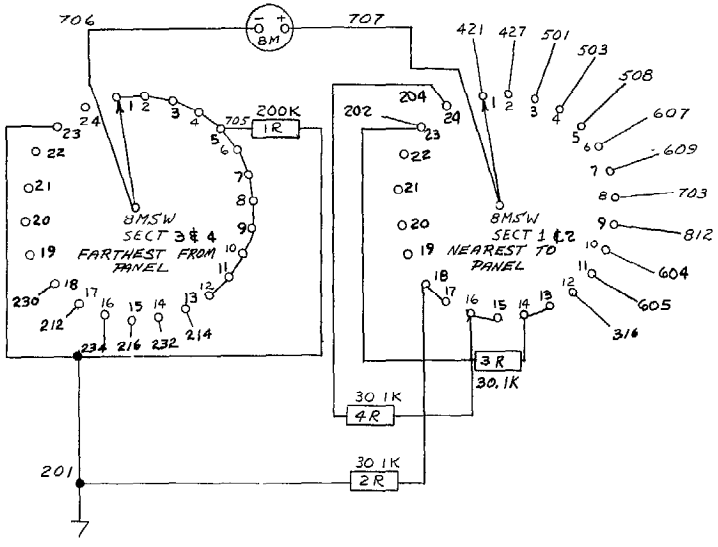
FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3





REGULATION AND STABILIZATION



AMPLIFIER

SWITCH POSITION	NUMBER	SYMBOL	FUNCTION	RACK LOCATION	APPROXIMATE OUTPUT VOLT RANGE	SHEET NO.
1	1	3SC	SIG. CONV.	0125	0 TO -5VDC	4
2	2	3SC	SIG. CONV.	0125	10VDC TO -10VDC	4
3	3	2LVG	MAN. REF.	0114	10VDC TO -5VDC	5
4	4	2LVG	AUTO REF.	0115	10VDC TO -5VDC	5
5	5	0'SPD	OVERSPEED	0113	10VDC TO -5VDC	5
6	6	1LVG	PRI. SPD.	0207	10VDC TO -7VDC	6
7	7	1LVG	SEC. SPD.	0208	10VDC TO -7VDC	6
8	8	1CR	COMP.&REG.	0211	10VDC TO -10VDC	7
9	9	2SA	SERVO AMP.	0214	7VDC TO -7VDC	8
10	10	2FVC	PRI. SPD.	0201	10VDC TO -7VDC	6
11	11	2FVC	SEC. SPD.	0204	0 TO -7VDC	6
12	12	2FVC	SPD. IND.	0222	0 TO -7VDC	3

POWER SUPPLY

					APPROXIMATE OUTPUT CURRENT RANGE	
13	LINE A	1SR	-22VDC	0316	5 AMPS	2
14	LINE B	1SR	-.2VDC	0318		2
15	LINE A	1SR	+24VDC	0322		2
16	LINE B	1SR	+24VDC	0328		2
17	LINE A	1SR	+30VDC	0308		2
18	LINE B	1SR	+30VDC	0310	5 AMPS	2

REVISIONS

1	C.L. Card Nov 22, 1972	
2	E.A. Clemmer May 14, 1973	
3	D.M. Wingler Aug 23, 1973	

PRINTS TO

	11
12P-DFB2	
FG-W	
12P-FS2	

MADE BY K. Templeton 6-7-72 ISSUED Bob Morgan June 12, 1972 APPROVALS

COMM. & CONT. DEVICES WAYNESBORO, VA

44C306284 CONT ON SHEET 8 SH NO 7

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GENERAL ELECTRIC

44C306284 CONT ON 9 SH NO 8

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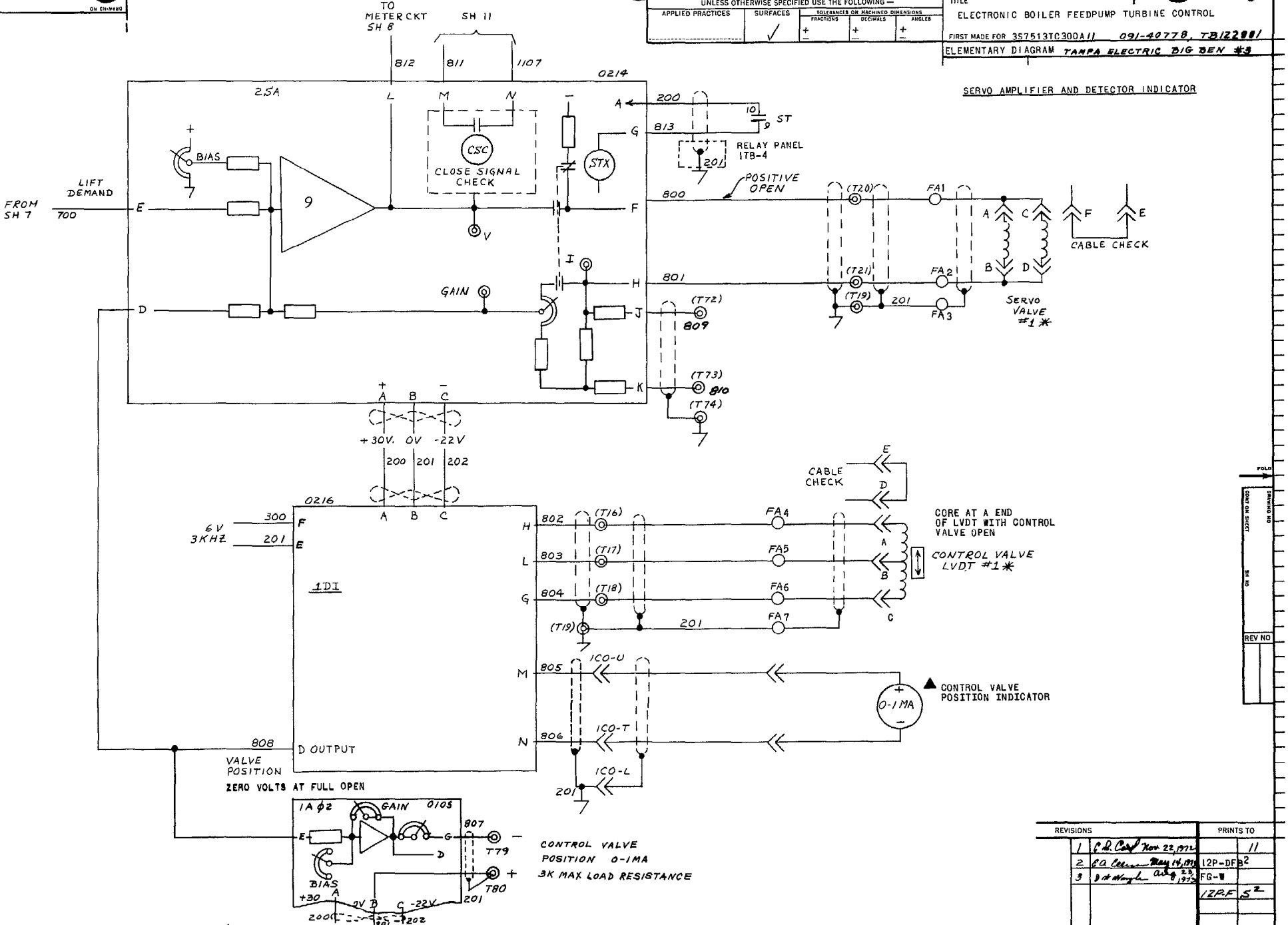
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
	✓			

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEN #3

SERVO AMPLIFIER AND DETECTOR INDICATOR



▲ INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL

REVISIONS		PRINTS TO	
1	E.A. Cook Nov 22, 1972		11
2	E.A. Cook May 14, 1973	12P-DF	2
3	D.A. Mayle Aug 28, 1973	FG-W	
		12RF	5 2

MADE BY L. Tangeltano 6-7-72
ISSUED Pat Mayes June 12, 1972
APPROVALS R.J. Culver
COMM. & CONT. DEVICES DIV OR DEPT WAYNESBORO, VA. LOCATION
44C306284 CONT ON SHEET 9 SH NO 8

DRAWING NO
SH NO
CONT ON SHEET
REV NO

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GENERAL ELECTRIC

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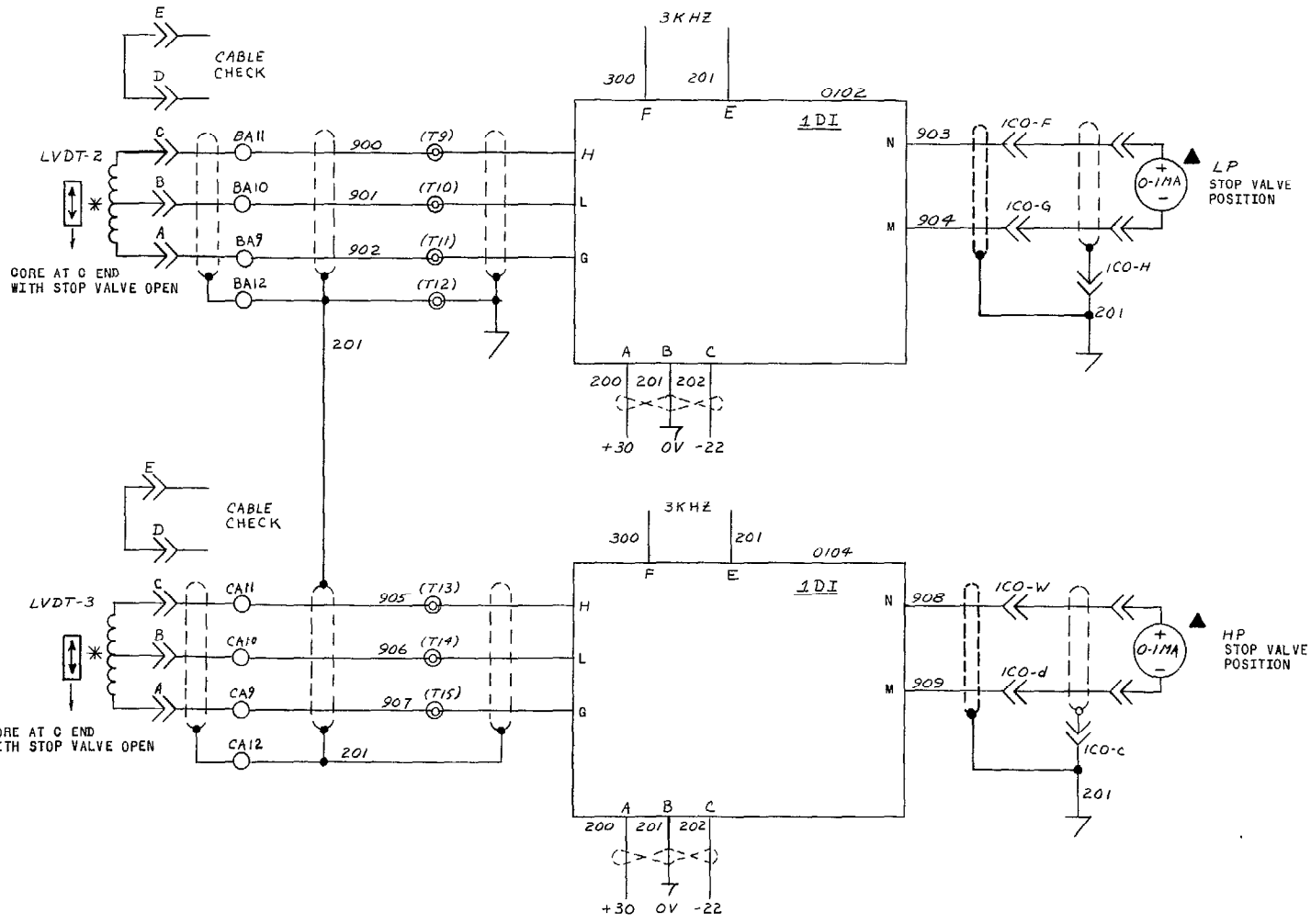
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3

STOP VALVE POSITION



▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL

REVISIONS		PRINTS TO	
1	C.A. Card Nov 22, 1972		11
		12P-DF	82
		FG-W	
		12R-F	82

MADE BY Eleanor 6/7/72 APPROVED

ISSUED Pat Mayer June 12, 1972

COMM. & CONT. DEVICES DIV OR DEPT
WAYNESBORO, VA LOCATION

44C306284 CONT ON SHEET 10 SH NO 9

44C306284 CONT ON SHEET 10 SH NO 9

REV NO

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GENERAL ELECTRIC

44C306284 CONT ON SHEET 11 SH NO 10

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APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

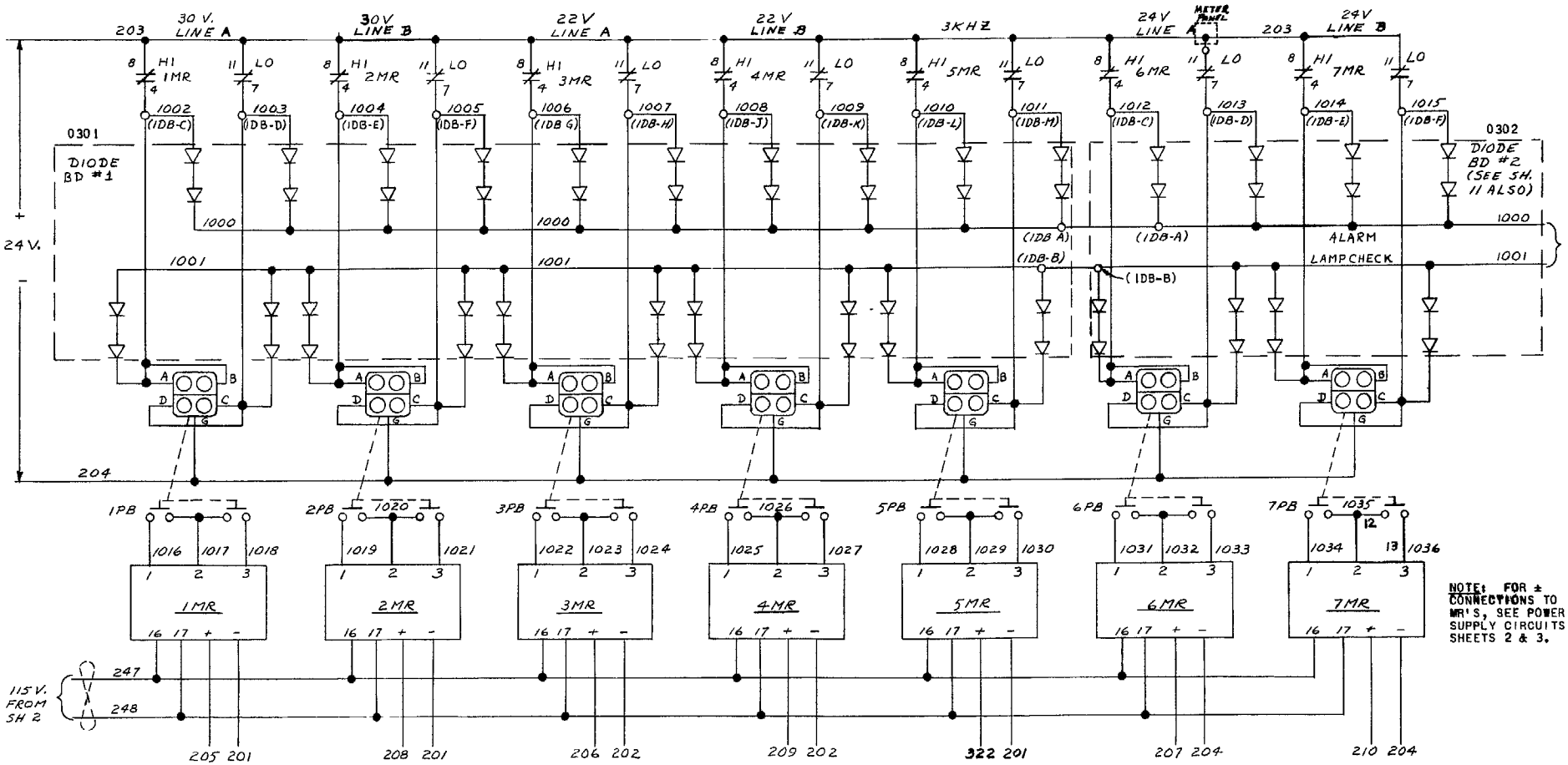
TITLE

ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3

MONITOR



NOTE: FOR ± CONNECTIONS TO MR'S, SEE POWER SUPPLY CIRCUITS SHEETS 2 & 3.

DRAWING NO 44C306284 CONT ON SHEET 11 SH NO 10 REV NO.

REVISIONS			PRINTS TO	
1	C.A. Carr	May 23, 1972		11
2	G.A. Clem	May 24, 1972	12P-DF8	2
			FG-W	
			12RF	5 2

MADE BY G.A. Clem 6/7/72 ISSUED Bob Morgan June 12, 1972

APPROVAL R.A. Carlson C.S. Mott

COMM. & CONT. DEVICES DIV OR DEPT WAYNESBORO, VA LOCATION

44C306284 CONT ON SHEET 11 SH NO 10

FF-609 F 12 681 PRINTED IN U.S.A.

GENERAL ELECTRIC

44C306284 CONT ON SHEET 12 SH NO 11

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

105X

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 091-40778, TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3

0302 DIODE BD #2

0303 DIODE BD. #3

DI (T40) 1108 (T41) 1109

(T42) 1109 (T43) 1111 STATION BATT 130 VDC

203 1000 1001 204 1100 1101 1102 1103 1104 1105 1106

24 V. TO SH 10

3P 4 3N 4 3CC 4 7MHSS 8 MLSS 7 8 4AHSS 5 4ALSS 5

(IDB-G) (IDB-H) (IDB-J) (IDB-C) (IDB-D) (IDB-E) (IDB-F) (IDB-B)

LAMP CHECK BPB

ALARM

1L 2L 3L 4L

+24 GND | -24 GND CABLE CHECK MAN HSS | MAN LSS AUTO HSS | AUTO LSS ALARM

3TB-1 RELAY PANEL

METER PANEL

LO 1MR HI TRIP LINE A, 30V
LO 2MR HI LINE B, 30V
LO 3MR HI LINE A, -22V
LO 4MR HI LINE B, -22V
320 LO 5MR HI 3KHZ
LO 6MR HI LINE A, 24V
LO 7MR HI LINE B, 24V
CC 1116

1055 1055X ALARM ITD 1 MINUTE TD PICKUP ITDX ST

TRIP 1107 CSC 811 MLSS 1118 ITDX 1117 ST

T44 T46 1173 1174 T45 1175

TI M30 (TT1) (T78) 1121 1170

M2LS M1LS A2LS A1LS

CABLE CHECK CIRCUIT

EHC TO CONTROL PANEL CABLE

M26 T3 3CO-F 3CO-X 2CO-m 1CO-K 203

417 T95 1149 1150 T97 1151 T98 1152 416 T100 1154 1153 T96 T99

438 439

FEED WATER CONTROLLER SIGNAL FAILURE

P 1119 1D 2D (ICB4) 3D 4D 5D 6D (T-GND) 344 7D 8D 1120 N CC MLSS MLSX MHSS MHSX ALSS AHSS HOLD

1R 2R

ICO-6

OPERATOR CONTROL

LOWER FAST C NC LOWER C NC RAISE C NC RAISE FAST C NC ICO-F

LOWER RAISE RAISE

ST LOWER FAST NO ICO-E LOWER NO ICO-D RAISE NO ICO-C RAISE NO ICO-B

2CB-1 1148 2CB-3 1149 1133

1123 9D 10D 15D 16D

1124 L 1125 R RF 1126 1130 1135

L R RF 1127 1131 1134 1128 1132 1129 LF

1138 TFSL H39 HD 12D TFHR H40 17D TFSO 18D

MHSS 1136 AHSS 1137 OTT

TURBINE FRONT STANDARD CONTROL

(T85) 1141 1143 (T86) (T87) 1142 (T88) 1157 1156 T89 (T90) 1158

▲ INDICATES ITEMS MOUNTED REMOTE ON CONTROL PANEL.

REVISIONS			PRINTS TO	
1	C.H. Card	Nov. 22, 1972		11
2	C.G. Clemens	May 14, 1973	12P-DFB2	
3	D.H. Mayle	Aug. 23, 1973	FG-W	
			12P-F-S2	

MADE BY C. Clemens 6/7/72 ISSUED Dist Mayle June 12, 1972

APPROVALS J. Lucas

COMM. & CONT. DEVICES WAYNESBORO, VA. DIV OR DEPT LOCATION

44C306284 CONT ON SHEET 12 SH NO 11

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5

GENERAL ELECTRIC

44C306284 CONT ON SHEET 13 SH NO 12

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

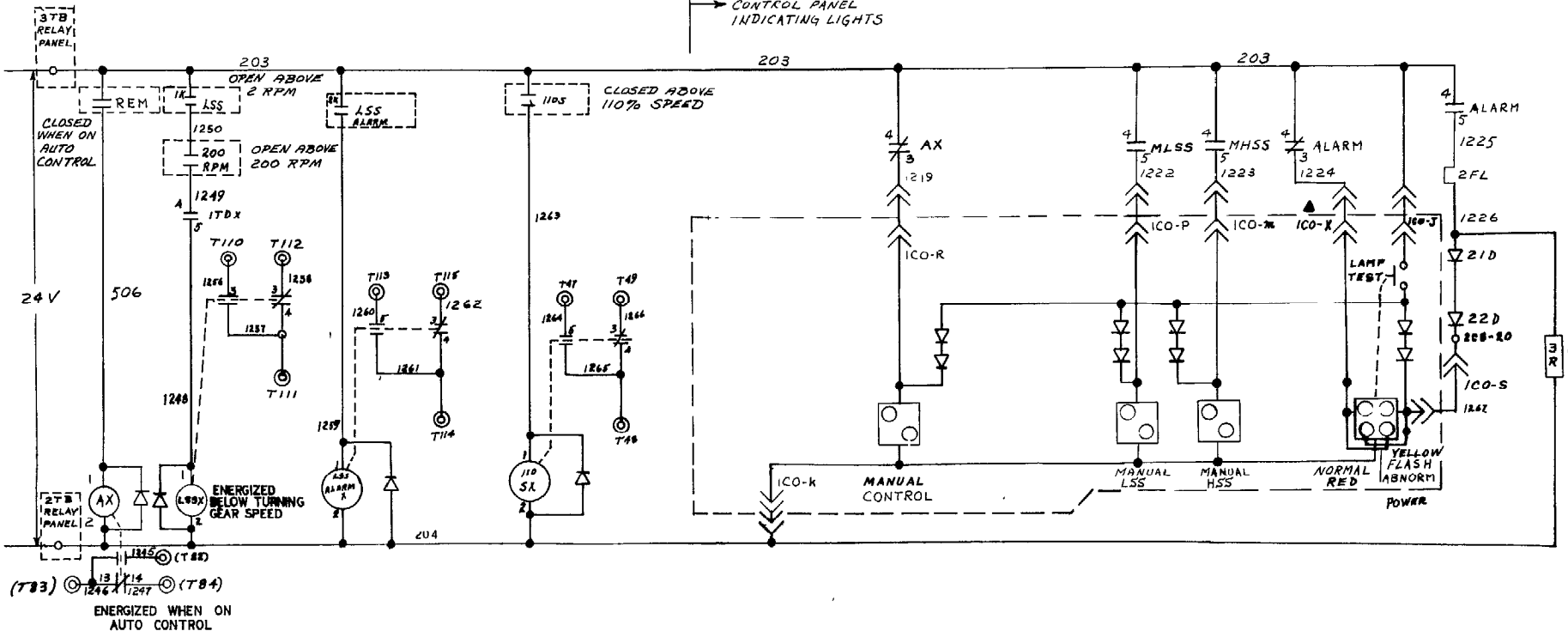
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

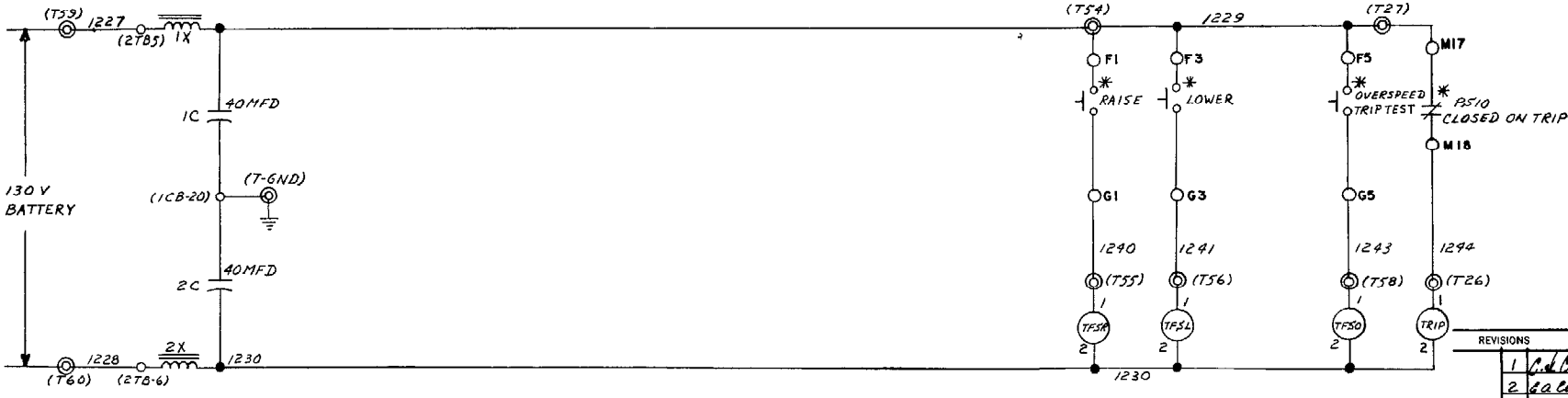
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 09/-40778, T3/122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3

OPERATOR'S CONTROL PANEL INDICATING LIGHTS





▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL.

REVISIONS

1	C.A. Reed	Nov. 22, 1972
2	G.A. Clem	May 14, 1973
3	D.W. Waugh	Aug 23, 1973

PRINTS TO

	//
12P-DR	B2
FG-W	
12P.F	S2

MADE BY	APPROVED	DIV OR DEPT	DRAWING NO
C.A. Clem 6/7/72	R.J. Taylor	COMM. & CONT. DEVICES	44C306284
ISSUED		LOCATION	CONT ON SHEET 13 SH NO 12
Pat Mayes June 12, 1972	C.S. Kuhn	WAYNESBORO, VA	

FF 909 F (2 68)
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GENERAL ELECTRIC

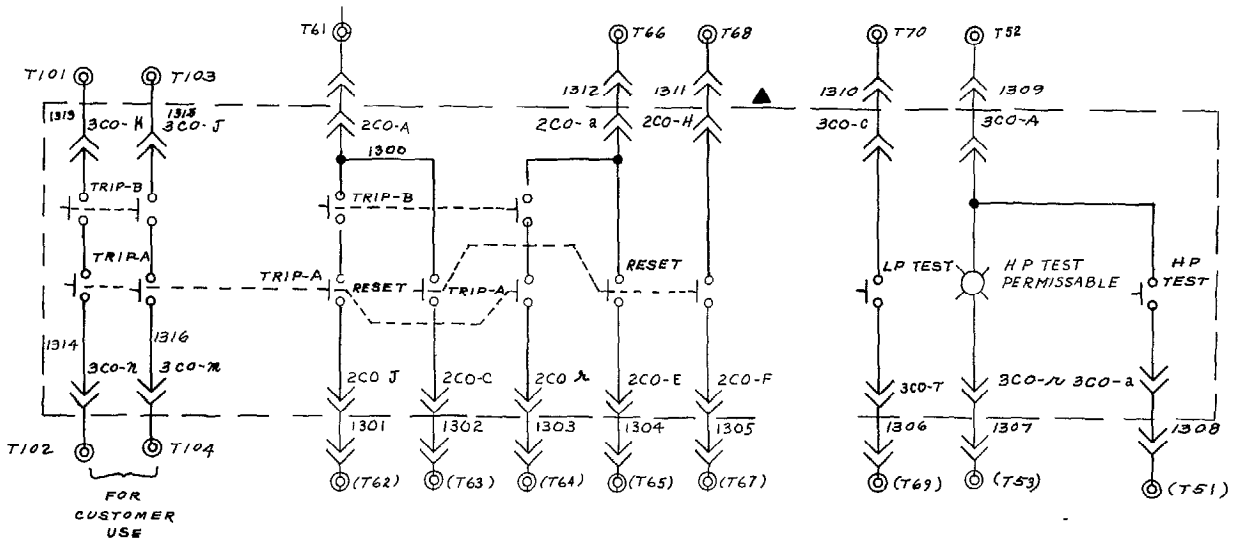
44C306284
CONT ON LET F SH NO 13

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSION: FRACTIONS ± | DECIMAL ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 09/-40778, TB122881

ELEMENTARY DIAGRAM TAMPA ELECTRIC BIG BEND #3



▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL.

REVISIONS			PRINTS TO
1	C.A. Carl Nov. 22, 1972		11
2	C.A. Carl May 14, 1973		12P-DFB 2
			FG-W
			12P-FS 2

MADE BY A. Templeton 6-7-72
ISSUED Dot Mayer June 12, 1972
APPROVALS
COMM. & CONT DEVICES DIV OR DEPT WAYNESBORO, VA LOCATION
44C306284 CONT ON SHEET F SH NO 13

FF 809 P 12 601 PRINTED IN U.S.A.

GENERAL ELECTRIC

44C306285 CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	√	±	±	±

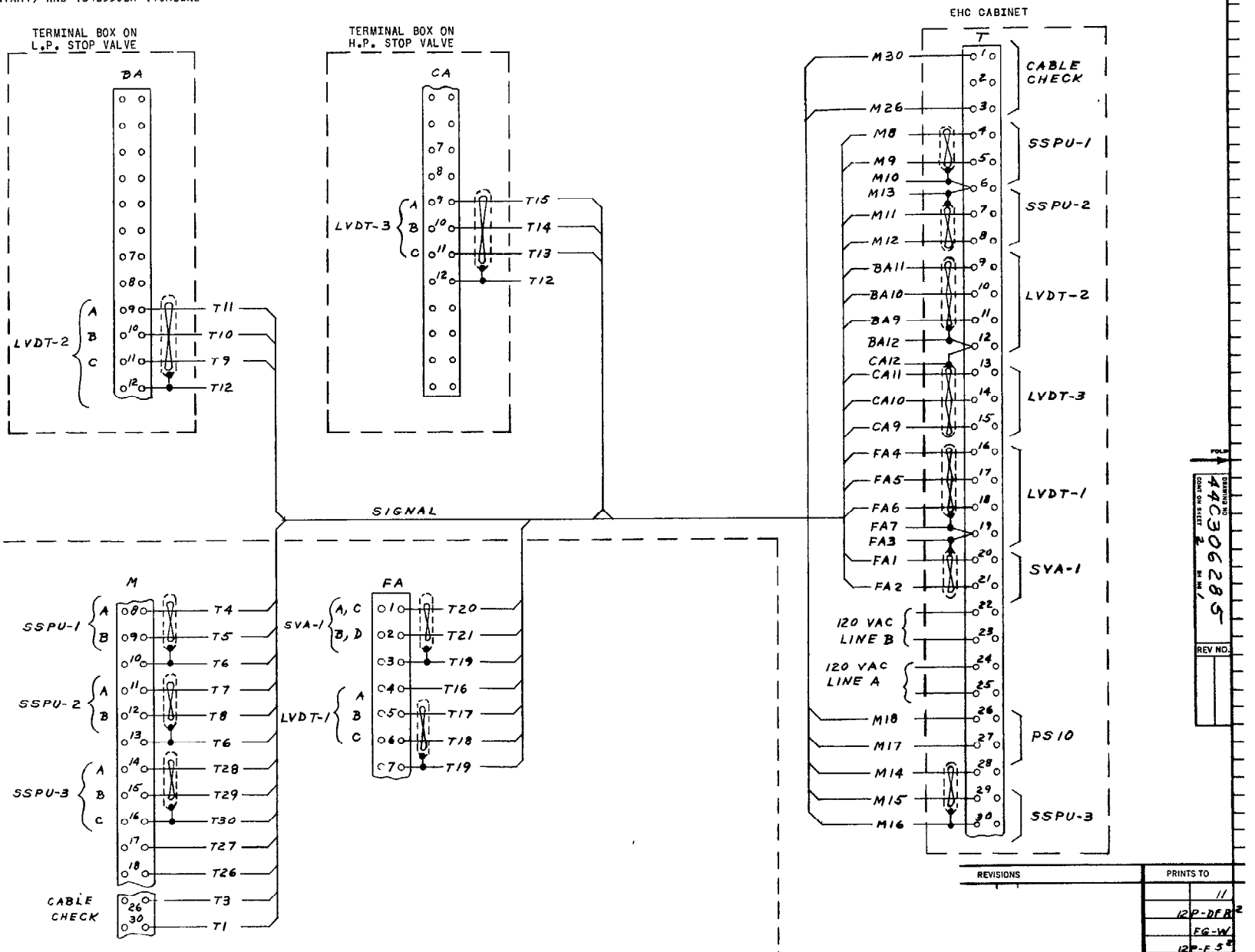
TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 TURBINE: 122881 091-40778

INTERCONNECTION DIAG. TAMPA ELECTRIC BIG BEND #3

NOTE:

1. REFER TO 44C306284 (EHC ELEMENTARY) AND 134B550BX (TURBINE WIRING DWG.) FOR REFERENCE.



REVISIONS

PRINTS TO
11
12P-DFR2
FG-W
12P-F52

MADE BY	APPROVALS			
E. Ellison 7/19/72	R. Rutger	DRIVE SYSTEMS PROD. DEPT. OR DEPT	44C306285	
ISSUED R. Moyer July 31, 1972	RCS	WAYNESBORO, VA. LOCATION	CONT ON SHEET 2	SH NO 1

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PRINTED IN U.S.A.

GENERAL ELECTRIC

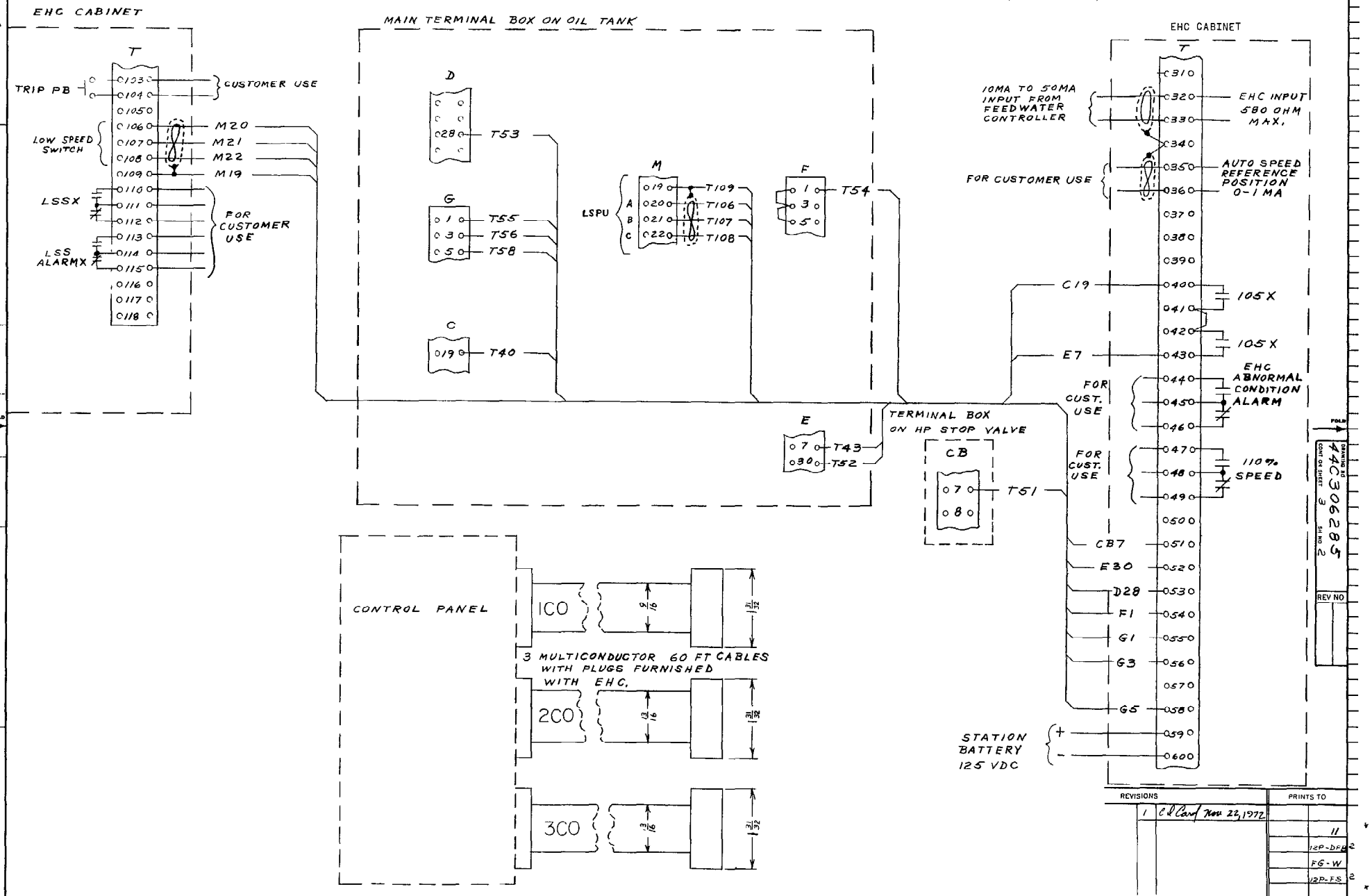
44C306285 CONT ON SHEET 3 SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A11 TURBINE: 122881 091-40778

INTERCONN. DIAG. TAMPA ELECTRIC BIG BEND #3



REVISIONS

1	C.D. Carr	Nov 22, 1972

PRINTS TO

11	
12P-DFP	2
FG-W	
12P-FS	2

MADE BY: Eleanor Ellison 7/19/72
ISSUED: D. Moyer July 31, 1972
APPROVALS: R.C.D.

DRIVE SYSTEMS PROD DEPT WAYNESBORO, VA.

44C306285 CONT ON SHEET 3 SH NO 2

DRAWING NO 44C306285 CONT ON SHEET 3 SH NO 2

REV NO

FORM FF 800 F 14 721 PRINTED IN U.S.A.

5

GENERAL ELECTRIC

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APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S75135C300A11 TURBINE: 122881 091-40778

INTERCONN. DIAG. TAMPA ELECTRIC BIG BEN #3

EHC CABINET

T
C2
E1
E2
E21
E22
C17
E13
BB7
C26
610
620
630
640
650
660
67 — RESET P.B.
68
690
700
710
72 — SERVO VALVE CURRENT 0-1 MA
73
74 — AUTO SPEED REFERENCE 0-1 MA
75
76
770
780
79 — FOR CUSTOMER USE (−) — 0-1 MA ANALOG CONTROL VALVE POSITION SIGNAL
80 — FOR CUSTOMER USE (+)
810
82
83 — AUTO CONTROL
84
85
86 — AUTO SIG. AT LSS
87
88
89 — AUTO SIG. AT HSS
90
91 — N10 — 0-1 MA TACH. SIGNAL
92 — N9
93 — FOR CUSTOMER USE — 0-1 MA SPEED SIGNAL
94
95
96 — MAN. REF. AT LSS
97
98
99 — MAN. REF. AT HSS
100
101 — FOR CUSTOMER USE — TRIP PB
102

CASE GND.

L.P. STOP VALVE T.B.
BB
7 — T69

MAIN TERMINAL BOX ON OIL TANK

E
1 — T62
2 — T63
3
4
13 — T67
21 — T64
22 — T65

N
9 — T92
10 — T91
11 — T92
0-1MA TACH TI-1

C
1 — T61
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17 — T66
18
19
20
21
22
23
24
25
26 — T70

REVISIONS | PRINTS TO: 11, 12P-NF1, FG-W, 12P-FS

MADE BY Eleanor Ellison 7/19/72
ISSUED D. Moyer July 31, 1972
APPROVALS R.C. Gilbert

DRIVE SYSTEMS PROD. DEPT. DIV. OR DEPT. WAYNESBORO, VA. LOCATION

44C306285 CONT ON SHEET FL SH NO 3

FF 8009 F 64 721 PRINTED IN U.S.A.

GENERAL ELECTRIC

44C306444 CONT ON SHEET 2 SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

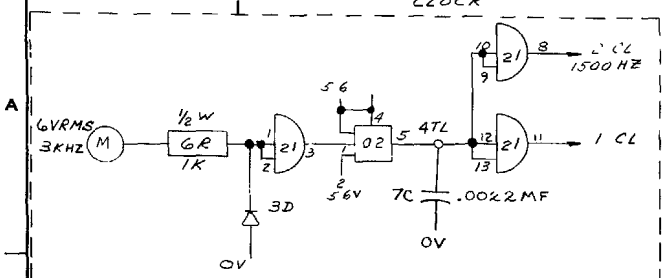
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	✓	±	±	±

TITLE

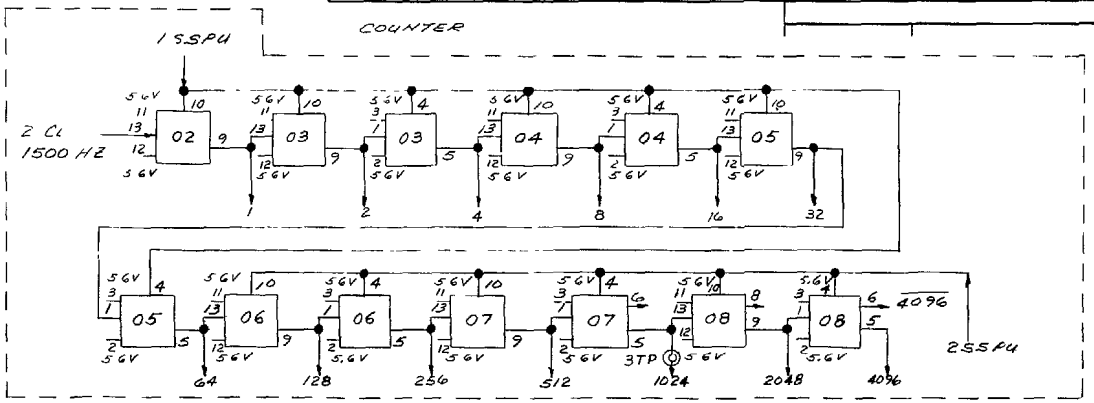
LOW SPEED SWITCH

FIRST MADE FOR 3575137C300A1

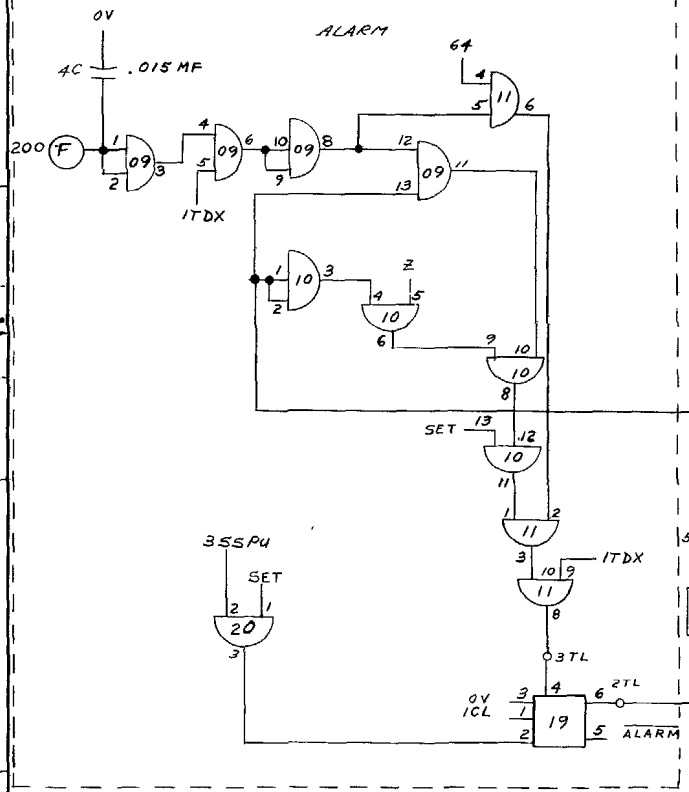
CLOCK



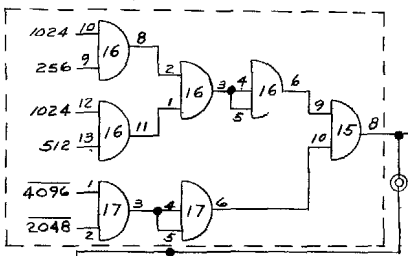
COUNTER



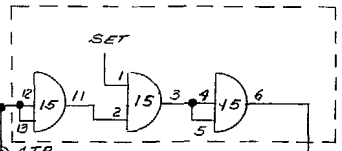
ALARM



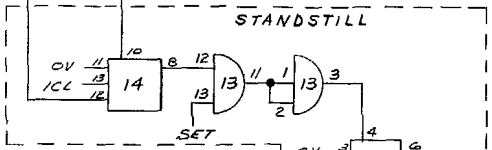
SPEED DETECTOR



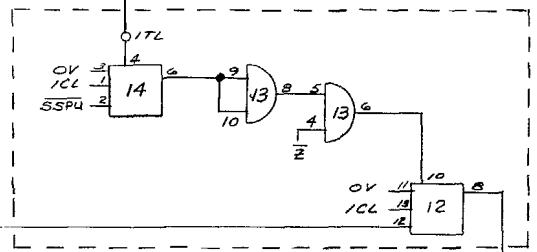
SPEED INHIBIT



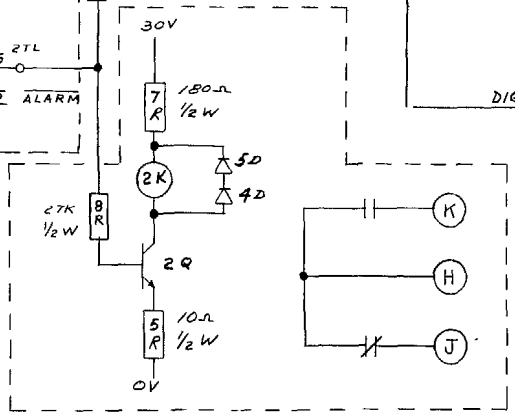
STANDSTILL



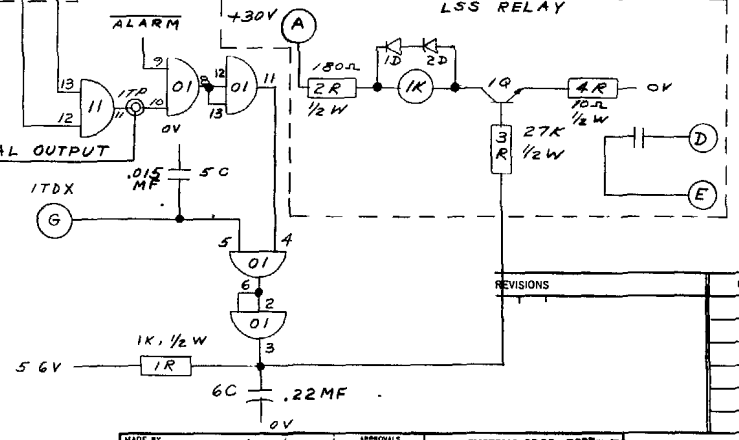
SPEED



ALARM RELAY



DIGITAL OUTPUT



LSS RELAY

REVISIONS

PRINTS TO 11

MADE BY	APPROVALS		
ISSUED B. Moyer Oct 6, 1972	R. S. Smith	DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.	44C306444 CONT ON SHEET 2 SH NO 1

FF 809 F 14 721 PRINTED IN U S A

GENERAL ELECTRIC

44C306444 CONT ON SHEET FL SH NO 2

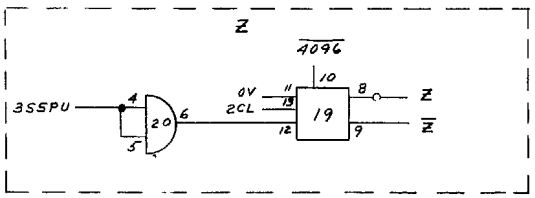
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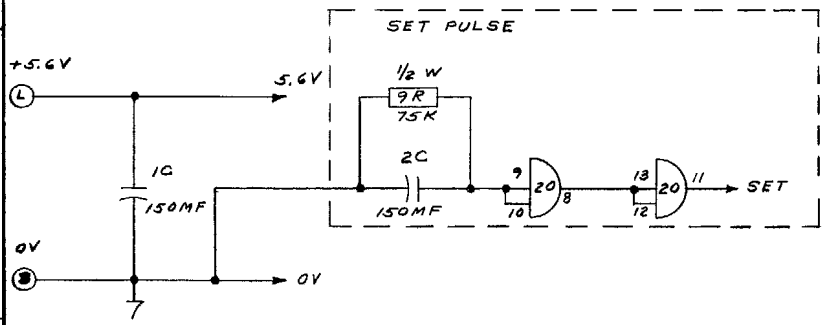
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

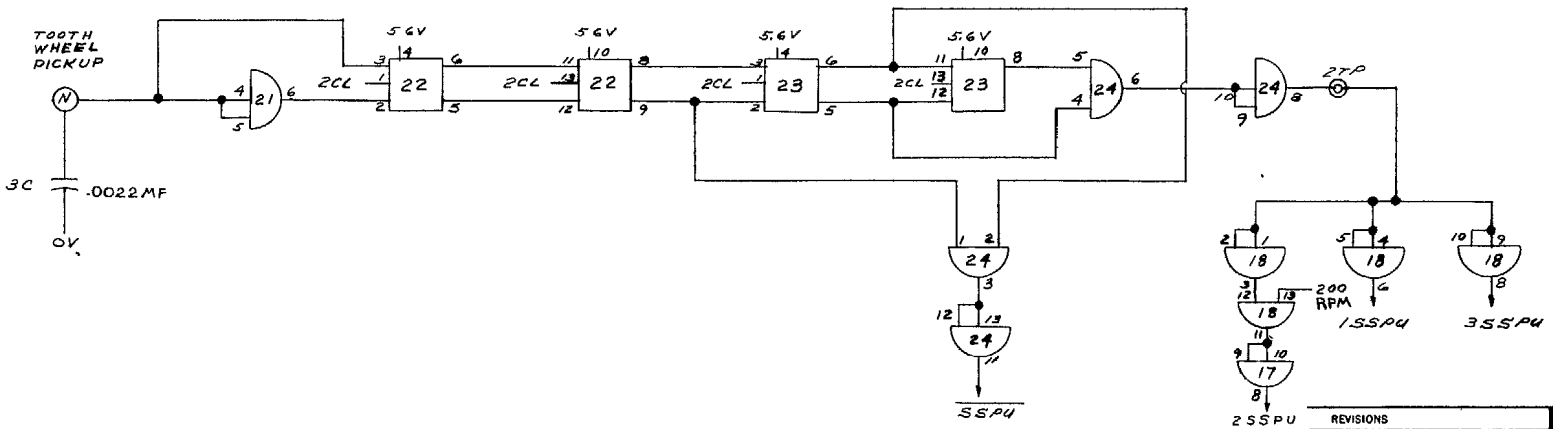
LOW SPEED SWITCH

FIRST MADE FOR 3S7513TC300A11





SHAFT SPEED PICK-UP PULSE



REVISIONS

PRINTS TO 11

MADE BY H. Tarpenter 8-7-72 ISSUED D. Moyer Oct. 6, 1972

APPROVALS RGS

DRIVE SYSTEMS PROD. DEPT. DIV OR WAYNESBORO, VA. LOCATION

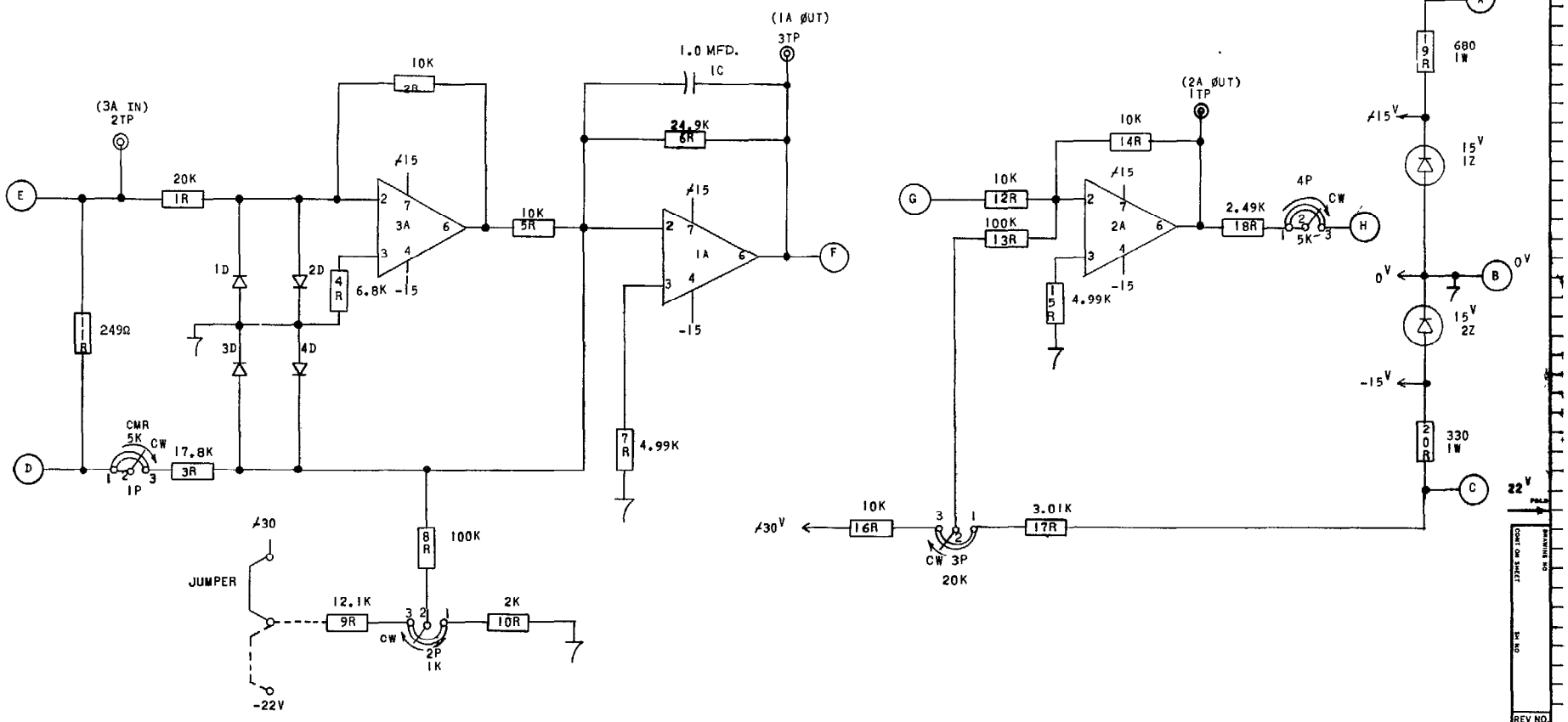
44C306444 CONT ON SHEET FL SH NO 2

FF 809 F (4 72) PRINTED IN U S A

GENERAL ELECTRIC 44C306492

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
7BQI BOILER CONTROL INTERFACE
4-20 MA INPUT
FIRST MADE FOR 44B331746-G03



NOTES:

1. 2TP INPUT

LSS	HSS
4MA	20MA

3TP OUTPUT

LSS	HSS
0V	5V

2. CONNECTIONS

A ±30 VDC
B 0V
C -22 VDC
D BOILER CONTROL COMMON
E INPUT FROM BOILER CONTROL
F OUTPUT
G EHC AUTO SPEED REFERENCE
H AUTO REF POSITION TO BOILER CONTROL

CONT ON SHEET
DRAWING NO
SH NO
REV NO

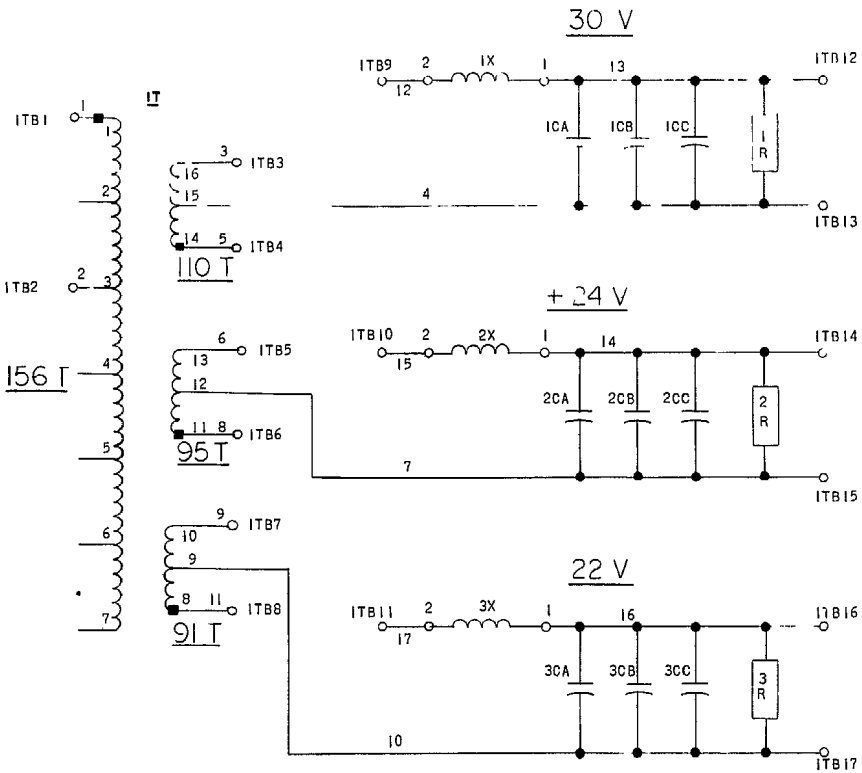
REVISIONS	PRINTS TO
	12P-DFB
	FG-W
	11

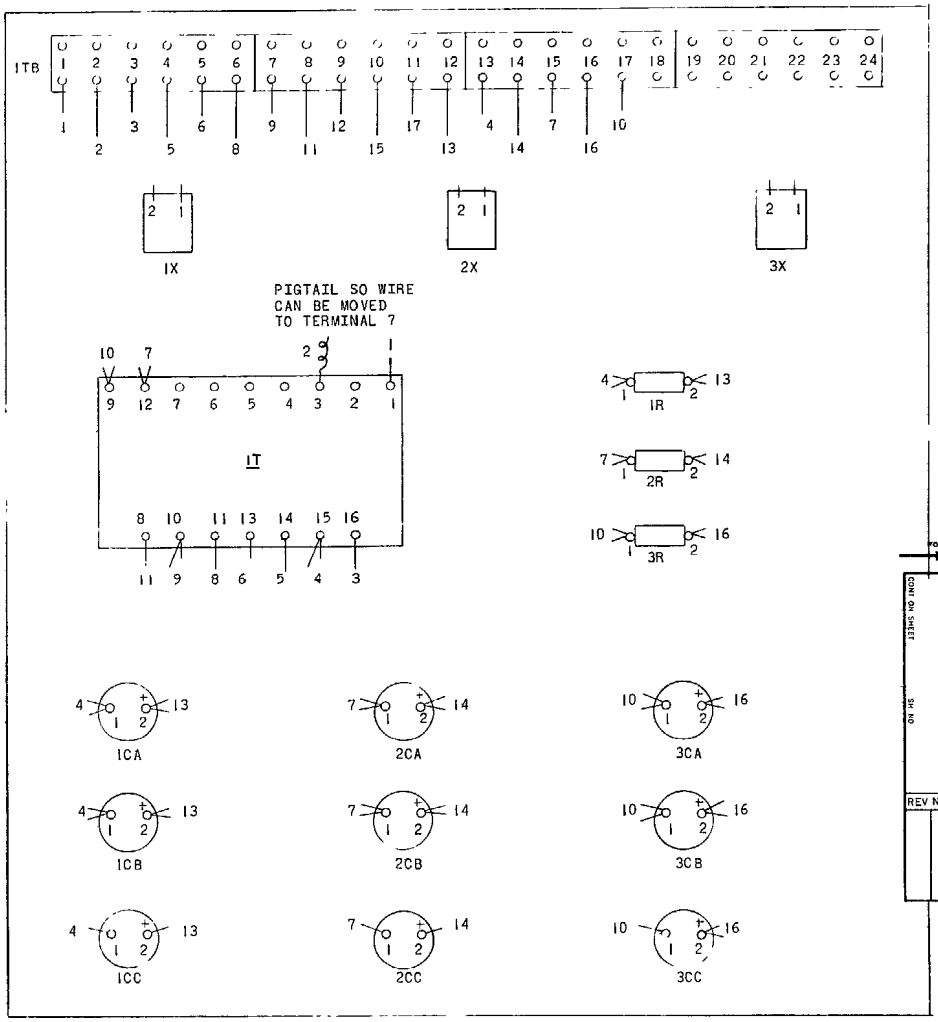
MADE BY D. Rumsford 10-23-72
ISSUED D. Moyer Nov. 2, 1972
APPROVED R. Weber
COMM. & CONT DEVICES, WAYNESBORO, VA
44C306492
CONT ON SHEET Final SH NO 1

3

GENERAL ELECTRIC

TITLE
LINE POWER SUPPLY
FIRST MADE FOR 3S7513KF300A5





CKT. NO.	WIRE SEQUENCE, WIRE M9729494P191B #18 WH.
1	ITB(1),IT(1)
2	ITB(2),IT(3)
3	ITB(3),IT(16)
4	ICC(1),ICB(1),ICA(1),IT(15),IR(1),ITB(13)
5	IT(14),ITB(4)
6	IT(13),ITB(5)
7	2CC(1),2CB(1),2CA(1),2R(1),IT(12),ITB(15)
8	IT(11),ITB(6)
9	IT(10),ITB(7)
10	3CC(1),3CB(1),3CA(1),3R(1),IT(9),ITB(17)
11	IT(8),ITB(8)
13	ICC(2),ICB(2),ICA(2),IR(2),ITB(12)
14	2CC(2),2CB(2),2CA(2),2R(2),ITB(14)
16	3CC(2),3CB(2),3CA(2),3R(2),ITB(16)

1X

CKT.	LEAD NO.	WIRE TO
13	1	ITB(12)
12	2	ITB(9)

2X

CKT.	LEAD NO.	WIRE TO
14	1	ITB(14)
15	2	ITB(10)

3X

CKT.	LEAD NO.	WIRE TO
16	1	ITB(16)
17	2	ITB(11)

REVISIONS		PRINTS TO
1	Nov 9, 1972	
2	May 10, 1973	
3	May 30, 1974	

Oct 20, 1972 D Mayer

DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.

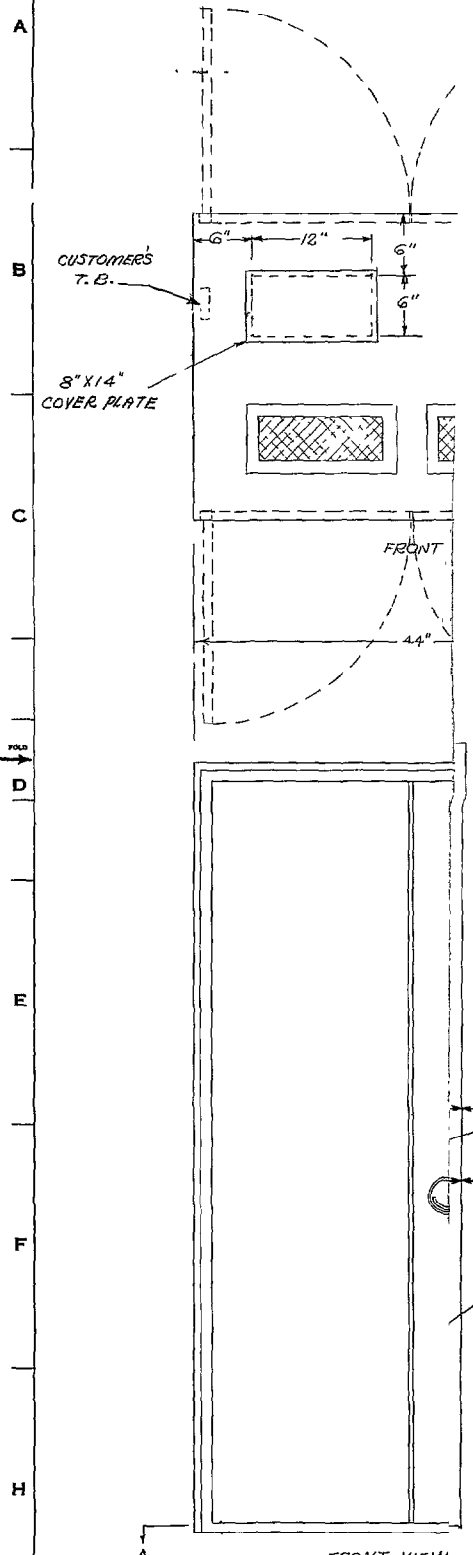
44C306526

GENERAL ELECTRIC | 44D300043 CONT ON SHEET SH NO

TITLE: ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3875I3TC300A2

CABINET OUTLINE



NOTES:

1. ONE CABINET REQUIRED FOR EACH BOILER FEED PUMP TURBINE.
2. ESTIMATED WT. 1500 LBS.
3. MAXIMUM AMBIENT TEMP. 40°C
4. FINISH OUTSIDE: ASA49 GRAY
   INSIDE : GLOSS WHITE LACQUER
   DOORS : BLUE AIR DRY LACQUER
5. CABINET HAS LOOSE PIN REMOVABLE HINGED DOORS.
6. DISCONNECTION OF CABINET/REMOTE CONTROL CABLE ENTRANCE EITHER FROM TOP ☐ OR BOTTOM ☐ AS CUSTOMER DESIRES. (CUSTOMER TO CHECK ONE BLOCK.)
7. CUSTOMER'S PROVISIONS FOR STRAPPING CABLE RUN NEXT TO CUSTOMER'S TERMINAL BOARDS.
8. IF NON-STANDARD FINISH REQUIRED SPECIFY ON ORDERING DRAWING.
9. POWER CONSUMPTION/DISSIPATION 750W.

REVISIONS			PRINTS TO
1		DEC 14 1967	11
2		MAR 7 1968	DP-IFB
3		NOV 29 1968	FS-W
4		FEB 21 1969	
5		APR 02 1969	
6			
7	C.A. Ellenberger 3.b.13.1973		

APPROVALS: C. Simkins NOV 21 1967 | J.R.M.

SPECIALTY CONTROL DIV OR DEPT | WAYNESBORO, VA LOCATION | 44D300043 FORM OR CATEG

44D300043 ON SHEET 2 SH NO

FORM 44D300043 ON SHEET

FORM 45-813-F (5-67) PRINTED IN U.S.A.

1 2 10

GENERAL ELECTRIC — 44D305954

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

REV NO | CONT ON SHEET | SH NO

TITLE: ENC BOILER FEEDPUMP ALIGNMENT DIAGRAM

FIRST MADE FOR 357613TC300A11, 091-40778

TAMPA ELECTRIC BENJ #3 TB 122881

PRIMARY CONTROL-5.00V
BACK-UP CONTROL-5.05V
AT RATED SPEED SSPU 4225 HZ

LSS ALARM X RELAY 0225

LOW SPEED SWITCH

SSPU-1 SSPU-2 SSPU-3

AMPLIFIER NOT CONTROLLING AT +10V.

1LVG-0209 PRIMARY BACKUP

BACK UP SPEED TEST

HP OIL SUPPLY | RAM | CONTROL VALVE | TURBINE | GEAR | SERVO VALVE | POS OPENS

HP STOP VALVE STEAM | LP STOP VALVE STEAM

1DI-0104 1DI 0102 NULL ZERO METER CAL

LVDT-1 LVDT-2 LVDT-3

REMOTE CONTROL PANEL

HP STOP VALVE POSITION | LP STOP VALVE POSITION

2FVC 0224 COARSE FINE

200 RPM

IVC 0221 OUTPUT REF LSSX RELAY

110% SPEED | 109% SPEED | 105% SPEED

IVC-0219 IVC-0218 OUTPUT DO-CCW REF

110SX RELAY | +30VDC | 105SX RELAY

3A02 1A 0110 3A02 1A 0108 2A

TI-1 0-1MA | 0-1MA

0-8000 RPM REMOTE CONT PANEL

NOT CONTROLLING AT + 10V

N CHSS

MAN D 3 0114 | AUTO D 4 0115

OUT-PUT

ROPOUT=LOCAL

IVC-0117 REF DO=CW

NTROL INDICATION

AX RELAY

POWER SUPPLIES

LPS 0227 LOW SPEED SW POWER SUPPLY 56V DC 12V DC

OSC-0217 METER LEVEL 6V RMS 3KHZ 5MA

3KHZ 6V

ISR-0308 INPUT LEVEL METER | +30V BUS BAR | 30 LINE A | 30 LINE B | F ISR-0310

0 VOLT BUS BAR

ISR-0316 | 22 LINE A | -22V BUS BAR | 22 LINE B | F ISR-0318

ISR-0324 | 24 LINE A | 24V | 24 LINE B | F ISR-0326

0328-F | 0326-G

120V LINE A 60HZ T24 T25

METER RELAYS

POWER PANEL

LINE B 120V, 60HZ T22 T23

POWER PANEL

REVISIONS

		PRINTS TO
1	Cathy Lewis June 26, 1972	1E
2	C.D. Card Nov. 22, 1972	12P-DMB2
3	W.M. Cook May 17, 1973	FG-W
4	Dick Mayer Aug 23, 1973	12P-F-G2

MADE BY R. Engelstad 6-7-72 | APPROVED R.J. Gilbert | ISSUED Dick Mayer June 12, 1972

GENER. & CONT. SERVICES DIV OR DEPT | WAYNESBORO, VA. LOCATION | 44D305954 CONT ON SHEET SH NO

5

GENERAL ELECTRIC

44D305988

TITLE

44D305988 CONTROL PANEL

FIRST MADE FOR 3S7513KL300A9

RPM

201 337 1CØ-V 1CØ-M 201 1CØ-N

CONNECTORS - DIODE BOARD (BACK VIEW)

PLACE 1CØ, 2CØ, 3CØ, LABELS ON FRONT & BACK OF BRACKETS.

1CØ

2CØ

3CØ

3CØ-K 3CØ-J 2CØ-A

1313 1315 1300

TRIP TRIP RESET TRIP

1314 1316 1301 1302

3CØ-m 3CØ-m 2CØ-J 2CØ-C

1206

1CØ-X 1CØ-S 1CØ-m

PWR/ALRM LAMP CHECK

HSS

NC NO 1CØ J

RF

1CØ-F 1CØ-B 1CØ-C

CABLE CHECK

1CØ-K 203 2CØ-m 3CØ-F

DB

1267 1223 1219 1222 1200

1CØ, 2CØ, 3CØ

⬡ INDICATES DIODE BOA

REVISIONS		PRINTS TO
1	D. Hampton Aug 9, 1973	11
		12-DEB
		FG-W

APPROVALS

Ellerman 5/1/73

eyer May 11, 1973 R Calvey

DRIVE SYSTEMS PROD DEPT. DEPT OR SECTION

WAYNESBORO, VA. LOCATION

44D305988 CONT ON SHEET SH NO

PF-813-P 10-71 PRINTED IN U.S.A.

1 2 10

2

GENERAL ELECTRIC

44D305989 CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

REV NO 44D305989 CONT ON SHEET SH NO

TITLE RELAY PANEL

FIRST MADE FOR 357513XF307A14

WIRE TABLE

CIRCUIT	WIRE SEQUENCE
201	RF,R,L,LF,ITDX,ITB
203	CC,MLSS,MLSS,MHSS,MHSS,ALSS,AHSS,ST, ALARM,P,P,N,AX,TRIP, 2CB,3TB
204	105X,CC,MLSS,MLSX,MHSS,MHSX,ALSS,AHSS, HOLD,OTT,LF,LF,ST,I10SX,ALARM,N,ITDX, AX,LSS,ALARMX,LSSX,ICB,ITD,2TB-1
204	2CB-2, 2TB
344	ICB-4,ICB-20
402	LF,RF
403	L,LF
405	R,RF
416	MHSS,MHSX
417	MLSS,MLSX
1106	ITD,ITDX
1107	TRIP,ST
1116	ITD,ITD,CC
1117	ST,ST,ITDX
1118	ITDX,MLSS
1119	ICB,P
1120	P,ICB
1121	RF,LF,ST
1124	L,R
1125	R,RF
1126	L,2CB
1127	L,RF
1128	RF,R
1129	R,L,LF
1130	R,2CB
1131	R,RF
1132	RF,R,L
1134	RF,R
1135	TFSO,MHSS
1136	MHSS,AHSS
1137	AHSS,OTT
1138	TFSO,TFSR,TFSL
1139	TFSL,2CB
1140	TFSR,2CB
1225	ALARM,2FL
1226	2FL,2CB
1230	TRIP,TFSO,TFSR,TFSL,ICB
1248	LSSX,ITDX

FACTORY NOTES:

1. USE #20 WIRE, M9729494P1618
2. DO NOT CORD-TIE WIRES. INTERCONNECTING WIRES TO BE ADDED AFTER PANEL IS MOUNTED IN CASE ASM.
3. CONNECT DIODES TO EACH RELAY SOCKET AS SHOWN IN RELAY SYMBOL.
4. WIRE CONNECTION SHOWN ON MASTER WIRE TABLE.

REACTOR WIRE TABLE

CIRCUIT	CONNECTIONS
1227	1X, 2TB
1228	2X, 2TB
1229	1X, ICB
1230	2X, ICB

REVISIONS		PRINTS TO
1	D. Wangler Aug 9, 1973	11
		12F-DEP
		FG-W

MADE BY Glenn Ellison 5/1/73 ISSUED R. Meyer May 11, 1973

APPROVALS R Culley

DRIVE SYSTEMS PROD. DEPT MANUFACTURING, VA.

44D305989 CONT ON SHEET SH NO

GENERAL ELECTRIC CO.
WAYNESBORO, VA.

REGULATOR SUMMARY 44X303673

COMPILED BY: R. J. Culver **DATE:** June 12, 1972 **SHEETS; SHEET** 1

CUSTOMER: MDTD For Tampa Electric Big Bend #3

TURBINES: 122881 REQN.: 091-40778

ITEM	MATERIAL	QUAN.
1	3S7513TC300A11 Electronic Feedpump Turbine Control	1
	Outline Drawing 44D300043	
	Interconnection Dwg. 44A306301	
	Elementary Dwg. 44C306284	
	Connection Dwg. 44C306285	
	Alignment Dwg. 44D305954	
	Alignment Instructions 44A306302	
	Renewal Parts 44A335415	
2	3S7513KL300A9 Control Panel	1
	Outline 44B306123	
3	Cabinet-Control Panel Power Cable	2
	~~XXXXXXXX~~ 44B306148	
4	Cabinet-Control Panel Control Cable	1
	44B300423	

REVISIONS 1/ Oct. 3, 1972 E.A. Ewing

DISTRIBUTION

11	
12P DFB	2
FG-W	
12P-FS	2

5

DRIVE SYSTEMS PRODUCT DEPARTMENT
GENERAL ELECTRIC COMPANY, WAYNESBORO, VIRGINIA 22980

GENERAL ELECTRIC

8-73 (50)